

# Information Storage and Retrieval Systems

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توصيف مقرر دراسي

| ١- بيانات المقرر                                                                                                                                                                                                                                                                                                                                                                                            |                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| الفرقة /المستوى : الفرقة الثانية / شعبة تسجيل طبي                                                                                                                                                                                                                                                                                                                                                           | اسم المقرر : نظم الحفظ and retrieval Systems |
| الرمز الكودي :                                                                                                                                                                                                                                                                                                                                                                                              | عدد الوحدات الدراسية :                       |
| Theory 24 Contact Hours                                                                                                                                                                                                                                                                                                                                                                                     |                                              |
| Practical 35 Contacts Hours                                                                                                                                                                                                                                                                                                                                                                                 |                                              |
| ٢- هدف المقرر:                                                                                                                                                                                                                                                                                                                                                                                              |                                              |
| This course will introduce the student to the fundamentals of information storage and retrieval (ISR) systems, including components, models, structures, information representation, vocabulary control, search strategies (query language/operation, indexing and search) . We will also cover some practical information retrieval (IR) systems new trends and issues such as electronic patient records. |                                              |
| ٣- المستهدف من تدريس المقرر :                                                                                                                                                                                                                                                                                                                                                                               |                                              |
| ILOS                                                                                                                                                                                                                                                                                                                                                                                                        | ١. المعلومات والمفاهيم :                     |
| Identify the dimensions of the information retrieval problem";<br>Describe the functions of an information retrieval system;<br>Identify the components of an information retrieval system;<br>Define the factors which optimize the information retrieval process;<br>Describe the current issues in information retrieval.                                                                                |                                              |
| <ul style="list-style-type: none"> <li>Interpret the accuracy of information retrieval Importance</li> <li>Analyze the problems that occur with information systems and who to report them to.</li> <li>Differentiate between information storing and retrieval systems</li> </ul>                                                                                                                          | ب- المهارات الذهنية :                        |
| <ul style="list-style-type: none"> <li>-Use information electronically for hospital information systems</li> <li>-Examine files quickly by naming/storing them appropriately.</li> <li>-Practice individual files can be password protected to prevent</li> <li>- Interpret of Storage of files when they are no longer in regular use.</li> </ul>                                                          | ج- المهارات المهنية الخاصة بالمقرر:          |
| <ul style="list-style-type: none"> <li>-Consider communication skills with patients</li> <li>- Retrieve recent data from websites</li> <li>- Respect patient's values and needs</li> <li>- Manage time effectively</li> </ul>                                                                                                                                                                               | د- المهارات العامة :                         |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <p><b>- Gather and organize information and presentation skills</b></p> <p><b>Introduction of Information Retrieval System</b><br/> <b>Basics of Information Retrieval Systems</b><br/> <b>How Does Information Retrieval Work</b><br/> <b>Information Representation</b><br/> <b>The Information Retrieval System Interface</b><br/> <b>Automated Searching</b><br/> <b>Information Retrieval System Capabilities</b><br/> <b>Information Visualization</b><br/> <b>Introduction to medical records</b><br/> <b>Electronic Health Record Systems</b><br/> <b>Electronic Medical Record Systems application</b><br/> <b>New trends on ISR</b></p> | <p>٤- محتوى المقرر:</p>                                           |
| <p><b>Teaching</b><br/> <b>Lectures</b><br/> <b>Practical classes</b><br/> <b>Self-directed learning</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>٥- أساليب التعليم والتعلم</p>                                  |
| <p><b>Individual guidance</b><br/> <b>Individual feedback</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>٦- أساليب التعليم والتعلم للطلاب<br/> ذوى القدرات المحدودة</p> |
| <p><b>Written Exam : Short essay - MCQ 90</b><br/> <b>Practical exam 40</b><br/> <b>Assignments 20</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>٧- تقويم الطلاب<br/> أ- الأساليب المستخدمة</p>                 |
| <ul style="list-style-type: none"> <li>• <b>Formative assessment:</b> quizzes throughout the course</li> <li>• <b>Continuous assessment:</b> home assignments, self-learning tasks,</li> <li>• <b>Summative assessment:</b> Final exam Written (Essay, MCQ)</li> </ul> <p>Practical exam<br/> <b>Mid-Term Examination (20 marks ) (10 marks assignments &amp; 10 marks quiz ) After 6 weeks from the start</b></p> <ul style="list-style-type: none"> <li>• <b>-Final-Term Examination ( 90 marks) At the end of the 12th week</b></li> </ul>                                                                                                     | <p>ب- التوقيت</p>                                                 |
| <p><b>Written Exam Final 90 Marks</b><br/> <b>Practical 40 Marks</b><br/> <b>Assignment &amp; Attendance 20 Marks</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>ج- توزيع الدرجات</p>                                           |
| <p>٨- قائمة الكتب الدراسية والمراجع :</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                   |
| <ul style="list-style-type: none"> <li>- Lecture Notes</li> <li>- Practical Notes</li> <li>- Recommended Books: <ul style="list-style-type: none"> <li>• Information Storage And Retrieval Systems Theory and Implementation Second or 3rd Edition, Gerald J. Kowalski, Mark T. Maybury</li> <li>• Heting Chu. Information Representation and retrieval in the digital age. Medford, NJ: Information Today Press 2003. (ASIS&amp;T monograph Series).</li> </ul> </li> </ul>                                                                                                                                                                      | <p>أ- مذكرات</p>                                                  |



|                                                                                                                                                                                                                                                                                                                                                                                                                            |                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <ul style="list-style-type: none"> <li>Charles T. Meadow, Bert R. Boyce, Donald H. Kraft, Carol L Barry. Text Information Retrieval Systems. Third Edition (Library and Information Science) (Library and Information Science). Academic Press 2007-01-23. ISBN: 0123694124/ 0-12-369412-4</li> </ul>                                                                                                                      |                                           |
| <p>INFORMATION STORAGE AND RETRIEVAL SYSTEMS Theory and Implementation</p> <p>Second or 3rd Edition, Gerald J. Kowalski, Mark T. Marbury</p>                                                                                                                                                                                                                                                                               | <p>ب- كتب ملزمة</p>                       |
| <p>1. Hating Chu. Information Representation and retrieval in the digital age. Medford, NJ: Information Today Press 2003. (ASIS&amp;T monograph Series).</p> <p>2. Charles T. Meadow, Bert R. Boyce, Donald H. Kraft, Carol L Barry. Text Information Retrieval Systems. Third Edition (Library and Information Science) (Library and Information Science). Academic Press 2007-01-23. ISBN: 0123694124/ 0-12-369412-4</p> | <p>ج- كتب مقترحة</p>                      |
| <p>Downie, J. Stephen. (2003). Music information retrieval. Annual Review of Information Science and Technology, 37, 295-340. Also available from <a href="http://musicir.org/downie_mir_arist37.pdf">http://musicir.org/downie_mir_arist37.pdf</a>. Ministry of health and population website <a href="http://www.mohp.gov.eg">Www. mohp.gov.eg</a></p>                                                                   | <p>د- دوريات علمية أو نشرات ..... الخ</p> |

## Course Description

This course will introduce the student to the fundamentals of information storage and retrieval (ISR) systems, including components, models, structures, information representation, vocabulary control, search strategies (query language/operation, indexing and search). We will also cover some practical information retrieval (IR) systems on the Web and for the libraries. A good familiarity with computers and some programming experience is highly desirable.

### Core Knowledge

- Understand the dimensions of the information retrieval "problem";
- Know the functions of an information retrieval system;
- Identify the components of an information retrieval system;
- Define the factors which optimize the information retrieval process;
- Examine current issues in information retrieval.

### Core Skills

- How to make sure information is accurate.
- The problems that occur with information systems and who to report them to.
- Knowing the advantages of storing information electronically
- Knowing how to find files quickly by naming/storing them appropriately.
- Be aware of individual files can be password protected to prevent
- Be aware of Storage of files when they are no longer in regular use.
- Be aware of All electronic files can be converted to a format by which all can be centrally indexed, searched, sorted and managed.
- Be aware about the importance of largely use in the medical industry for electronic sharing and storage of images, e.g. X-rays, scans.

## Course Overview

| ID               | Topics                                                    | Methods of Teaching/ Training with<br>Number of Total Hours per Topic |            |                      |          |     |
|------------------|-----------------------------------------------------------|-----------------------------------------------------------------------|------------|----------------------|----------|-----|
|                  |                                                           | Interactive<br>Lecture                                                | Field Work | Class<br>Assignments | Research | Lab |
| 1                | Introduction of Information Storage and Retrieval Systems | 2                                                                     |            |                      |          |     |
| 2                | What is information?                                      | 2                                                                     |            | 1                    |          | 1   |
| 3                | Basics of Information Storage and Retrieval               | 2                                                                     |            | 3                    |          | 1   |
| 4                | Structure of ISR                                          | 2                                                                     |            | 1                    |          | 1   |
| 5                | ISR representation                                        | 2                                                                     |            | 6                    |          | 1   |
|                  | Retrieval Techniques and Query representation             | 2                                                                     |            | 4                    |          | 2   |
| 6                | Retrieval Approaches                                      | 2                                                                     |            | 1.5                  |          | 1   |
| 7                | Information Retrieval Models                              | 2                                                                     |            | 1.5                  |          | 1   |
| 8                | IR Systems                                                | 2                                                                     |            | 1                    |          | 1   |
| 9                | New Trends in ISR                                         | 2                                                                     |            | 1                    |          | 1   |
| 10               | New Trends in ISR                                         | 2                                                                     |            | 1                    |          | 1   |
| 11               | The future of information retrieval                       | 2                                                                     |            | 1                    |          | 1   |
| TOTAL HOURS (36) |                                                           | 24                                                                    |            | 20                   |          | 12  |



# Chapter 1

## Introduction of Information Retrieval Systems

### Introduction of Information Retrieval Systems

- 1.1 Definition of Information Retrieval System
- 1.2 Objectives of Information Retrieval Systems
- 1.3 Functional Overview
- 1.4 Relationship to Database Management Systems
- 1.5 Digital Libraries and Data Warehouses
- 1.6 Summary

This chapter defines an Information Storage and Retrieval System (called an Information Retrieval System for brevity) and differentiates between information retrieval and database management systems. Tied closely to the definition of an Information Retrieval System are the system objectives. It is satisfaction of the objectives that drives those areas that receive the most attention in development. For example, academia pursues all aspects of information systems, investigating new theories, algorithms and heuristics to advance the knowledge base. Academia does not worry about response time, required resources to implement a system to support thousands of users nor operations and maintenance costs associated with system delivery. On the other hand, commercial institutions are not always concerned with the optimum theoretical approach, but the approach that minimizes development costs and increases the salability of their product. This text considers both view points and technology states. Throughout this text, information retrieval is viewed from both the theoretical and practical viewpoint.

The functional view of an Information Retrieval System is introduced to put into perspective the technical areas discussed in later chapters. As detailed algorithms and architectures are discussed, they are viewed as subfunctions within a total system. They are also correlated to the major objective of an Information Retrieval System which is minimization of human resources required in the

finding of needed information to accomplish a task. As with any discipline, standard measures are identified to compare the value of different algorithms. In information systems, precision and recall are the key metrics used in evaluations. Early introduction of these concepts in this chapter will help the reader in understanding the utility of the detailed algorithms and theory introduced throughout this text.

There is a potential for confusion in the understanding of the differences between Database Management Systems (DBMS) and Information Retrieval Systems. It is easy to confuse the software that optimizes functional support of each type of system with actual information or structured data that is being stored and manipulated. The importance of the differences lies in the inability of a database management system to provide the functions needed to process “information.” The opposite, an information system containing structured data, also suffers major functional deficiencies. These differences are discussed in detail in Section 1.4.

### 1.1. Definition of Information Retrieval System

An Information Retrieval System is a system that is capable of storage, retrieval, and maintenance of information. Information in this context can be composed of text (including numeric and date data), images, audio, video and other multi-media objects. Although the form of an object in an Information Retrieval System is diverse, the text aspect has been the only data type that lent itself to full functional processing. The other data types have been treated as highly informative sources, but are primarily linked for retrieval based upon search of the text. Techniques are beginning to emerge to search these other media types (e.g., EXCALIBUR’s Visual RetrievalWare, VIRAGE video indexer). The focus of this book is on research and implementation of search, retrieval and representation of textual and multimedia sources. Commercial development of pattern matching against other data types is starting to be a common function integrated within the total information system. In some systems the text may only be an identifier to display another associated data type that holds the substantive information desired by the system’s users (e.g., using closed captioning to locate video of interest.) The term “user” in this book represents an end user of the information system who has minimal knowledge of computers and technical fields in general.

The term “item” is used to represent the smallest complete unit that is processed and manipulated by the system. The definition of item varies by how a specific source treats information. A complete document, such as a book, newspaper or magazine could be an item. At other times each chapter, or article may be defined as an item. As sources vary and systems include more complex processing, an item may address even lower levels of abstraction such as a contiguous passage of text or a paragraph. For readability, throughout this book the terms “item” and “document” are not in this rigorous definition, but used interchangeably. Whichever is used, they represent the concept of an item. For most of the book it is best to consider an item as text. But in reality an item may be a combination of many modals of information. For example a video news program could be considered an item. It is composed of text in the form of closed captioning, audio text provided by the speakers, and the video images being displayed. There are multiple “tracks” of information possible in a single item. They are typically correlated by time. Where the text discusses multimedia information retrieval keep this expanded model in mind.

An Information Retrieval System consists of a software program that facilitates a user in finding the information the user needs. The system may use standard computer hardware or specialized hardware to support the search subfunction and to convert non-textual

sources to a searchable media (e.g., transcription of audio to text). The gauge of success of an information system is how well it can minimize the overhead for a user to find the needed information. Overhead from a user's perspective is the time required to find the information needed, excluding the time for actually reading the relevant data. Thus search composition, search execution, and reading non-relevant items are all aspects of information retrieval overhead.

The first Information Retrieval Systems originated with the need to organize information in central repositories (e.g., libraries) (Hyman-82). Catalogues were created to facilitate the identification and retrieval of items. Chapter 3 reviews the history of cataloging and indexing. Original definitions focused on "documents" for information retrieval (or their surrogates) rather than the multi-media integrated information that is now available (Minker-77, Minker-77.)

As computers became commercially available, they were obvious candidates for the storage and retrieval of text. Early introduction of Database Management Systems provided an ideal platform for electronic manipulation of the indexes to information (Rather-77). Libraries followed the paradigm of their catalogs and references by migrating the format and organization of their hardcopy information references into structured databases. These remain as a primary mechanism for researching sources of needed information and play a major role in available Information Retrieval Systems. Academic research that was pursued through the 1980s was constrained by the paradigm of the indexed structure associated with libraries and the lack of computer power to handle large (gigabyte) text databases. The Military and other Government entities have always had a requirement to store and search large textual databases. As a result they began many independent developments of textual Information Retrieval Systems. Given the large quantities of data they needed to process, they pursued both research and development of specialized hardware and unique software solutions incorporating Commercial Off The Shelf (COTS) products where possible. The Government has been the major funding source of research into Information Retrieval Systems. With the advent of inexpensive powerful personnel computer processing systems and high speed, large capacity secondary storage products, it has become commercially feasible to provide large textual information databases for the average user. The introduction and exponential growth of the Internet along with its initial WAIS (Wide Area Information Servers) capability and more recently advanced search servers (e.g., INFOSEEK, EXCITE) has provided a new avenue for access to terabytes of information (over 800 million indexable pages -Lawrence-99.) The algorithms and techniques to optimize the processing and access of large quantities of textual data were once the sole domain of segments of the Government, a few industries, and academics. They have now become a needed capability for large quantities of the population with significant research and development being done by the private sector. Additionally the volumes of nontextual information are also becoming searchable using specialized search capabilities. Images across the Internet are searchable from many web sites such as WEBSEEK, DITTO.COM, ALTAVISTA/IMAGES. News organizations such as the BBC are processing the audio news they have produced and are making historical audio news searchable via the audio transcribed versions of the news. Major video organizations such as Disney are using video indexing to assist in finding specific images in their previously produced videos to use in future videos



or incorporate in advertising. With exponential growth of multi-media on the Internet capabilities such as these are becoming common place. Information Retrieval exploitation of multi-media is still in its infancy with significant theoretical and practical knowledge missing.

## 1.2. Objectives of Information Retrieval Systems

The general objective of an Information Retrieval System is to minimize the overhead of a user locating needed information. Overhead can be expressed as the time a user spends in all of the steps leading to reading an item containing the needed information (e.g., query generation, query execution, scanning results of query to select items to read, reading non-relevant items). The success of an information system is very subjective, based upon what information is needed and the willingness of a user to accept overhead. Under some circumstances, needed information can be defined as all information that is in the system that relates to a user's need. In other cases it may be defined as sufficient information in the system to complete a task, allowing for missed data. For example, a financial advisor recommending a billion dollar purchase of another company needs to be sure that all relevant, significant information on the target company has been located and reviewed in writing the recommendation. In contrast, a student only requires sufficient references in a research paper to satisfy the expectations of the teacher, which never is all inclusive. A system that supports reasonable retrieval requires fewer features than one which requires comprehensive retrieval. In many cases comprehensive retrieval is a negative feature because it overloads the user with more information than is needed. This makes it more difficult for the user to filter the relevant but non-useful information from the critical items. In information retrieval the term "relevant" item is used to represent an item containing the needed information. In reality the definition of relevance is not a binary classification but a continuous function. From a user's perspective "relevant" and "needed" are synonymous. From a system perspective, information could be relevant to a search statement (i.e., matching the criteria of the search statement) even though it is not needed/relevant to user (e.g., the user already knew the information). A discussion on relevance and the natural redundancy of relevant information is presented in Chapter 11.

The two major measures commonly associated with information systems are precision and recall. When a user decides to issue a search looking for information on a topic, the total database is logically divided into four segments shown in Figure 1.1. Relevant items are those documents that contain information that helps the searcher in answering his question. Non-relevant items are those items that do not provide any directly useful information. There are two possibilities with respect to each item: it can be retrieved or not retrieved by the user's query. Precision and recall are defined as:

$$\text{Precision} = \frac{\text{Number\_Retrieved\_Relevant}}{\text{Number\_Total\_Retrieved}}$$

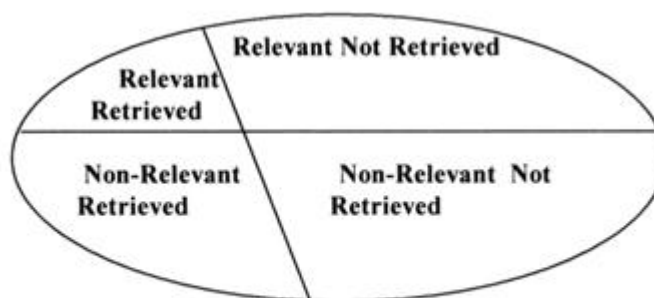


Figure 1.1 Effects of Search on Total Document Space

$$\text{Recall} = \frac{\text{Number\_Retrieved\_Relevant}}{\text{Number\_Possible\_Relevant}}$$

where Number\_Possible\_Relevant are the number of relevant items in the database. Number\_Total\_Retrieved is the total number of items retrieved from the query. Number\_Retrieved\_Relevant is the number of items retrieved that are relevant to the user's search need. Precision measures one aspect of information retrieval overhead for a user associated with a particular search. If a search has a 85 per cent precision, then 15 per cent of the user effort is overhead reviewing non-relevant items. Recall gauges how well a system processing a particular query is able to retrieve the relevant items that the user is interested in seeing. Recall is a very useful concept, but due to the denominator, is non-calculable in operational systems. If the system knew the total set of relevant items in the database, it would have retrieved them. Figure 1.2a shows the values of precision and recall as the number of items retrieved increases, under an optimum query where every returned item is relevant. There are "N" relevant items in the database. Figures 1.2b and 1.2c show the optimal and currently achievable relationships between Precision and Recall (Harman-95). In Figure 1.2a the basic properties of precision (solid line) and recall (dashed line) can be observed. Precision starts off at 100 per cent and maintains that value as long as relevant items are retrieved. Recall starts off close to zero and increases as long as relevant items are retrieved until all possible relevant items have been retrieved. Once all "N" relevant items have been retrieved, the only items being retrieved are non-relevant. Precision is directly affected by retrieval of non-relevant items and drops to a number close to zero. Recall is not effected by retrieval of non-relevant items and thus remains at 100 per



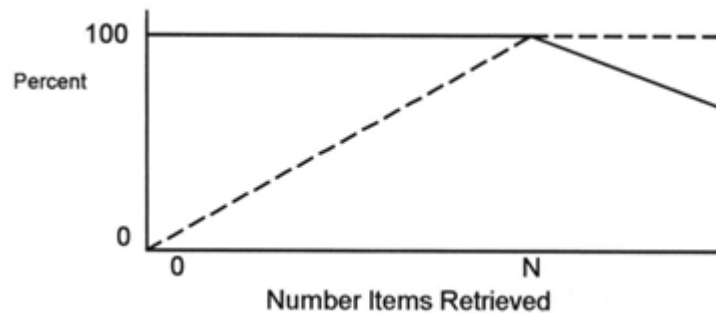


Figure 1.2a Ideal Precision and Recall



Figure 1.2b Ideal Precision/Recall Graph

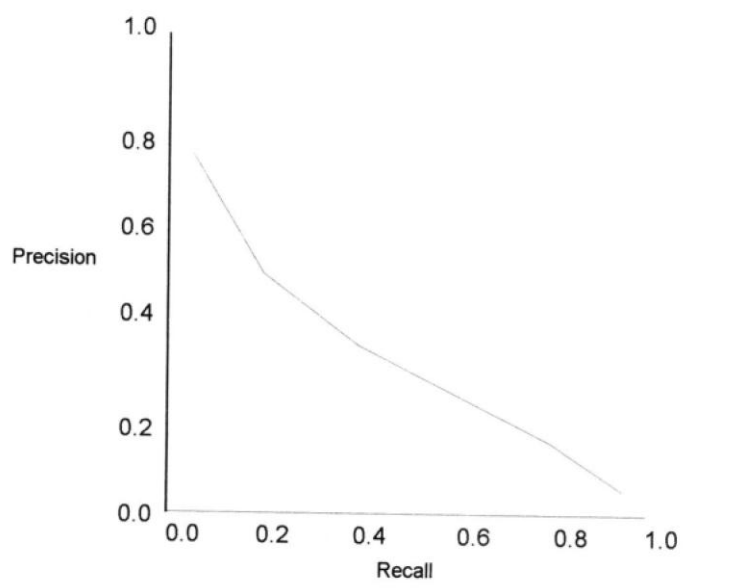


Figure 1.2c Achievable Precision/Recall Graph

Cent once achieved. Precision/Recall graphs show how values for precision and recall change within a search results file (Hit file) as viewed from the most relevant to least relevant item. As with Figure 1.2a, in the ideal case every item retrieved is relevant. Thus precision stays at 100 per cent (1.0). Recall continues to increase by moving to the right on the x-axis until it also reaches the 100 per cent (1.0) point. Although Figure 1.2c stops here, continuation stays at the same x-axis location (recall never changes) but precision

decreases down the y-axis until it gets close to the x-axis as more non-relevant are discovered and precision decreases. Figure 1.2c is from the latest TREC conference (see Chapter 11) and is representative of current capabilities.

To understand the implications of Figure 1.2c, it is useful to describe the implications of a particular point on the precision/recall graph. Assume that there are 100 relevant items in the data base and from the graph at precision of .3 (i.e., 30 per cent) there is an associated recall of .5 (i.e., 50 per cent). This means there would be 50 relevant items in the Hit file from the recall value. A precision of 30 per cent means the user would likely review 167 items to find the 50 relevant items.

The first objective of an Information Retrieval System is support of user search generation. There are natural obstacles to specification of the information a user needs that come from ambiguities inherent in languages, limits to the user's ability to express what information is needed and differences between the user's vocabulary corpus and that of the authors of the items in the database. Natural languages suffer from word ambiguities such as homographs and use of acronyms that allow the same word to have multiple meanings (e.g., the word "field" or the acronym "U.S."). Disambiguation techniques exist but introduce significant system overhead in processing power and extended search times and often require interaction with the user.

Many users have trouble in generating a good search statement. The typical user does not have significant experience with nor even the aptitude for Boolean logic statements. The use of Boolean logic is a legacy from the evolution of database management systems and implementation constraints. Until recently, commercial systems were based upon databases. It is only with the introduction of Information Retrieval Systems such as Retrieval Ware, TOPIC, AltaVista, Info seek and INQUERY that the idea of accepting natural language queries is becoming a standard system feature. This allows users to state in natural language what they are interested in finding. But the completeness of the user specification is limited by the user's willingness to construct long natural language queries. Most users on the Internet enter one or two search terms.

Multi-media adds an additional level of complexity in search specification. Where the modal has been converted to text (e.g., audio transcription, OCR) the normal text techniques are still applicable. But query specification when searching for an image, unique sound, or video segment lacks any proven best interface approaches. Typically they are achieved by having presorted examples of known objects in the media and letting the user select them for the search (e.g., images of leaders allowing for searches on "Tony Blair".) This type specification becomes more complex when coupled with Boolean or natural language textual specifications.

In addition to the complexities in generating a query, quite often the user is not an expert in the area that is being searched and lacks domain specific vocabulary unique to that particular subject area. The user starts the search process with a general concept of the information required, but not has a focused definition of exactly what is needed. A limited knowledge of the vocabulary associated with a particular area along with lack of

focus on exactly what information is needed leads to use of inaccurate and in some cases misleading search terms. Even when the user is an expert in the area being searched, the ability to select the proper search terms is constrained by lack of knowledge of the author's vocabulary. All writers have a vocabulary limited by their life experiences, environment where they were raised and ability to express themselves. Other than in very technical restricted information domains, the user's search vocabulary does not match the author's vocabulary. Users usually start with simple queries that suffer from failure rates approaching 50% (Nordlie-99).

Thus, an Information Retrieval System must provide tools to help overcome the search specification problems discussed above. In particular the search tools must assist the user automatically and through system interaction in developing a search specification that represents the need of the user and the writing style of diverse authors (see Figure 1.3) and multi-media specification.

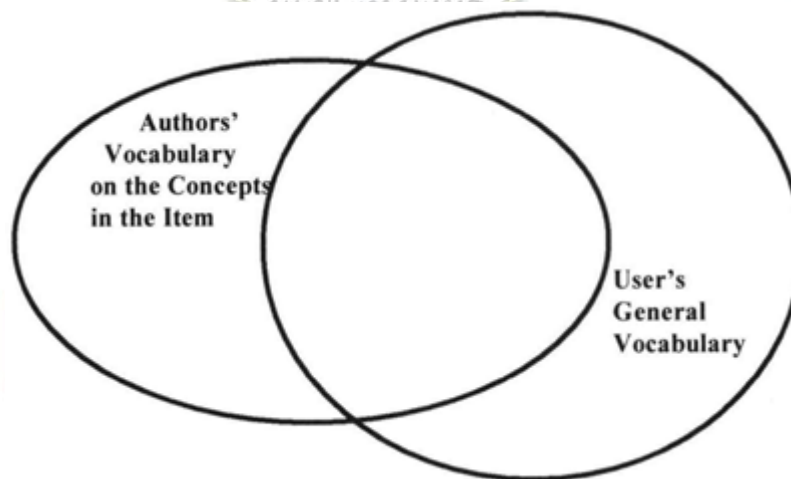


Figure 1.3 Vocabulary Domains

In addition to finding the information relevant to a user's needs, an objective of an information system is to present the search results in a format that facilitates the user in determining relevant items. Historically data has been presented in an order dictated by how it was physically stored. Typically, this is in arrival to the system order, thereby always displaying the results of a search sorted by time. For those users interested in current events this is useful. But for the majority of searches it does not filter out less useful information. Information Retrieval Systems provide functions that provide the results of a query in order of potential relevance to the user. This, in conjunction with user search status (e.g., listing titles of highest ranked items) and item formatting options, provides the user with features to assist in selection and review of the most likely relevant items first. Even more sophisticated techniques use item clustering, item summarization and link analysis to provide additional item selection insights (see Chapter 8.) Other features such as viewing only "unseen" items also help a user who cannot complete the item review process in one session. In the area of Question/Answer systems that is coming into focus in Information Retrieval, the retrieved items are not returned to the user. Instead the answer to their question - or a short segment of text that contains the answer - is what is returned.



This is a more complex process than summarization since the results need to be focused on the specific information need versus general area of the user's query. The approach to this problem most used in TREC - 8 was to first perform a search using existing algorithms, then to syntactically parse the highest ranked retrieved items looking for specific passages that answer the question. See Chapter 11 for more details.

Multi-media information retrieval adds a significant layer of complexity on how to display multi-modal results. For example, how should video segments potentially relevant to a user's query be represented for user review and selection? It could be represented by two thumbnail still images of the start and end of the segment, or should the major scene changes be represented (the latter technique would avoid two pictures of the news announcer versus the subject of the video segment.)

### 1.3. Functional Overview

A total Information Storage and Retrieval System is composed of four major functional processes: Item Normalization, Selective Dissemination of Information (i.e., "Mail"), archival Document Database Search, and an Index Database Search along with the Automatic File Build process that supports Index Files. Commercial systems have not integrated these capabilities into a single system but supply them as independent capabilities. Figure 1.4 shows the logical view of these capabilities in a single integrated Information Retrieval System. Boxes are used in the diagram to represent functions while disks represent data storage.

#### 1.3.1. Item Normalization

The first step in any integrated system is to normalize the incoming items to a standard format. In addition to translating multiple external formats that might be received into a single consistent data structure that can be manipulated by the functional processes, item normalization provides logical restructuring of the item. Additional operations during item normalization are needed to create a searchable data structure: identification of processing tokens (e.g., words), characterization of the tokens, and stemming (e.g., removing word endings) of the tokens. The original item or any of its logical subdivisions is available for the user to display. The processing tokens and their characterization are used to define the searchable text from the total received text. Figure 1.5 shows the normalization process.

Standardizing the input takes the different external formats of input data and performs the translation to the formats acceptable to the system. A system may have a single format for all items or allow multiple formats. One example of standardization could be translation of foreign languages into Unicode. Every language has a different internal binary encoding for the characters in the language. One standard encoding that covers English, French, Spanish, etc. is ISO-Latin. There are other internal encodings for other language groups such as Russian (e.g., KOI-7, KOI-8), Japanese, Arabic, etc. Unicode is an evolving international standard based upon 16 bits (two bytes) that will be able to represent



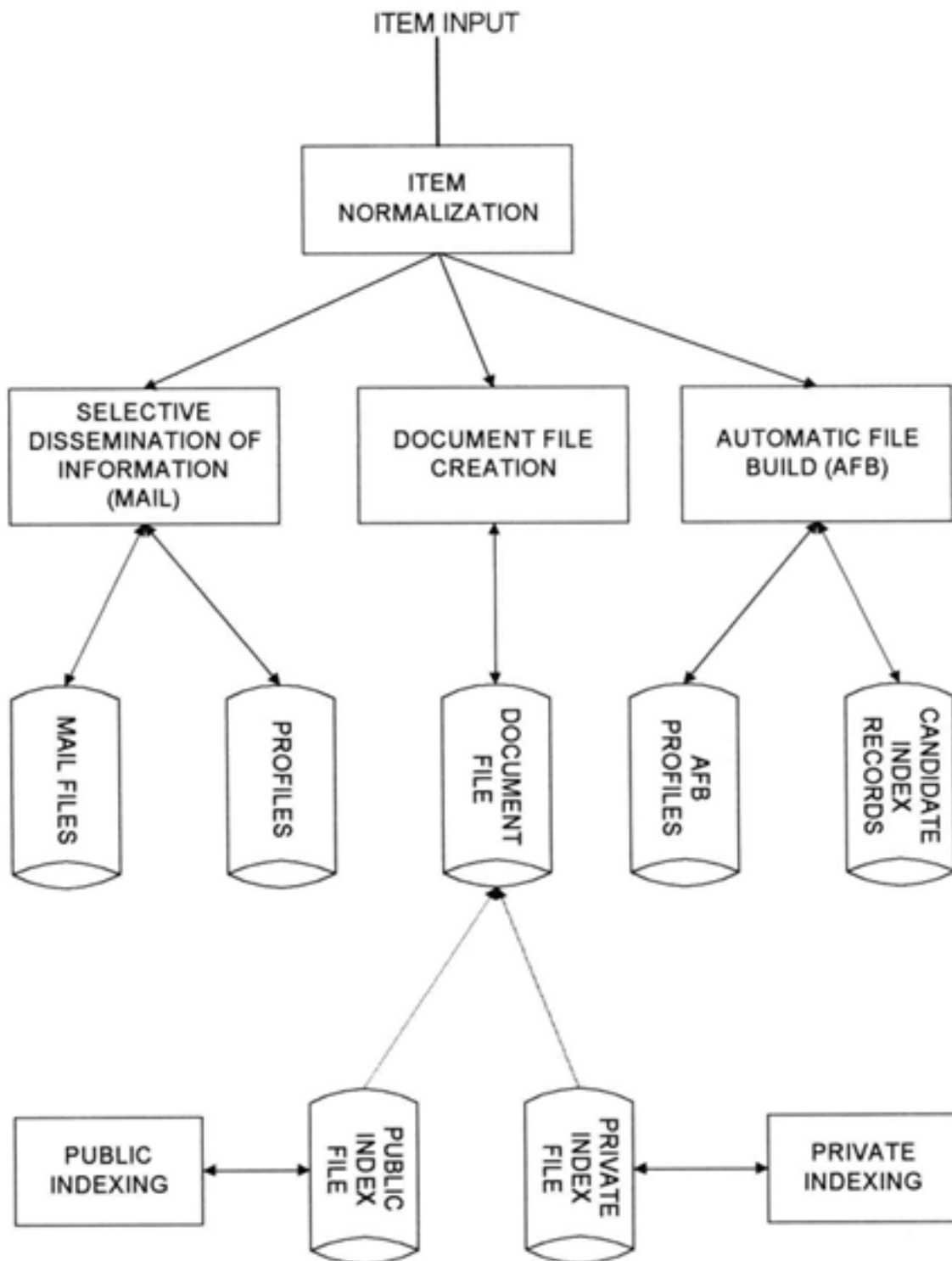


Figure 1.4 Total Information Retrieval Systems

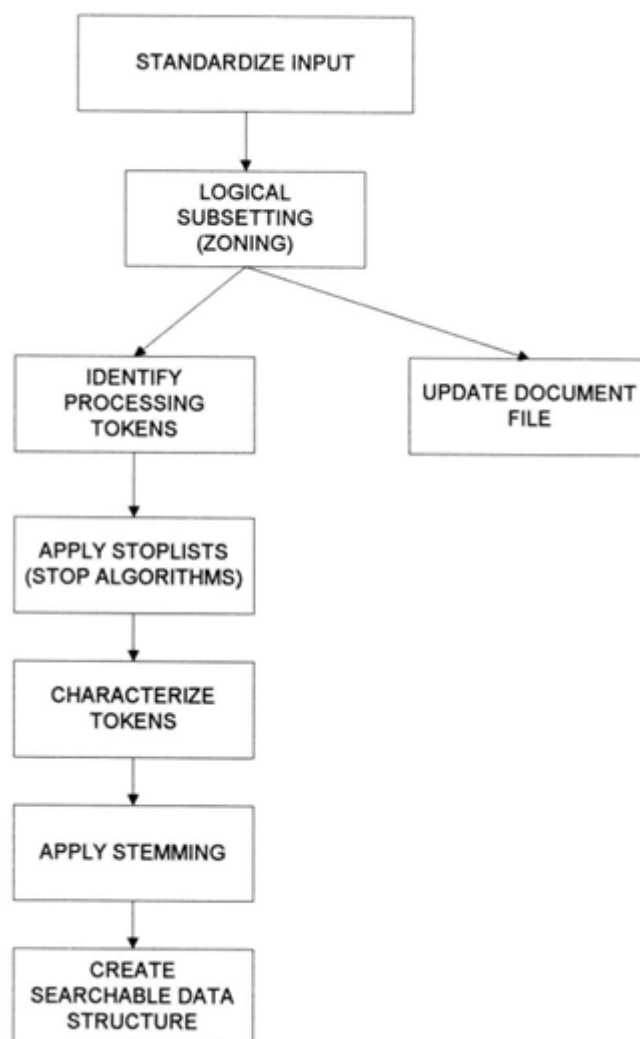


Figure 1.5 The Text Normalization Process

all languages. Unicode based upon UTF-8, using multiple 8-bit bytes, is becoming the practical Unicode standard. Having all of the languages encoded into a single format allows for a single browser to display the languages and potentially a single search system to search them. Of course such a search engine would have to have the capability of understanding the linguistic model for all the languages to allow for correct tokenization (e.g., word boundaries, stemming, word stop lists, etc.) of each language.

Multi-media adds an extra dimension to the normalization process. In addition to normalizing the textual input, the multi-media input also needs to be standardized. There are a lot of options to the standards being applied to the normalization. If the input is video the likely digital standards will be either MPEG-2, MPEG-1, AVI or Real Media. MPEG (Motion Picture Expert Group) standards are the most universal standards for higher quality video where Real Media is the most common standard for lower quality video being used on the Internet. Audio standards are typically WAV or Real Media (Real Audio). Images vary from

JPEG to BMP. In all of the cases for multi-media, the input analog source is encoded into a digital format. To index the modal different encodings of the same input may be required (see Section 1.3.5 below). But the importance of using an encoding standard for the source that allows easy access by browsers is greater for multi-media than text that already is handled by all interfaces.

The next process is to parse the item into logical sub-divisions that have meaning to the user. This process, called “Zoning,” is visible to the user and used to increase the precision of a search and optimize the display. A typical item is sub-divided into zones, which may overlap and can be hierarchical, such as Title, Author, Abstract, Main Text, Conclusion, and References. The term “Zone” was selected over field because of the variable length nature of the data identified and because it is a logical sub-division of the total item, whereas the term “fields” has a connotation of independence. There may be other source-specific zones such as “Country” and “Keyword.” The zoning information is passed to the processing token identification operation to store the information, allowing searches to be restricted to a specific zone. For example, if the user is interested in articles discussing “Einstein” then the search should not include the Bibliography, which could include references to articles written by “Einstein.” Zoning differs for multi-media based upon the source structure. For a news broadcast, zones may be defined as each news story in the input. For speeches or other programs, there could be different semantic boundaries that make sense from the user’s perspective.

Once a search is complete, the user wants to efficiently review the results to locate the needed information. A major limitation to the user is the size of the display screen which constrains the number of items that are visible for review. To optimize the number of items reviewed per display screen, the user wants to display the minimum data required from each item to allow determination of the possible relevance of that item. Quite often the user will only display zones such as the Title or Title and Abstract. This allows multiple items to be displayed per screen. The user can expand those items of potential interest to see the complete text.

Once the standardization and zoning has been completed, information (i.e., words) that are used in the search process need to be identified in the item. The term processing token is used because a “word” is not the most efficient unit on which to base search structures. The first step in identification of a processing token consists of determining a word. Systems determine words by dividing input symbols into three classes: valid word symbols, inter-word symbols, and special processing symbols. A word is defined as a contiguous set of word symbols carefully selected. Examples of word symbols are alphabetic characters and numbers. Examples of possible inter-word symbols are blanks, periods and semicolons. The exact definition of an inter-word symbol is dependent upon the aspects of the language domain of the items to be processed by the system. For example, an apostrophe may be of little importance if only used for the possessive case in English, but might be critical to represent foreign names in the database. Based upon the required accuracy of searches and language characteristics, a trade off is made on the selection of inter-word symbols. Finally there are some symbols that may require special processing. A hyphen can be used many ways, often left to the taste and judgment of the

writer (Bernstein-84). At the end of a line it is used to indicate the continuation of a word. In other places it links independent words to avoid absurdity, such as in the case of “small business men.” To avoid interpreting this as short males that run businesses, it would properly be hyphenated “small-business men.” Thus when a hyphen (or other special symbol) is detected a set of rules are executed to determine what action is to be taken generating one or more processing tokens.

Next, a Stop List/Algorithm is applied to the list of potential processing tokens. The objective of the Stop function is to save system resources by eliminating from the set of searchable processing tokens those that have little value to the system. Given the significant increase in available cheap memory, storage and processing power, the need to apply the Stop function to processing tokens is decreasing. Nevertheless, Stop Lists are commonly found in most systems and consist of words (processing tokens) whose frequency and/or semantic use make them of no value as a searchable token. For example, any word found in almost every item would have no discrimination value during a search. Parts of speech, such as articles (e.g., “the”), have no search value and are not a useful part of a user’s query. By eliminating these frequently occurring words the system saves the processing and storage resources required to incorporate them as part of the searchable data structure. Stop Algorithms go after the other class of words, those found very infrequently.

Zipf (Zipf-49) postulated that, looking at the frequency of occurrence of the unique words across a corpus of items, the majority of unique words are found to occur a few times. The rank-frequency law of Zipf is:

$$\text{Frequency} * \text{Rank} = \text{constant}$$

where Frequency is the number of times a word occurs and rank is the rank order of the word. The law was later derived analytically using probability and information theory (Fairthorne-69). Table 1.1 shows the distribution of words in the first TREC test database (Harman-93), a database with over one billion characters and 500,000 items. In Table 1.1, WSJ is Wall Street Journal (1986-89), AP is AP Newswire (1989), ZIFF - Information from Computer Select disks, FR -Federal Register (1989), and DOE - Short abstracts from Department of Energy.

The highly precise nature of the words only found once or twice in the database reduce the probability of their being in the vocabulary of the user and the terms are almost never included in searches. Eliminating these words saves on storage and access structure (e.g., dictionary - see Chapter 4) complexities. The best technique to eliminate the majority of these words is via a Stop algorithm versus trying to list them individually. Examples of Stop algorithms are:

Stop all numbers greater than “999999” (this was selected to allow dates to be searchable)

Stop any processing token that has numbers and characters intermixed



The algorithms are typically source specific, usually eliminating unique item numbers that are frequently found in systems and have no search value.

In some systems (e.g., INQUIRE DBMS), inter-word symbols and Stop words are not included in the optimized search structure (e.g., inverted file structure, see Chapter 4) but are processed via a scanning of potential hit documents after inverted file search reduces the list of possible relevant items. Other systems never allow interword symbols to be searched.

| Source                               | WSJ     | AP      | ZIFF    | FR      | DOE     |
|--------------------------------------|---------|---------|---------|---------|---------|
| Size in Mbytes                       | 295     | 266     | 251     | 258     | 190     |
| Median number terms/record           | 182     | 353     | 181     | 313     | 82      |
| Average number terms/record          | 329     | 375     | 412     | 1017    | 89      |
| Number Unique Terms                  | 156,298 | 197,608 | 173,501 | 126,258 | 186,225 |
| Number of Terms Occurring Once       | 64,656  | 89,627  | 85,992  | 58,677  | 95,782  |
| Average number terms occurrences > 1 | 199     | 174     | 165     | 106     | 159     |

**Table 1.1 Distribution of words in TREC Database (from TREC-1 Conference Proceedings, Harmon-93)**

The next step in finalizing on processing tokens is identification of any specific word characteristics. The characteristic is used in systems to assist in disambiguation of a particular word. Morphological analysis of the processing token's part of speech is included here. Thus, for a word such as "plane," the system understands that it could mean "level or flat" as an adjective, "aircraft or facet" as a noun, or "the act of smoothing or evening" as a verb. Other characteristics may classify a token as a member of a higher class of tokens such as "European Country" or "Financial Institution." Another example of characterization is if upper case should be preserved. In most systems upper/lower case is not preserved to avoid the system having to expand a term to cover the case where it is the first word in a sentence. But, for proper names, acronyms and organizations, the upper case represents a completely different use of the processing token versus it being found in the text. "Pleasant Grant" should be recognized as a person's name versus a "pleasant grant" that provides funding. Other characterizations that

are typically treated separately from text are numbers and dates.

Once the potential processing token has been identified and characterized, most systems apply stemming algorithms to normalize the token to a standard semantic representation. The decision to perform stemming is a tradeoff between precision of a search (i.e., finding exactly what the query specifies) versus standardization to reduce system overhead in expanding a search term to similar token representations with a potential increase in recall. For example, the system must keep singular, plural, past tense, possessive, etc. as separate searchable tokens and potentially expand a term at search time to all its possible representations, or just keep the stem of the word, eliminating endings. The amount of stemming that is applied can lead to retrieval of many non-relevant items. The major stemming algorithms used at this time are described in Chapter 4. Some systems such as RetrievalWare, that use a large dictionary/thesaurus, looks up words in the existing dictionary to determine the stemmed version in lieu of applying a sophisticated algorithm.

Once the processing tokens have been finalized, based upon the stemming algorithm, they are used as updates to the searchable data structure. The searchable data structure is the internal representation (i.e., not visible to the user) of items that the user query searches. This structure contains the semantic concepts that represent the items in the database and limits what a user can find as a result of their search. When the text is associated with video or audio multi-media, the relative time from the start of the item for each occurrence of the processing token is needed to provide the correlation between the text and the multi-media source. Chapter 4 introduces the internal data structures that are used to store the searchable data structure for textual items and Chapter 5 provides the algorithms for creating the data to be stored based upon the identified processing tokens.

### 1.3.2. Selective Dissemination of Information

The Selective Dissemination of Information (Mail) Process (see Figure 1.4) provides the capability to dynamically compare newly received items in the information system against standing statements of interest of users and deliver the item to those users whose statement of interest matches the contents of the item. The Mail process is composed of the search process, user statements of interest (Profiles) and user mail files. As each item is received, it is processed against every user's profile. A profile contains a typically broad search statement along with a list of user mail files that will receive the document if the search statement in the profile is satisfied. User search profiles are different than ad hoc queries in that they contain significantly more search terms (10 to 100 times more terms) and cover a wider range of interests. These profiles define all the areas in which a user is interested versus an ad hoc query which is frequently focused to answer a specific question. It has been shown in recent studies that automatically expanded user profiles perform significantly better than human generated profiles (Harman-95).

When the search statement is satisfied, the item is placed in the Mail File(s) associated with the profile. Items in Mail files are typically viewed in time of receipt order and automatically deleted after a specified time period (e.g., after one month) or upon

command from the user during display. The dynamic asynchronous updating of Mail Files makes it difficult to present the results of dissemination in estimated order of likelihood of relevance to the user (ranked order). This is discussed in Chapter 2.

Very little research has focused exclusively on the Mail Dissemination process. Most systems modify the algorithms they have established for retrospective search of document (item) databases to apply to Mail Profiles. Dissemination differs from the ad hoc search process in that thousands of user profiles are processed against one item versus the inverse and there is not a large relatively static database of items to be used in development of relevance ranking weights for an item.

Both implementers and researchers have treated the dissemination process as independent from the rest of the information system. The general assumption has been that the only knowledge available in making decisions on whether an incoming item is of interest is the user's profile and the incoming item. This restricted view has produced suboptimal systems forcing the user to receive redundant information that has little value. If a total Information Retrieval System view is taken, then the existing Mail and Index files are also potentially available during the dissemination process. This would allow the dissemination profile to be expanded to include logic against existing files. For example, assume an index file (discussed below) exists that has the price of oil from Mexico as a value in a field with a current value of \$30. An analyst will be less interested in items that discuss Mexico and \$30 oil prices than items that discuss Mexico and prices other than \$30 (i.e., looking for changes). Similarly, if a Mail file already has many items on a particular topic, it would be useful for a profile to not disseminate additional items on the same topic, or at least reduce the relative importance that the system assigns to them (i.e., the rank value).

Selective Dissemination of Information has not yet been applied to multimedia sources. In some cases where the audio is transformed into text, existing textual algorithms have been applied to the transcribed text (e.g., the DARPA's TIDES Portal), but little research has gone into dissemination techniques for multimedia sources.

### 1.3.3. Document Database Search

The Document Database Search Process (see Figure 1.4) provides the capability for a query to search against all items received by the system. The Document Database Search process is composed of the search process, user entered queries (typically ad hoc queries) and the document database which contains all items that have been received, processed and stored by the system. It is the retrospective search source for the system. If the user is on-line, the Selective Dissemination of Information system delivers to the user items of interest as soon as they are processed into the system. Any search for information that has already been processed into the system can be considered a "retrospective" search for information. This does not preclude the search to have search statements constraining it to items received in the last few hours. But typically the searches span far greater time periods. Each query is processed against the total document database. Queries differ from profiles in that they are typically short and focused on a specific area of interest. The



Document Database can be very large, hundreds of millions of items or more. Typically items in the Document Database do not change (i.e., are not edited) once received. The value of much information quickly decreases over time. These facts are often used to partition the database by time and allow for archiving by the time partitions. Some user interfaces force the user to indicate searches against items received older than a specified time, making use of the partitions of the Document database. The documents in the Mail files are also in the document database, since they logically are input to both processes.

#### 1.3.4. Index Database Search

When an item is determined to be of interest, a user may want to save it for future reference. This is in effect filing it. In an information system this is accomplished via the index process. In this process the user can logically store an item in a file along with additional index terms and descriptive text the user wants to associate with the item. It is also possible to have index records that do not reference an item, but contain all the substantive information in the index itself. In this case the user is reading items and extracting the information of interest, never needing to go back to the original item. A good analogy to an index file is the card catalog in a library. Another perspective is to consider Index Files as structured databases whose records can optionally reference items in the Document Database. The Index Database Search Process (see Figure 1.4) provides the capability to create indexes and search them. The user may search the index and retrieve the index and/or the document it references. The system also provides the capability to search the index and then search the items referenced by the index records that satisfied the index portion of the query. This is called a combined file search. In an ideal system the index record could reference portions of items versus the total item.

There are two classes of index files: Public and Private Index files. Every user can have one or more Private Index files leading to a very large number of files. Each Private Index file references only a small subset of the total number of items in the Document Database. Public Index files are maintained by professional library services personnel and typically index every item in the Document Database. There is a small number of Public Index files. These files have access lists (i.e., lists of users and their privileges) that allow anyone to search or retrieve data. Private Index files typically have very limited access lists.

To assist the users in generating indexes, especially the professional indexers, the system provides a process called Automatic File Build shown in Figure 1.4 (also called Information Extraction). This capability processes selected incoming documents and automatically determine potential indexing for the item. The rules that govern which documents are processed for extraction of index information and the index term extraction process are stored in Automatic File Build Profiles. When an item is processed it results in creation of Candidate Index Records. As a minimum, certain citation data can be determined and extracted as part of this process assisting in creation of Public Index Files. Examples of this information are author(s), date of publication, source, and references. More complex data, such as countries an item is about or corporations referenced, have



high rates of identification. The placement in an index file facilitates normalizing the terminology, assisting the user in finding items. It also provides a basis for programs that analyze the contents of systems trying to identify new information relationships (i.e., data mining). For more abstract concepts the extraction technology is not accurate and comprehensive enough to allow the created index records to automatically update the index files. Instead the candidate index record, along with the item it references, are stored in a file for review and edit by a user prior to actual update of an index file.

The capability to create Private and Public Index Files is frequently implemented via a structured Database Management System. This has introduced new challenges in developing the theory and algorithms that allow a single integrated perspective on the information in the system. For example, how to use the single instance information in index fields and free text to provide a single system value of how the index/referenced item combination satisfies the user's search statement. Usually the issue is avoided by treating the aspects of the search that apply to the structured records as a first level constraint identifying a set of items that satisfy that portion of the query. The resultant items are then searched using the rest of the query and the functions associated with information systems. The evaluation of relevance is based only on this later step. An example of how this limits the user is if part of the index is a field called "Country." This certainly allows the user to constrain his results to only those countries of interest (e.g., Peru or Mexico). But because the relevance function is only associated with the portion of the query associated with the item, there is no way for the user to ensure that Peru items have more importance to the retrieval than Mexican items.

#### 1.3.5. Multimedia Database Search

Chapter 10 provides additional details associated with multi-media search against different modalities of information. From a system perspective, the multimedia data is not logically its own data structure, but an augmentation to the existing structures in the Information Retrieval System. It will reside almost entirely in the area described as the Document Database. The specialized indexes to allow search of the multi-media (e.g., vectors representing video and still images, text created by audio transcription) will be augmented search structures. The original source will be kept as normalized digital real source for access possibly in their own specialized retrieval servers (e.g., the Real Media server, ORACLE Video Server, etc.) The correlation between the multi-media and the textual domains will be either via time or positional synchronization. Time synchronization is the example of transcribed text from audio or composite video sources. Positional synchronization is where the multi-media is localized by a hyperlink in a textual item. The synchronization can be used to increase the precision of the search process. Added relevance weights should be assigned when the multi-media search and the textual search result in hits in close proximity. For example when the image of Tony Blair is found in the section of a video where the transcribed audio is discussing Tony Blair, then the hit is more likely than when either event occurs independently. The same would be true when the JPEG image hits on Tony Blair in a textual paragraph discussing him in an HTML item.

Making the multi-media data part of the Document Database also implies that the

linking of it to Private and Public Index files will also operate the same way as with text.

#### 1.4. Relationship to Database Management Systems

There are two major categories of systems available to process items: Information Retrieval Systems and Data Base Management Systems (DBMS). Confusion can arise when the software systems supporting each of these applications get confused with the data they are manipulating. An Information Retrieval System is software that has the features and functions required to manipulate “information” items versus a DBMS that is optimized to handle “structured” data. Information is fuzzy text. The term “fuzzy” is used to imply the results from the minimal standards or controls on the creators of the text items. The author is trying to present concepts, ideas and abstractions along with supporting facts. As such, there is minimal consistency in the vocabulary and styles of items discussing the exact same issue. The searcher has to be omniscient to specify all search term possibilities in the query.

Structured data is well defined data (facts) typically represented by tables. There is a semantic description associated with each attribute within a table that well defines that attribute. For example, there is no confusion between the meaning of “employee name” or “employee salary” and what values to enter in a specific database record. On the other hand, if two different people generate an abstract for the same item, they can be different. One abstract may generally discuss the most important topic in an item. Another abstract, using a different vocabulary, may specify the details of many topics. It is this diversity and ambiguity of language that causes the fuzzy nature to be associated with information items. The differences in the characteristics of the data are one reason for the major differences in functions required for the two classes of systems.

With structured data a user enters a specific request and the results returned provide the user with the desired information. The results are frequently tabulated and presented in a report format for ease of use. In contrast, a search of “information” items has a high probability of not finding all the items a user is looking for. The user has to refine his search to locate additional items of interest. This process is called “iterative search.” An Information Retrieval System gives the user capabilities to assist the user in finding the relevant items, such as relevance feedback (see Chapters 2 and 7). The results from an information system search are presented in relevance ranked order. The confusion comes when DBMS software is used to store “information.” This is easy to implement, but the system lacks the ranking and relevance feedback features that are critical to an information system. It is also possible to have structured data used in an information system (such as TOPIC). When this happens the user has to be very creative to get the system to provide the reports and management information that are trivially available in a DBMS.

From a practical standpoint, the integration of DBMS's and Information Retrieval Systems is very important. Commercial database companies have already integrated the two types of systems. One of the first commercial databases to integrate the two systems into a single view is the INQUIRE DBMS. This has been available for over fifteen years. A more current example is the ORACLE DBMS that now offers an imbedded capability called



CONVECTIS, which is an informational retrieval system that uses a comprehensive thesaurus which provides the basis to generate “themes” for a particular item. CONVECTIS also provides standard statistical techniques that are described in Chapter 5. The INFORMIX DBMS has the ability to link to Retrieval Ware to provide integration of structured data and information along with functions associated with Information Retrieval Systems.

### 1.5. Digital Libraries and Data Warehouses

Two other systems frequently described in the context of information retrieval are Digital Libraries and Data Warehouses (or Data arts). There is significant overlap between these two systems and an Information Storage and Retrieval System. All three systems are repositories of information and their primary goal is to satisfy user information needs. Information retrieval easily dates back to Vannevar Bush’s 1945 article on thinking (Bush-45) that set the stage for many concepts in this area. Libraries have been in existence since the beginning of writing and have served as a repository of the intellectual wealth of society.

As such, libraries have always been concerned with storing and retrieving information in the media it is created on. As the quantities of information grew exponentially, libraries were forced to make maximum use of electronic tools to facilitate the storage and retrieval process. With the worldwide interneting of libraries and information sources (e.g., publishers, news agencies, wire services, radio broadcasts) via the Internet, more focus has been on the concept of an electronic library. Between 1991 and 1993 significant interest was placed on this area because of the interest in U.S. Government and private funding for making more information available in digital form (Fox-93).

During this time the terminology evolved from electronic libraries to digital libraries. As the Internet continued its exponential growth and project funding became available, the topic of Digital Libraries has grown. By 1995 enough research and pilot efforts had started to support the 1<sup>ST</sup> ACM International Conference on Digital Libraries (Fox-96).

There remain significant discussions on what is a digital library. Everyone starts with the metaphor of the traditional library. The question is how the traditional library functions change as they migrate into supporting a digital collection. Since the collection is digital and there is a worldwide communications infrastructure available, the library no longer must own a copy of information as long as it can provide access. The existing quantity of hardcopy material guarantees that we will not have all digital libraries for at least another generation of technology improvements. But there is no question that libraries have started and will continue to expand their focus to digital formats. With direct electronic access available to users the social aspects of congregating in a library and learning from librarians, friends and colleagues will be lost and new electronic collaboration equivalencies will come into existence (Wiederhold-95).

Indexing is one of the critical disciplines in library science and significant effort has gone into the establishment of indexing and cataloging standards. Migration of many of the library products to a digital format introduces both opportunities and challenges. The full text of items available for search makes the index process a value added effort as described



in Section 1.3. Another important library service is a source of search intermediaries to assist users in finding information. With the proliferation of information available in electronic form, the role of search intermediary will shift from an expert in search to being an expert in source analysis. Searching will identify so much information in the global Internet information space that identification of the “pedigree” of information is required to understand its value. This will become the new refereeing role of a library.

Information Storage and Retrieval technology has addressed a small subset of the issues associated with Digital Libraries. The focus has been on the search and retrieval of textual data with no concern for establishing standards on the contents of the system. It has also ignored the issues of unique identification and tracking of information required by the legal aspects of copyright that restrict functions within a library environment. Intellectual property rights in an environment that is not controlled by any country and their set of laws has become a major problem associated with the Internet. The conversion of existing hardcopy text, images (e.g., pictures, maps) and analog (e.g., audio, video) data and the storage and retrieval of the digital version is a major concern to Digital Libraries. Information Retrieval Systems are starting to evolve and incorporate digitized versions of these sources as part of the overall system. But there is also a lot of value placed on the original source (especially printed material) that is an issue to Digital Libraries and to a lesser concern to Information Retrieval systems. Other issues such as how to continue to provide access to digital information over many years as digital formats change have to be answered for the long term viability of digital libraries.

The term Data Warehouse comes more from the commercial sector than academic sources. It comes from the need for organizations to control the proliferation of digital information ensuring that it is known and recoverable. Its goal is to provide to the decision makers the critical information to answer future direction questions. Frequently a data warehouse is focused solely on structured databases. A data warehouse consists of the data, an information directory that describes the contents and meaning of the data being stored, an input function that captures data and moves it to the data warehouse, data search and manipulation tools that allow users the means to access and analyze the warehouse data and a delivery mechanism to export data to other warehouses, data marts (small warehouses or subsets of a larger warehouse), and external systems.

Data warehouses are similar to information storage and retrieval systems in that they both have a need for search and retrieval of information. But a data warehouse is more focused on structured data and decision support technologies. In addition to the normal search process, a complete system provides a flexible set of analytical tools to “mine” the data. Data mining (originally called Knowledge Discovery in Databases - KDD) is a search process that automatically analyzes data and extract relationships and dependencies that were not part of the database design. Most of the research focus is on the statistics, pattern recognition and artificial intelligence algorithms to detect the hidden relationships of data. In reality the most difficult task is in preprocessing the data from the database for processing by the algorithms. This differs from clustering in information retrieval in that clustering is based upon known characteristics of items, whereas data mining does not depend upon known relationships. For more detail on data mining see the November 1996

*Communications of the ACM* (Vol. 39, Number 11) that focuses on this topic.

### 1.6. Summary

Chapter 1 places into perspective a total Information Storage and Retrieval System. This perspective introduces new challenges to the problems that need to be theoretically addressed and commercially implemented. Ten years ago commercial implementation of the algorithms being developed was not realistic, allowing theoreticians to limit their focus to very specific areas. Bounding a problem is still essential in deriving theoretical results. But the commercialization and insertion of this technology into systems like the Internet that are widely being used changes the way problems are bounded. From a theoretical perspective, efficient scalability of algorithms to systems with gigabytes and terabytes of data, operating with minimal user search statement information, and making maximum use of all functional aspects of an information system need to be considered. The dissemination systems using persistent indexes or mail files to modify ranking algorithms and combining the search of structured information fields and free text into a consolidated weighted output are examples of potential new areas of investigation.

The best way for the theoretician or the commercial developer to understand the importance of problems to be solved is to place them in the context of a total vision of a complete system. Understanding the differences between Digital Libraries and Information Retrieval Systems will add an additional dimension to the potential future development of systems. The collaborative aspects of digital libraries can be viewed as a new source of information that dynamically could interact with information retrieval techniques. For example, should the weighting algorithms and search techniques discussed later in this book vary against a corpus based upon dialogue between people versus statically published material? During the collaboration, in certain states, should the system be automatically searching for reference material to support the collaboration?

## EXERCISES

1. The metric to be minimized in an Information Retrieval System from a user's perspective is user overhead. Describe the places that the user overhead is encountered from when a user has an information need until when it is satisfied. Is system complexity also part of the user overhead?
2. Under what conditions might it be possible to achieve 100 per cent precision and 100 per cent recall in a system? What is the relationship between these measures and user overhead?
3. Describe how the statement that "language is the largest inhibitor to good communications" applies to Information Retrieval Systems.
4. What is the impact on precision and recall in the use of Stop Lists and Stop Algorithms?
5. Why is the concept of processing tokens introduced and how does it relate to a word? What is the impact of searching being based on processing tokens versus the original words in an item.
6. Can a user find the same information from a search of the Document file that is generated by a Selective Dissemination of Information process (Hint - take into consideration the potential algorithmic basis for each system)? Document database search is frequently described as a "pull" process while dissemination is described as a "push" process. Why are these terms appropriate?
7. Does a Private Index File differ from a standard Database Management System (DBMS)? (HINT - there are both structural and functional differences) What problems need to address when using a DBMS as part of an Information retrieval System?
8. What is the logical effect on the Document file when a combined file search of both a Private Index file and Document file is executed? What is returned to the user?
9. What are the problems that need resolution when the concept of dissemination profiles expands to including existing data structures (e.g., Mail files and/or Index files)?
10. What is the difference between the concept of a "Digital Library" and an Information Retrieval System? What new areas of information retrieval research may be important to support a Digital Library?



## Chapter 2

# What Is Information?

### 2.1. What Is Information?

To know what information retrieval is, we must first know what *information* is. There is no fully satisfactory definition. We examine the question in some detail in Chapter 2. Temporarily, consider the oversimplification that information is something (1) that is represented by a set of *symbols*, (2) whose symbols are *organized* or fit into some *structure*, and (3) whose symbols and organization can be to some extent understood by *users* of the symbols who read them. We mean the broadest sense of *read*: seeing and understanding the printed word or graphic images, hearing sounds that convey words or codes, music, etc., feeling texture, tasting food or other substances ingested, or smelling anything. All these can be means by which to convey information.

*Symbol*, like *information*, is a word with many meanings. It can be any recognizable image, whether in print, paint, sound, or even odor. Little is lost if we insist that information can be represented only by symbols. Indeed, the computer scientist Edsger Dijkstra (1989, p. 1401) has stated that “when all is said and done, the only thing computers can do for us is to manipulate symbols.” The symbols can be as simple as a single pulse of sound or electromagnetic energy, essentially a bit, or they can be as complex as a complete symphonic score or a textbook. Between these extremes, and certainly the more common kind of entity to which we normally apply this word, are a national flag, a word, a punctuation mark, or any string of printed characters. We leave for later discussion whether these symbols need such qualities as significance, novelty, truth, or authority to be *information*.

As soon as we get beyond the most elementary of symbols, such as A or 71, then information, or its symbolic representation, must have structure to be understood. ACT, CAT, ATC, and TCA all have quite different meanings (including none, for some readers), as do FOOD FISH and FISH FOOD. The string 12345 can have any of the several possible meanings, even if the reader knows it represents a date (is it JANUARY 23, 1945 or DECEMBER 3, 1945, American style, or 12 MARCH 1945, European style?).

To a large extent, information retrieval, or IR, is concerned with three concepts: how to represent information, how to interpret the structure of its symbols, and how to tell when one set of symbols has the same or a similar meaning as another. All present formidable problems for computers, and equally formidable challenges for the humans involved in the process, for it is the humans who give the symbols *meaning* and *significance*.

## 2.2. What Is Information Retrieval?

In brief, IR involves finding some desired information in a *store* of information or a *database*. The term *database* does not have a generally accepted, precise definition. It may be taken as a collection of records where each record is about some one entity and all the records of the database have some common attributes. A library card catalog is a database, as is a newspaper, and the personnel files of a company. A broad but useful distinction between types of databases is that some are highly structured. This means that the kind of information contained is well defined, as is any coding used to represent it. If information on any given subject is to be found, it will be known in advance whether or not it is in records of the database. For example, a database covering inventory of automobiles at a dealership will certainly include type of engine, various extras, and color of the car, it is unlikely to include day of the week the car was assembled; there is no point looking for that information, although it is sometimes mentioned in reviews as important to the probable reliability of the car. At the other extreme, if records of the database are in the form of natural language text, a searcher can never be sure whether or not the type of information sought is present or, if present, how it will be represented or phrased. Thus, there are highly structured databases and what are often called unstructured ones, and there are others in between. Implicit in IR is the concept of *selectivity*; to exercise selectivity usually requires that a price be paid in effort, time, money, or all three. Information recovery is not the same as IR unless there has been a search and selection process. Copying a complete disk file is not retrieval in our sense. Watching the news on CNN all day with the viewer exercising no control over what is being shown, once the channel was selected, is not retrieval in our sense. A library is the best example of an institution devoted to selective retrieval. One does not go there to read the entire collection. One goes to look for something selectively, often something that will satisfy a set of highly individualized information needs.

We can categorize the types of information retrieval systems (IRS) as:

1. *Table look-up*—This is the simplest form, so simple that it is not usually included as a type of IR. It assumes a table sorted on one or more fields. Searches are done only by looking for a match on a sort key, then, if found, retrieving other fields in the table. Looking up a telephone number in a typical

telephone directory is an example. We search by name and retrieve address or phone number.

2. *Search for partial match on a field value*—This, which can be done with many database systems or spreadsheet software, allows for a search for a character string within a field rather than the entire content of the field. In an alphabetical telephone directory it could eliminate the troublesome problem of wondering whether the “vons” or “vanders” come before, or following, the family name, i.e. is it von Trapp or Trapp, von?

3. *Searching the content of a record for a match or partial match on any set of symbols, using Boolean algebra to specify the search criteria*—This is what we commonly do when using traditional or World Wide Web (WWW) search engines.

IR is a communication process. In one sense it is a means by which authors or creators of records communicate with readers, but indirectly and with a possibly long time lag between creation of a message or text and its delivery to the IRS user. Sometimes the IRS or librarian conducting a search may pass on information about the probable relevance or value of what is retrieved, thereby adding information to the set of retrieved items. The records of a database are created and assembled without knowledge of exactly who will read them, or under what circumstances. The languages and channels of such a communication system are quite different from other well-known models, such as broadcasting or point-to-point communication.

Is IR a computer activity? It is not strictly necessary that it be, but as a practical matter, that is what we usually imply by the term, and this book is primarily about that usage. The computer system, consisting of both hardware and software, is what we call the IRS. The term IRS may include the database, but whether or not it does may depend on the context in which the expression is used. However, all the important principles of IR apply to purely non-mechanical IR as well. For example, a nonmechanized library functions as a retrieval system, in this case the staff or users are the instruments of the search and decision making. King Henry II of England, in 1160, rode with his entourage from palace to palace, permanently on tour. Written records had to be carried along. Records, such as memos or letters, were stitched one to the previous one, creating an ever-growing scroll carried on the backs of mules. It grew so large that it was called the Great Roll of the Pipe. A new one was started on the first day of each year (Duggan, 1940, pp. 150-151).

We will primarily be discussing *interactive* IR. Today, this is found in three primary modes: (1) A local system operating in a single computer, in which are stored both retrieval software and a database, often on CD-ROMs. An example is an encyclopedia, stored on a disk together with its search software. (2) A central service, consisting of a number of databases with usually a single system for searching them, operated remotely from most users and requiring a telecommunications network for access. This was the primary online mode until the WWW or Web came along. (3) A search engine operating on the Web, which does not have its own databases but finds and searches “sites” owned by others.

A remote retrieval service is operated through the medium of a computer in which the data are stored in files or databases and in which software interprets users’ requests for information, finds it, and controls its transmission to the requester. There is relatively little difference between local and remote systems in terms of operation from the user’s point of view. Both are interactive in that there is almost continuous conversation between the user and the database computer. There tends to be a big difference between WWW pages whose internal structure may vary considerably from site to site and the more consistently structured records of conventional databases. Any of them is a long way from King Henry’s pipe.

Yet another view of the question of the mechanical nature of IR is that we do not usually think of library catalogs, printed indexes, or encyclopedias as mechanical aids, but they are, even if they are older technologies. When first created, they may have seemed as strangely mechanical as today’s computers do to some present-day information users. An encyclopedia is a collection of relatively small works bundled together with extensive



indexes, citations, and cross references. It is not designed to be read from beginning to end, but to be used as a mechanism for finding articles of interest, often only after much searching.

While the selection of a record from an inventory file or a file of bank depositor accounts can be considered IR, the term is more commonly applied to the searching of files of text records or records descriptive of text or graphics. The uncertainty about how the content should be queried underlies much of the activity of IR. When querying is restricted to use of a single search key, such as a part or account number, there is not much challenge.

Blair (1990, p. vii) has stated that the “central problem of Information Retrieval is how to represent documents for retrieval.” We are inclined to think that this is becoming less true as more and more databases consist of the complete text of documents, while previously they consisted of index or classification terms that represented the documents’ content. Blair’s statement is critical for descriptions of graphic images represented as alpha-numeric codes and values.

Not only is the original, natural language of the text being used in the databases, but natural language is being used, increasingly, as the means of a user’s telling the retrieval system what is wanted. This is called the *query*; whether it is oral or written in natural language or any of the many artificial languages developed for the purpose. While other than Web queries often tend toward complexity, Spink *et al.* (2001) looked at over a million Web queries, from over 200,000 users of the Excite engine on September 16, 1997. They found the mean number of terms per query was between 2.3 and 2.4. Hence, natural language texts can be searched using just a few words; how successfully is another question.

It appears that the central problem is becoming how to match, compare, or relate the user’s often simplistic request for information with the texts stored in the database. As Liddy *et al.* (1995, p. 106) put it, “[A] successful retrieval system must retrieve on the basis of what people mean in their query, not just what they say.” On the other hand, the query might be considered a form of document representation, in that it represents characteristics of documents to be retrieved. If so, we can agree with Blair that it is a question of major importance how best to phrase this query, i.e., how to represent what we want to find in documents.

### 2.3. How Does Information Retrieval Work?

The process of selectively searching for information in a database can be viewed as starting from at least two different points. One starting point is the *user*, or prospective user, of information. We will henceforth consider *user* to mean a person who has information need and who uses an IRS to attempt to satisfy that need. At the other starting point, a person or agency makes a decision to collect information and organize and store it for later search and retrieval. There are many complex steps before stored information and user come together.

We present here a model of the retrieval process and will use it as a basis for all the more detailed discussions to come. The reader should be aware that there are other models and even other ideas about what is meant by a model of the IR process (Bookstein, 1983; Boyce and Kraft, 1985; Croft, 1982; Harter, 1986; Kochen, 1974; Korfhage, 1997; Lancaster, 1968; Maron and Kuhns, 1960; Salton and McGill, 1983; Soergel, 1985; van Rijsbergen, 1979, 1986). Generally, the purpose of any of them is to set a framework within which the reader can interpret various aspects of the entire process.

The base model is illustrated in Fig. 1.1. This is an overall view of IR showing that it is a process that depends on separate actions of at least two groups. A user may be an individual or part of a community of some sort. It may be hard to tell from where a need for information arose. Some individual develops a personal need, even though he or she may not know exactly what is needed at this point. When, eventually, the information is found (or even found not to exist) typically, some new information is created and contributed to the world. In order to carry out their selection function database producers study what kind of information is available and what is likely to be needed by their clients. Then they collect and use it to create the records of a database. Back now to the user, whose information need may be conveyed to an intermediary, a person whose profession it is to assist users. The provision of information by the database producer and its search and retrieval by the user are asynchronous. Both will fare better if they have some understanding of each other. The user should know what producers' policies are, and the producers should know of users' general needs and characteristics. Figures 1.2-1.4 show this three- or four-way interaction (user-intermediary-IRS-creation of new records) in more detail.

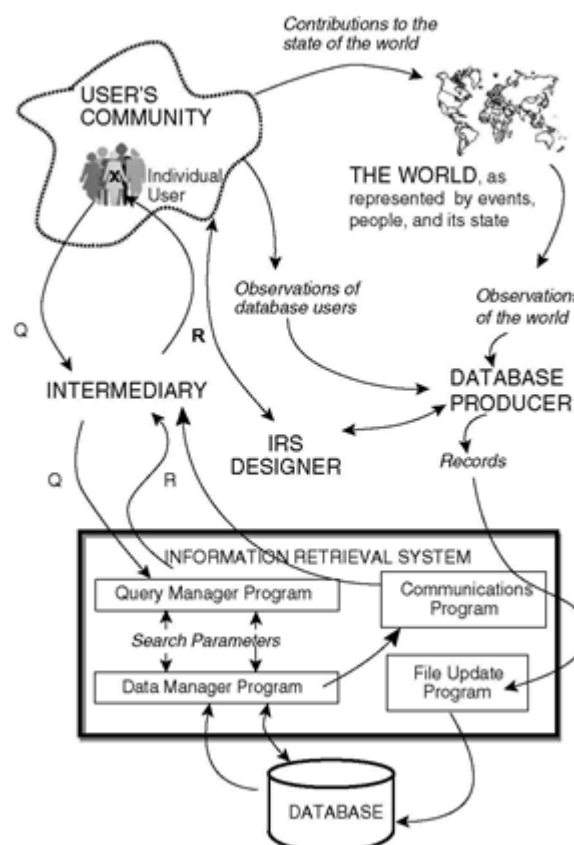


Figure 2.1

General view of an IRS: the system is cyclic, with information generally flowing in a clockwise direction in this diagram. Information about the world comes from a community of users or potential users, who also represent the creators of information, and this is used, in turn, to affect the world, including the users. The software producer provides the mechanism for storage and retrieval. Note that each major segment operates asynchronously but with some degree of dependence on the others.

### 2.3.1. The User Sequence

The user does not exist in a vacuum. He or she is a usually member of a community or organization, has a job, mission, or interest, and interacts with other people to accomplish some goal. The communication between a user and an IRS is illustrated in Fig. 1.2. Paragraphs below are keyed to numbers in the figure.

- 1- *The user's need for information*—somehow and for some reason, such as being assigned to a new project or case, the user senses a lack of information.

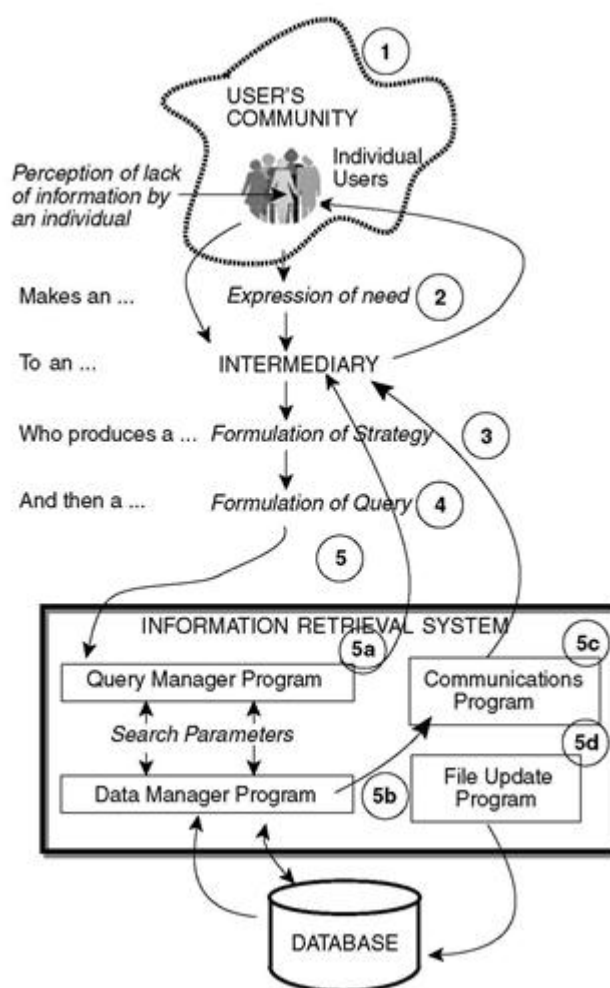


Figure 2.2



Communication between the user and the IRS: from the time a user perceives a lack of information until the receipt of an acceptable response from an IRS, there may be many steps, some involving intermediaries, and many repeated iteratively.

Belkin (1980) called this an anomalous state of knowledge (or ASK), a gap, or discontinuity in the person's knowledge. This term means nothing more than that the person recognizes the lack of or need for data, information, or knowledge. This is the start of the IR process.

Recognition that a lack of knowledge exists does not necessarily mean that the user knows *what* information is lacking. A physician or attorney working on a difficult case may know that information is needed, but may not know at the start what tests to run, questions to ask, obscure laws or journals to read, or precedents to examine.

In everything that is to follow about how to do IR and to evaluate what is done, it is important to realize that *nothing in the entire process is more important to the outcome than the user's own understanding of his or her information need and ability to recognize what satisfies that need.*

Some people need information but get along without it. Others seek it out by going to some kind of information source, which may be a co-worker, a library, a fortune teller, a stock broker, or an online database search service.

2- *Translating or expressing the need*—The user's next step is to express the information need. The user traditionally has gone to some information agency, where there is a person or machine to whom the need will be conveyed. If it is a person, he or she is called a *search intermediary* or simply an *intermediary*. However, since the growth of the Internet and the development of Web-based search engines the user is most likely to supply her own expression of this need to a Web search engine like Google or Yahoo rather than express it to an intermediary. What has come to be known as the end user search is now certainly the most common occurrence in electronic searching. Nonetheless we shall discuss the intermediary role, since it is still of some significance. Later (Chapter 9) we shall show that this function can be assisted to some extent by a computer program. The *expression* of need is a translation of the need, vague as it may be, into language meaningful to both the user and, if present, the intermediary. The translation is not always going to work well, for a number of reasons.

An actual example was the case of an engineer who asked a librarian for information about "processing of jewelry." What he wanted, it turned out later, was information about the use of laser technology to drill holes in diamonds. Did he not know the difference? Did he equate JEWELRY PROCESSING with DIAMOND DRILLING in his own mind? Or did he phrase it this way because he thought the librarian would require a simpler explanation or that the jewelry industry would be the best source of information?

Another, not uncommon, example involves a search for an explanation of a disease in lay terms, from a database containing only research information. The request is likely to be

worded along the lines of finding information about such and such a disease. The user's true goal, understanding the nature of the disease rather than complex details of treatment or cause, may not be stated as a need because the user does not understand that it is necessary to say something so obvious. He or she is asked the subject and states it as the name of the disease. The physician looking for details of treatment information may well use the same terms as the layman—the name of the disease—and for the same reason, namely that it is not clear anything else is required or not. Both will then get the same output from the IR process, but one will likely deem it satisfactory and the other, useless. An intermediary faced with such a different judgment of the usefulness of returned material will try to renegotiate the question in order to come closer to the user's need. A Web search engine will provide a user with a large return of documents from a large number of different sources, but their order will depend upon term occurrence counts of the specified terms, plus numbers of in-links and recorded click counts. The user must then determine how deeply to investigate the set of retrieved documents and whether to initiate a reformulated expression.

For a user to explain his or her information needs to the intermediary does not require that person to use library classification codes, thesaurus classes, or keywords. It involves one human being speaking to another, in natural language, although possibly laden with technical jargon. If this step is not done well, the retrieved results are almost inevitably going to be poor because the intermediary is likely to look for the wrong thing. The first two steps (recognition of an ASK and translation of it into a statement of information need) almost never are the subjects of formal training of end users, with some exceptions in such fields as law, chemistry, and library science.

- 3- Formulating a search strategy—The formal statement of need is not always uttered. If a user is going to do his or her own searching, say in the card catalog (still found in small libraries) or even online, then he or she will probably not take the time to write down a formal statement of need. In a library, the user may even go directly to a known stack area and start browsing the books in the hope of finding what is wanted.

The search strategy is the general plan for finding the needed information. If a trained information specialist such as a reference librarian is involved, that person usually plans the strategy, but the very decision to go to that person may have resulted from a strategy conceived by the user.

There can be levels of strategy. Some writers use the term to refer to very explicit and detailed aspects of a search, such as whether the previously mentioned engineer with the drilling problem should have asked to retrieve records in a database containing the words (DIAMOND OR JEWELRY) AND DRILLING or just DIAMOND AND DRILLING. This level of query formulation detail fits the usual meaning of *tactic* rather than strategy. The latter term usually means a broad plan.

- 4- Formulating a query—This step in a typical retrieval process entails composing a formulation of the information need in terms suitable for the retrieval

system to use. For a computer system, this may mean writing the equivalent of a computer program—a sequence of commands intended to produce the desired information. In a library, the step involves selecting classification codes or journals and indexes to investigate. While most Web search engines will have the capability of processing fairly sophisticated search tactics, the typical end user will simply provide two or three terms from their initial expression of the need (Spink et al., 2001).

The *query* is this precise formulation of the search. There are two important aspects of it: (1) the terms, words, codes, or values used to describe records the user wants to retrieve, and (2) the logic relating different sets of values. In our example, DIAMOND and DRILLING are terms and they are linked by the operator AND. As we noted, in the early history of IR, the purpose of the query was to express the index terms or other surrogates for the desired documents. Today, the query is more likely to have to represent the relatively uncontrolled, actual set of words used in the documents themselves. This change is both a help and a hindrance. To those who knew the various classification or subject heading systems, their nonuse creates ambiguities. To those who never knew them, there is no longer the need to struggle to find an appropriate code in order to do retrieval.

- 5- The IR system—This is the mechanical system, consists of the machinery and the software than operates it. The main program subsystems we have termed the query manager, which accepts and translates the user's question; the data manager, which handles the storage, search, and retrieval; of records; the communications manager, which controls all interactions with users, database producers, and other retrieval systems; and, finally, a file maintenance manager, which handles additions and corrections to the files and deletions.

### 2.3.2. The Database Producer Sequence

Traditionally, databases were created in response to a perceived user need and, in some cases, perceived commercial opportunity. When viewed as databases, Web sites may be created for a variety of reasons, ranging from vanity to advertising.

- 1- *The traditional sequence*—The user's information need arises from that person's organization or community. A different organization has made a decision to create an information store or database: perhaps it is a telephone directory, a file of citations and abstracts of articles published in medical journals, or a file of sales volumes and prices of shares on a stock exchange.

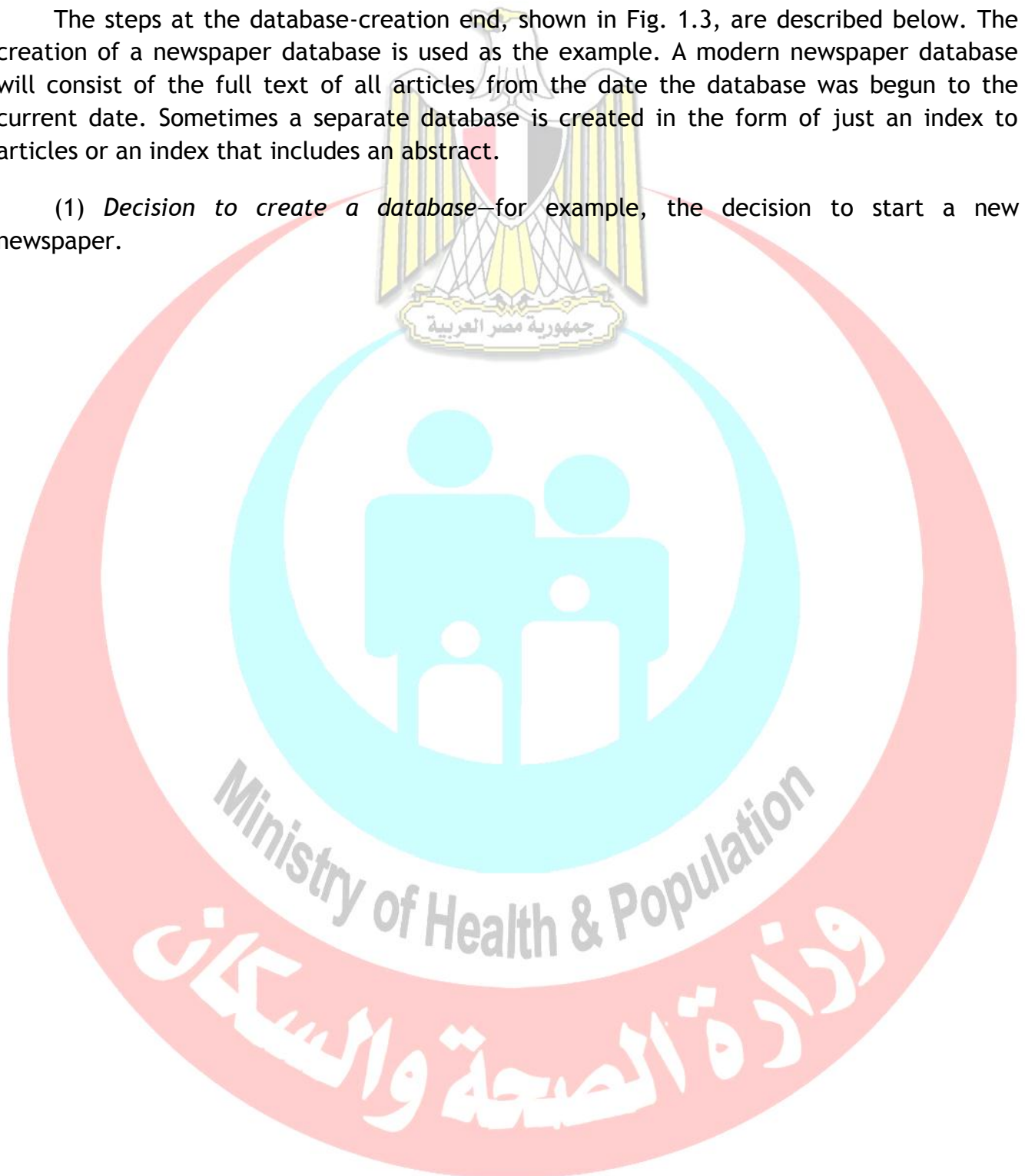
The decision to collect the information has to be followed by decisions on what the scope or coverage and the content of the record should be (Fidel, 1987; Harter, 1986; Marshall, 1990). For example, a designer has to specify which stocks or stock exchanges should be included, how up-to-date information should be, and whether and how to represent such items of information as stock symbol, number of shares, price, journal title, abstract, or article title. These, of course, are not likely to occur in the same database, but database producers may have to deal with a wide variety of subject matters. Regarding timeliness, newspapers usually do no better than yesterday's closing stock prices, while



online financial information systems can be up to the minute with a stock exchange's records. The database producer must also see to the maintenance of the database, i.e., providing new records, calling for deletion of old or erroneous ones, and providing information needed to modify a record (mistakes *do* get made).

The steps at the database-creation end, shown in Fig. 1.3, are described below. The creation of a newspaper database is used as the example. A modern newspaper database will consist of the full text of all articles from the date the database was begun to the current date. Sometimes a separate database is created in the form of just an index to articles or an index that includes an abstract.

(1) *Decision to create a database*—for example, the decision to start a new newspaper.



## Chapter 3

# Representation of Information

### 3.1. Information to Be Represented

There are so many kinds of information structures, ranging from a bit or pixel, at one extreme, to an encyclopedia or complete library at the other, that it is difficult to suggest there is a base unit of information. In one sense, the base is the *bit* or binary digit representing either of the values 0 or 1, and not being further subdividable. It could also be the *letter*, as the base unit of text, or the *phoneme*, the base sound unit of spoken language. A relatively newly defined unit of information is the *meme*. This term was coined by the biologist Richard Dawkins to denote “a unit of cultural transmission . . . . Examples . . . are tunes, ideas, catch-phrases, clothes fashions . . . .” (Dawkins, 1989, p. 192) The idea is that a meme is not just an idea of one person, but an idea that has widely penetrated a culture. Today, the World Wide Web is one. A pet word or phrase used to identify one member of a particular family is so only within the limited culture of the family. A *pixel* can be the base unit of a picture, typified by the dot in a newspaper photograph. A pixel has coordinates, and either color or shade of gray. It is not restricted to the binary choices, black or white. The basic unit of sound could be a single cycle of sound of a given wavelength and amplitude. For us to hear a monotonic sound, such as from striking a single piano key once, requires a sequence of cycles of the same wavelength. Other sounds are made up of combinations of such elementary tones. Whatever the elementary form, a datum is made up of elementary symbols combined according to some rule, depending on the type of symbol. For letters being grouped into words, that structure is called *morphology*. Individual data are then combined into higher order structures that may range from arrays of numbers to book-length texts or motion pictures. The higher order groupings are also governed by rules called *syntax* in written language and increasingly in other forms of language.

Information comes to us in the form of messages that are made up of symbols. This applies to a learned paper about a new development in astrophysics, a stock market quotation, or a waft of an alluring perfume from a companion. The information is not the symbols—the words, numbers, or aroma—but it is derived from these symbols by an interpreter, in these cases, the reader or the date.

This is a rather technical view of data. Another view, more oriented toward content and meaning, is to assume that the basis of information is an *entity* about which we have information. The entity may be an event described in a news article, or a scene portrayed in a painting, or a person about whom we have medical observations. Each type of information we have about an entity is called an *attribute*, which has a *value*. In the case of a medical record, the attributes will be well defined: name, date of birth, date of visit,

blood type, complaint, diagnosis, etc. The corresponding values may be JOHN Q. CITIZEN, 15 MARCH 1966, 13 JULY 1789, A POS, etc. Note that the nature of the patient's complaint or the physician's diagnosis may be poorly defined, but the meaning or intent of an attribute called complaint or diagnosis is usually expressed more clearly. In the case of the news article, the attributes are less clearly defined, and the exact attributes to be represented will depend on the nature of the event, but the classic ones are the five W's: *who*, *what*, *where*, *when*, and *why*. The descriptions of these attributes are values in the form of strings of text, and they will vary widely from article to article.

A painting is so complex an entity that its attributes are not completely defined. Artists, critics, and viewers may all perceive different attributes. In one sense, we may say that they are a set of pixels, each of which has a location and a color. This view breaks the painting down too far for most viewers, who cannot get the sense of the overall entity from this sort of mechanical description, while they can get the sense of the news event from the five Ws.

Text, or written natural language, is sometimes referred to as unstructured in the Information Retrieval (IR) context. It *is* unstructured when compared to a bank statement where elements such as date, account holder's name, and amount of deposits always appear in a known location. If the bank manager were to write a letter to a depositor, giving all the relevant facts about status of the account, it is not at all clear exactly where in the document the amount of the latest deposit should be found, nor even how it should be represented—numerals or words. Hence, in IR, we tend to use occurrence of single words or specific sequences of words, as search terms, leaving us always open to false drops owing to lack of context.

We give the name *record* to the set of information about an entity. A record was once a precisely defined concept in data processing, when it might have referred to all the information contained in a physical subdivision of memory, such as a punched card, a fixed-length set of characters on a magnetic tape, or a segment of random access memory. This view is no longer current with modern developments in hardware and software, but it probably is current in most people's thinking. Today, the set of information about an entity may be dispersed among several locations, with the different pieces being assembled as needed into an *ad hoc* record. Modern records need not be of uniform size for all entities. For example, the complete record about a student maintained by a university may consist of many individual records in different forms, such as original application for enrollment, letters from references, record of courses taken and grades received, medical records, books on loan from the library, and financial account. At least temporarily, we refer to records in a somewhat imprecise way as the complete *set of recorded values of attributes* of an *entity*. The entity may be another record (a library catalog record referring to a book, itself a record), but it is more likely to be something outside the information system, such as a person, event, or painting.

A collection of records constitutes a *file* or *database*. In some cases it is important that the database contain records about all the entities in some population, say all the patients in a hospital or all the employees of a company. In other cases, we know this is not



possible. In spite of the *New York Times*' motto, "All the news that's fit to print," it cannot print all the news, unless there is a very narrow interpretation of what is fit to print. One of the bases of the *Times*' readers' respect for the paper is the manner in which stories are selected for printing from among all those that could have been printed. *File* tends to have a precise meaning in modern computer science, such as a set of data assigned a name and storage location by the computer's operating system or to a set of records, while *database* may be used as a synonym for file or to mean a set of related files.

In future chapters we shall reconsider the question of the structure of information. For now, we turn our attention to the question of what information to represent in a record, and how. We should point out at this time, however, that we must not be too dogmatic about what is an entity and what is an attribute. An entity can be a person. A person has attended a school; this fact is then used as an attribute of the person. Information (a set of attributes) about that school refers to the school as the entity, not to the person. But this is conceptually no different from saying that there is a subject of an essay, expressed by a paragraph within it, that there is a subject of that paragraph expressed by a sentence within it, and a subject of that sentence expressed by a clause within it. So an entity may be the subject of a record, and another entity may be part of the description of the first.

It is easy to decide to represent a person's name. It is another question to decide exactly *how* to represent the person variously identified as:

John Fitzgerald Kennedy,

Kennedy, John F.,

Kennedy JF, or

The 35th President of the United States of America.

These are just a few of the possibilities. To the human reader the first three are virtually identical. To a typical computer program searching for a particular string of symbols, they are not. The fourth version, not a name at all, nevertheless uniquely identifies the person named John F. Kennedy, but not to people who do not keep track of the sequence numbers of presidents of the United States.

Dates are another example of how complexity can come out of apparent simplicity. The strings:

October 14, 1988 14 October 1988 10/14/88 14/10/88 881014

can all mean the same thing, but even to the human reader there is some ambiguity here. European usage for October 14, 1988 is 14/10/88 and the American is 10/14/88. The date 000704 is the cause of great anxiety as the 20th century ended because it could refer either to the year 1900 or 2000 and we have no basis for deciding which without some contextual information.

Most users of libraries know that it is one thing to have a well-defined attribute called

subject, but that it is quite another for the user to figure out how the library has classified the subject being searched for. There may also be more than one value for the subject attribute for any given book. How many descriptions to use and how to represent them are database design decisions. The book *The Expedition of Humphry Clinker* by Tobias Smollet was classified in the following Library of Congress classes by different libraries: PR3694, PR3690, PR3691, PR3693, and PZ3. The first four of these codes all refer to different aspects of Tobias, himself. The PZ3 code is a result of an inconsistency in the Library of Congress classification, whereby English language fiction is classed not only in PR (for English) as well as PS (for American), but also in PZ. In the other P classes all material by and about a particular author is found together. The English language fiction itself, however, may be separated from the material about it and placed into PZ providing a separate reading collection (Immroth, 1971, pp. 208-209). It is easy to see how differences can arise. There is no right or wrong here, only differences of opinion. The incidence of such differences is decreasing since shared cataloging systems have come into use.

As we noted in connection with the attributes of a painting, we have no well-established, general method of determining the attributes of a graphic image comparable to the fairly standardized set of attributes we have developed over time to describe a body of text. Whether in books in a library or in letters in a law office, we tend to describe text in terms of title, subject, date, and author. Then, depending on who are the expected users, there may be such attributes as: author's affiliation, case number, or publisher.

The attribute case number has an interesting aspect. In a large law firm handling civil litigation, it would be useful and perhaps necessary to maintain an index of cases showing all the parties to a case, whether plaintiffs or defendants, to assure that different parts of the firm do not accidentally undertake to represent both sides of a case. The simplistic attribute case number does not suffice.

While it is possible to describe a graphic image in terms of pixel placement and color, and sometimes possible to find the various shapes represented, we cannot expect any database that contains only representations of pixels to tell us what picture in a collection contains the image of a dog or shows the use of broad brush strokes.

The situation with regard to sound is similar. It is possible to record sound digitally and store it along with some descriptive text. There are indexing methods for music. Some of these represent only the opening bars of a piece. If the searcher remembers only a passage in the middle, it is unlikely to be found by one of these indexes. In Chapter 4 we show a few rather elemental methods of representing sound in an index. We can, to some extent, convert spoken words to characters representing words, but no known information retrieval system (IRS) would search for word occurrences in an unconverted file of sound recordings. The time required would be prohibitive.

The designer of a record has to decide what information (attributes) ideally should be represented about each entity in a database. He or she must also decide what entities are to be represented: "All the news," a selection of the news referring to certain people, a selection of the news about certain types of events, or a selection that presents those

events meeting the local criteria for political correctness. Selecting the books to include in a library's collection presents a similar decision problem. Finally, the designer must decide how each attribute is to be represented: as codes, numbers, text, digitized pictures, etc.

In the first round of decision making, prime consideration must be given to the information that users of the database will probably want. What they want is conditioned by what they know about the subjects of the database, and what they may come to learn from using it. For example, although a corporate personnel department might be content to identify employees by name, programmers and accountants might insist on a unique identifying number so that changes of name or multiple occurrences of the same name do not lead to errors in payroll and tax accounting. This is almost the inverse of the case number discussion above.

Technical computer and communication considerations may then come into play. Perhaps users of a library's catalog would like the entire table of contents of each book to be represented in the catalog card or its computer equivalent. Many would like the complete text of every book and article in a library to be computer searchable. Even storage of the table of contents is not yet a practical reality, and the full text on a searchable medium is unlikely to be seen for years to come, because of both technical and copyright considerations. However, memories are now large enough and we are seeing more and more large texts on the World Wide Web, which may encourage publishers to venture into new forms of publishing.

Librarians at Harvard, Stanford Universities, the University of Oxford, in England, and the New York Public Library have agreed to let Google scan and digitize some of their libraries' books and add them to the company's searchable online index. The University of Michigan is specifically including those still protected by copyright, and scanning has been underway there since 2004. In a similar project, the Internet Archive will digitize only works in the public domain, and make them available on free search engines. The Association of American Publishers has sued to stop the Google book-scanning project. Whether Google's claim that the project represents a fair use under the law will hold up in court remains to be seen.

### 3.2. Types of Representation

Information must be represented as symbols. Within a computer, the symbols representing the words of a text are bytes made up of strings of bits. Most bytes have graphical representations for printing. While the set of 7-bit bytes is fully adequate for representing English language text, if we are interested also in other languages, more than 26 letters, and upper and lower cases of each, are needed (e.g., the non-English characters ä, á, X, ñ). Then there are musical notes and mathematical symbols. Because the total is limited, not all symbols of interest to all users can appear. Because the number of characters is limited, the selection of a means of representation is critically important. The representation must be comprehensible both to computer programs and to human users.

A database designer must decide how each attribute will be represented, as numbers, letter strings, codes, or text strings (which usually implies use of upper and lower case



letters, numbers, and punctuation). Codes, measures, or text can range from precisely selective to indicative of a broad class of values. The choice should be up to the designer, but what matters is how the codes or values function in practice—do they separate non-identical entities and bring together similar ones?

There is virtually a continuum of methods for representing information, from natural-language text to highly compact, previously defined codes, and to pictures and sounds. The methods vary mainly in terms of discriminating power, identification of similarity, descriptiveness, ambiguity, and conciseness, which we shall discuss at greater length in Section 3.4. A different approach, altogether, is to represent the *words* not the information, itself, on the assumption that the words constitute a fair representation of the information. We heard confirmation, in the many hearings concerning President Clinton, that simple-sounding words may mean different things to different people. In this case it was the word *is* (Clinton's Grand Jury Testimony, Part 4, 1998). But, when we are trying to find documents that discuss certain topics, then the mere occurrence of certain words is a strong indicator that the topic was discussed.

What is the best representation to use for any given database or query language? There is virtually no answer. Use natural language? Should it be highly precise, laden with mathematical and scientific terms? Highly metaphoric, to get a valid impression across to the reader, but little detail? Should the message be limited to standardized expressions? What, then, about new concepts? A designer must consider the ultimate user, if such is known; the persons who collect the information; storage space required; and search algorithms that take advantage of numeric or linguistic aspects of the representation, such as ability to stem words or move up and down in a hierarchy. This kind of uncertainty leads to textbooks that are hard to understand, book classification systems that are relatively easy for librarians to use but not necessarily for users, financial databases that require use of a stock "ticker" symbol not necessarily known to first-time users, and at one time limited the amount of text that could be stored in a large bibliographic database to abstracts only because computer storage of the day could not handle full facts in large quantities. In our discussion here, queries are considered a form of document. Representational issues are quite similar for both.

### 3.2.1. Natural Language

Natural language is the language we "naturally" speak. It is contrasted with artificial languages, such as BASIC or JAVA, that are consciously designed and usually highly restrictive in vocabulary and syntax. Natural language has no limit to its vocabulary and no complete set of rules to describe its syntax, or grammar. While we have some generally accepted vocabularies, it is also true that many words have more than one meaning, and several different words can have the same meaning. An author (anyone who creates a text, even an oral one, not just one who writes books) is free to define new words or temporarily change the meaning of established ones, or to switch from one natural language to another with or without explicit warning to readers, as in these examples:

A device that modulates outgoing signals and demodulates incoming signals is called a

*modem*. (Defining a new word)

Henceforth, we shall use the term *text* to be synonymous with *natural language message*. (Redefining an existing word)

As they say in French, *c'est la vie*, and there are similar expressions in Spanish, Italian, *et cetera*. (Using different languages in a single text, only one of which is explicitly identified)

The primary advantages of natural language as a means of communication are that (1) it can be understandable to large groups of users without special training and (2) just about any nuance of meaning can be expressed. This does not mean that untrained people can read chemical journals any more than they can read medical X-ray pictures. It does mean that any user familiar with the vocabulary of chemistry can read the text, but if there is a reference within it to a particular *Chemical Abstracts* Registry Number, even many chemists will not know what that code means.

Another advantage is that natural language and its users are fault tolerant. The song titled *Yes, We Have No Bananas*, the common expression that inverts its own literal meaning, "I could care less," (which actually means "I *couldn't* care less.") or the just plain ungrammatical, "I ain't got none," are all readily understood by most native speakers of English.

The disadvantages of text are primarily (1) the lack of conciseness, (2) ambiguity, the possibility that the text will not yield a single meaning to all readers, and (3) the difficulty of interpretation of text by a computer. Can a computer understand the topic of the sentence "To be or not to be, that is the question?" Probably not, but it can do some text interpretation. If the text is of a fairly straightforward, explanatory nature, whose author has not used unusual imagery or metaphor, computers have been fairly successful at "understanding" what the subject is. This usually means selecting the words most important to the subject, or recognizing that two or more texts are on the same or similar subjects. When computers can do this, they can relieve human users of much work in retrieving information. It may be precisely by the use of unexpected images, as in poetry, that an author is best able to convey a meaning, but such information is likely to be missed by a computer.

### 3.2.2. Restricted Natural Language

When it is necessary for a computer as well as humans to interpret a text, restricting the vocabulary or syntax can alleviate many problems. Like natural language, restricted natural language is not well defined. It can be any sublanguage for which the vocabulary or syntax or both are significantly limited, yet for which the meanings remain more or less as in natural language. Reasonably descriptive texts can be written if all sentences are of the form *noun*, *simple verb*, *noun*, and we can easily allow an optional adjective before either noun, giving, for example:

Computers understand restricted language. People understand complex language.

Computers misinterpret natural text.\* Most humans comprehend natural language.

The asterisked sentence is an example of how a restricted language can be misleading. Computers do not always misunderstand natural text, but they may do so and will probably do so more often than people will. This is actually a conditional sentence, but we cannot express such a subtle conditional or probabilistic concept in this primitive language.

The restricted natural language technique is used with command languages for computer software. A common form of language for use with a database search system permits such sentences as PRINT NAME AND ADDRESS FOR SALARY  $\geq$  65000. The program that recognizes this might allow variation in the order of occurrence of clauses: FOR SALARY  $\geq$  65000 PRINT NAME AND ADDRESS, but might probably not recognize the meaning of GIVE ME THE NAMES AND ADDRESSES OF EMPLOYEES WHO MAKE AT LEAST 65K.

Human users seem to like the flexibility of being able to write clauses in any order. Many like the fact that the resulting sentence reads like an English

sentence. If there were a syntactic error, a computer program might be quite adept at identifying where the error lay, something it probably could not do unless a restrictive syntax were in force.

The Association européenne des constructeurs de matériel aéronautique has developed a form of “simplified English” for use in aircraft maintenance documentation. (AECMA, 1995) Commercial airplanes are manufactured in only a few countries, but used and serviced in many. A guide was developed to reduce the complexities of English for non-native speakers. It is not a new language. It is an attempt to reduce ambiguity. For example, we commonly use an expression such as “follow the instructions.” But, *follow* is formally defined as meaning only “to come after.” The preferred form is “obey the instructions.” Or, instead of “This regulation *extends* to all units,” say “This regulation *is applicable* to all units.” Basically, the guide simply urges good English usage, and it could only help, hardly hurt, text-based information retrieval. Nevertheless, it does restrict expressiveness somewhat.

### 3.2.3. Artificial Language

When the information to be represented is limited in variability, it can be represented in highly compact and unambiguous form, cutting storage requirements and vastly simplifying the computer programs that must interpret it. While user training in the use of the language is required, the chance of usage errors can be minimized.

An example is a command language for a computer system, whether for general purpose computing, as C++ or BASIC, or specialized use, as with the database systems APPROACH, or DIALOG. Command languages have a syntax, and there is usually at least some degree of context dependence. This means that a symbol does not always carry the same meaning, wherever or however it occurs, but that context can determine meaning. For example, many such languages use quotation marks, as we do in natural language, to alter interpretation. WRITE (“X + y”) in a computer programming language typically means to write the symbols “X”, “+” and “y”, while WRITE (X + y) means to write the sum of the



values of variables X and Y.

In the DIALOG IR command language, SELECT COMPUTER means to retrieve all records containing the word COMPUTER as a value of any of several tacitly understood attributes. The command EXPANDCOMPUTER means to look COMPUTER up in a dictionary and display some of the words that precede and follow it alphabetically. The command EXPAND SELECT COMPUTER is ambiguous in English because it could either consist of a phrase with two commands (an error) or a request to EXPAND the phrase SELECT COMPUTER, which is syntactically acceptable even if not very meaningful. DIALOG would sense only the latter meaning and would look in its dictionary for the phrase SELECT COMPUTER. If a mistake had been made (the first interpretation) DIALOG would not catch it, and the user might be baffled by the system's response, which is meaningless

in this assumed context. It is something like what happens when a wrong number is dialed on the telephone, but the caller does not immediately recognize the initial response and believes the connection is correct. Then, the question, "Is Ms. So-and-so in?" can receive a baffling response when the answerer denies that this person, who is expected to be well known, is even to be found at that number.

The very precision of meaning of artificial language may require expenditure of great effort by users to learn error-free usage. It is characteristic that small errors are not tolerated. In effect, the use of artificial language tends to represent a shift of responsibility from computer software and records designers to users.

#### 3.2.4. Codes, Measures, and Descriptors

These are individual strings of symbols such as a Dewey Classification Code (used to identify the subject matter of a library book), a single word (descriptive of a book's subject), a color code (for describing an automobile), or a real number (patient's temperature). Some other examples are:

A *bit map* is an array of bits. Outside the field of computer graphics it means that each bit has an assigned meaning that can be represented by the value 0 or 1. In graphics, it refers to a means of representing a picture. The map 011100 in a personnel file might compactly indicate a person who is Not a Union Member (first 0); Salaried (1); In the Savings Bond Program (1); In the Retirement Program (1); Not in the Stock Purchase Program (0); and Not a Full-time Employee (0).

A *menu*, in computer terms, more or less as it is in restaurant terms, is a list of options offered to a user. Selection of an option is often done by entry of a code, typically the line number of the choice or the first character of the text describing the choice, or pointing to the choice with a mouse. It is almost unheard of to require that a user type in the entire text of a choice instead of a code representing that choice (but it has been done).

A list of frequently used phrases is used in voice radio communications, to be transmitted as codes to speed transmission and reduce error, for example, ROGER, OVER,

or 10-4. Each of these has a precise, assigned meaning within a community of users.

Descriptors, in the sense of words or phrases used to describe the subject content of a text or graphic, are discussed in Chapter 4. They are usually drawn from a limited list of allowable terms, as are codes.

The advantage of strictly limited attribute values is that they tend to be unambiguous and concise. If we ask for the number on hand of a particular item in the warehouse, both the number and the item description will be well identified and unambiguous in meaning, although, in fact, the information could be wrong. For example, 1.21 divided by 0.31 gives a quotient of 3.903225 . . . , which appears quite correct to the uninformed. But, as data this is misleading. If the first two numbers are known only to two decimal places, the quotient

cannot be known to six or more, even though there is the appearance of greater precision. Similarly, a linear measurement made with a tape measure subjected to stretching or heat-caused expansion may not be as accurate as it appears.

It is possible that the appearance of precision of value or definition could lure users into asking the wrong question, say for number sold when the real goal is to find rate of demand or projected sales. The difference, which can be critical, is that what is actually sold may not be what the customers initially wanted, but agreed to accept if what they wanted was not available. It could result in restocking second-choice items and not reordering what was really wanted. Classification codes may represent little more than the opinion of one classifier, not “truth.” Knowledgeable users, in such cases, learn that errors and ambiguities do occur, and they look for alternate possibilities.

When records are designed to appear well defined and structured, and something goes wrong with the data, there may be more confusion than if the data did not *appear* to be so precise. What is an air traveler to think when an incoming flight is posted on a display board as due to arrive one hour *after* that same flight is shown as scheduled to depart the airport? Or what does a department store customer do when an automated store directory gives directions such as, “Go down two flights, then turn right” that are all wrong because the machine is located on the wrong floor? These may appear far-fetched, but they have been encountered by the authors.

### 3.2.5. Mathematical Models of Text

Another approach to document representation is to focus on what words or other symbols are present on the assumption that these represent the meaning. The simplest method is simply to list the attributes of the document and the values of these attributes. Then, this list is searched instead of the document. When an attribute may contain more than one word or string of non-blank characters, it is useful to record the relative position of each word within the text. This way, if the words NEW and ORLEANS are in a text, we can quickly determine whether they constitute the phrase NEW ORLEANS or are merely part of a discussion about something new in the city of Orleans.

A frequently used model is called the *vector space model* (Salton and McGill, 1983, pp. 120-123). It represents a *text* rather than a complete document, in that it normally does not recognize attributes, other than words occurring in the one attribute that contains text, i.e., not author, date, etc. The model assumes an  $n$ -dimensional space, one dimension for each possible word or term. This means that, in general usage,  $n$  would have a value in the thousands or hundreds of thousands, the number of words in a natural language. Each text is represented as a vector in this space. Each component of the vector (distance along one axis) is either 0 or 1, depending on whether the corresponding term is absent from or present in the text. In its simplest form each text vector terminates at a point in the vector space within the unit *hypercube*, as shown in Fig. 3.1. A hypercube is

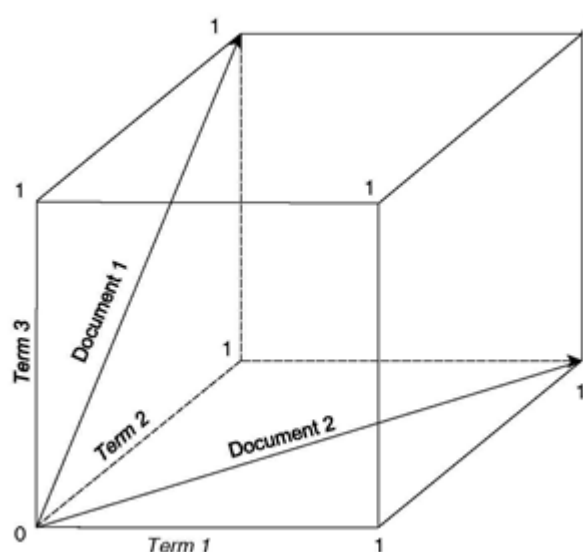


Figure 3.1

Document vectors on the vertices of the *unit hypercube*: each text is represented by a vector whose component, along each term axis, is either 0 or 1. Thus, all vectors must terminate only at one of the corners of the cube. In real life the number of dimensions is far greater than can be illustrated on a two-dimensional page.

simply a geometric construct analogous to a three-dimensional cube, but having  $n$  dimensions. Since each possible term is either present or absent, all vectors reach a point at a vertex of the hypercube, as shown in the figure.

A variation permits the use of a measure of significance or frequency of occurrence of the term to be used instead of the binary present-absent value. In this case, a vector can terminate at any point within the hypercube. See Fig. 3.2.

In either case, the use of text vectors allows us to compute a measure of similarity of one text to another. Similarity, here, means similarity of vocabulary which is a reasonable approximation to similarity of meaning. The measure is based on the angle between two vectors. The smaller the angle, the greater the similarity. This will be discussed at greater length in Chapters 9 and 10.



### 3.3. Characteristics of Information Representations

When we select a mode of representation of an attribute, such as natural language or a code, we are selecting a tool, albeit an information tool. We want it to perform certain tasks: (1) to discriminate between different entities, (2) to identify similarity among entities, (3) to allow accurate description of entities, and (4) to minimize ambiguity in interpretation. These desiderata may conflict with one another.

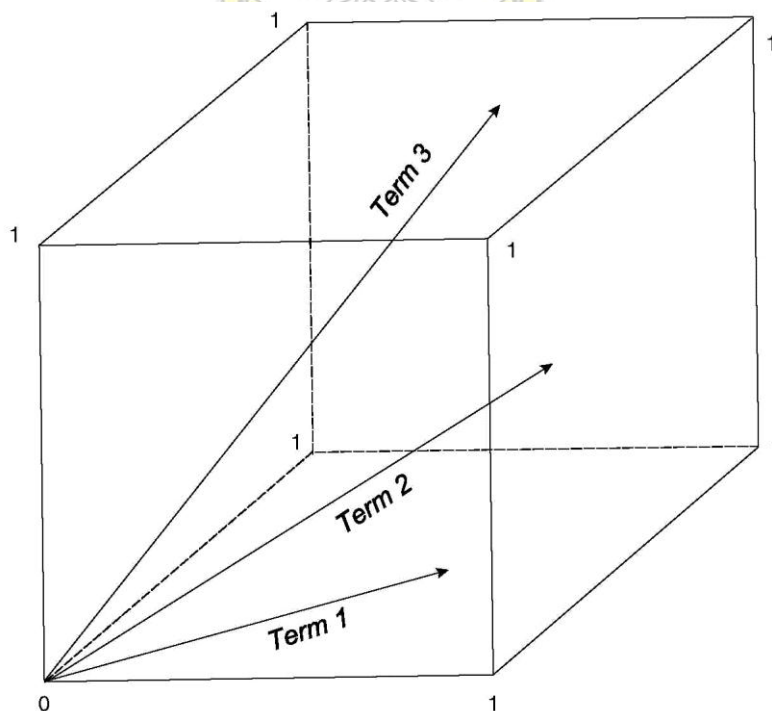


Figure 3.2

Document vectors in the unit hypercube: each document is again represented by a vector whose component length may be anywhere in the range 0.0 to 1.0. Therefore, the vectors may terminate anywhere within the hypercube.

#### 3.3.1. Discriminating Power

The ultimate in discriminating or resolving power is a unique code value, one that applies to one and only one entity in a database, thereby clearly separating the one entity from all the others. Because a person's name cannot be relied on to have this characteristic, large employers, schools, prisons, and tax authorities issue numbers that are unique in the community in which they are used. By mistake or fraud such codes are occasionally not unique, but the effort is made, and the instance of error is usually low. The U.S. social security number, intended as a unique identifier, does not always function as such. When, early in the history of Social Security, a facsimile of a social security card was included in a line of men's wallets, over 5000 people reported the number thereon as their own. Moreover, the Social Security Administration has reported that over four million people have two or more numbers (*Records, Computers, and the Rights of Citizens*, 1973, p. 112). Hence, with probability about as high as we can get in information systems design, we can accept that different values of an attribute imply different referents, that equal

values in different records imply that the records refer to the same entity, *and* that some errors will

be made. A well-designed, complete information system should have some procedures for recognizing errors and for handling them when they do occur.

### 3.3.2. Identification of Similarity

Classification codes can show both similarity and dissimilarity. When entities are not truly unique, we would like to be able to tell when there are differences among them. Dewey decimal classification codes show both the similarity and difference among entities. If two code values are 004.692 (electronic mail) and 004.693 (electronic bulletin boards), then we can see at a glance that the entities are alike, in that they share the attribute of being in classes 004 (data processing), 004.6 (interfacing and communication), and 004.69 (specific kinds of computer communication), yet are not identical because they are in different subdivisions of the 004.69 class.

### 3.3.3. Descriptiveness

This is a characteristic related to uniqueness. We use descriptive power to show *how* one entity differs from the others, but at the same time to show similarities. Of importance, then, is completeness of description, accurately describing all the important information about an entity. Of course, importance will vary with the beholder. A language with great descriptive power gives its user the choice of what aspects to stress. Natural language, as we have noted, offers the ultimate in expressiveness in words and enables us to avoid the problems caused by overly simplistic classifications or codes. Library classification codes may or may not be highly descriptive of book content; it depends on whether users understand them.

### 3.3.4. Ambiguity

The partner of descriptiveness is often ambiguity. Novelists and poets can be descriptive in ways many of us cannot approach, but they can also be easily misunderstood. Where a community of users shares a common understanding of a set of symbols (a hospital's operating room team, pilots and air traffic controllers, waiters and cooks in a restaurant) terse language can be used with little probability of error. Where there is no prior basis for common understanding, meanings must somehow be established within the text of the communication, and here is where the poet can surpass and the inarticulate writer simply fail to communicate.

*Ambiguity* refers to lack of uniqueness of meaning. Precise writing is not necessarily concise, but it is unambiguous. "The temperature is five degrees Celsius" is ambiguous to some extent because the reader does not know whether "five" means "5" an integer between 4 and 6, or "5.00," a number between 4.995 and 5.005. Codes have the capacity for precision, but do not always attain it, as color codes often really identify a dye lot rather than describing a color in a way that conveys a visual image to readers of the words in a file.

### 3.3.5. Conciseness

This refers to the number of symbols used to represent a value: “2” is more concise than “two.” Also, “Mt.” is a concise and common representation of “Mount” but this conciseness comes at a cost when the abbreviation is used for sorting, as in a telephone directory. In that case, the publishers must recognize the abbreviation, temporarily replace it with the full word, sort, then go back to the abbreviation for printing. People not fluent in English may find it difficult to find Mt. Sinai Hospital or St. Paul’s Church in the directory because they are not listed as spelled. Further, highly concise codes may not be particularly useful for expressing fine points of variability. The value of conciseness is mostly economic, but it also affects how well a message holds a reader’s attention.

### 3.4. Relationships Among Entities and Attribute Value

The composition of symbols can be used to make clear the relationship between attribute values and the entities to which they refer.

#### 3.4.1. Hierarchical Codes

A hierarchical code identifies a node in a hierarchical or tree-like structure (Fig. 3.3), in this case a portion of the Dewey decimal classification. Such a strictly hierarchical structure allows each node to have one and only one “parent” or superordinate (except for the highest node in the structure) and any number of “children” or subordinates. Records or nodes sharing a common parent are known as “siblings.”

If the reader or user knows the structure, then identifying a node means that its relationship with all other nodes is understood. In a geographic context, identifying the Province of Ontario implies that it is included within the nation of Canada and the continent of North America, and that it contains the cities of Toronto, Kingston, and Hamilton. If the term ONTARIO were not part of a hierarchy, then we would not know if, standing alone, it indicated the province in Canada, the town in New York, the city in California, or the Great Lake.

#### 3.4.2. Measurements

A measurement is a comparison of some attribute with a reference value or scale. (Boyce *et al.*, 1994) There are, of course, internationally accepted standards for measuring length, time, and mass. To measure the height of a person is to make a statement of the ratio of the person’s height to the standard of



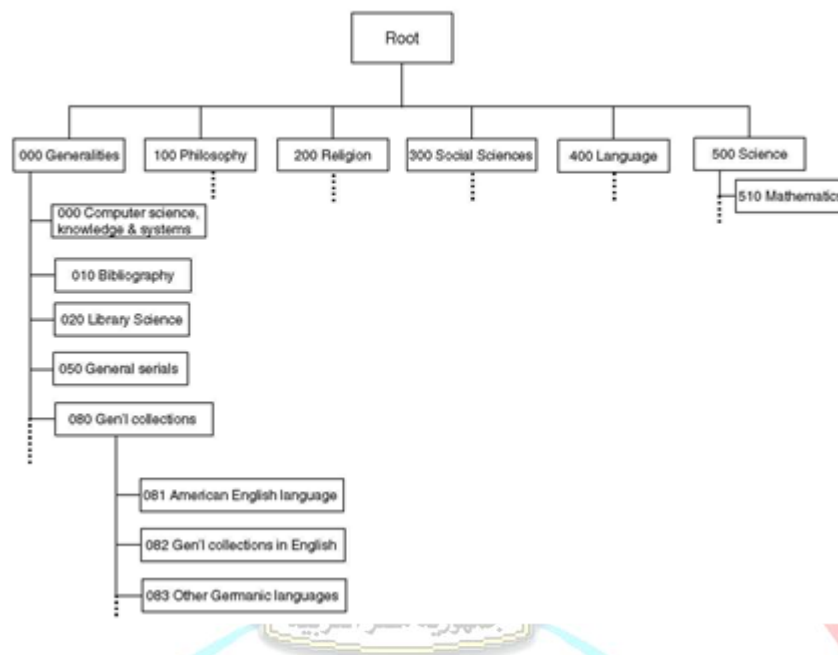


Figure 3.3

Basic hierarchical structure: perhaps the most common organizational method in use, each element has a single “parent,” or superordinate element, except for the first element. Any element may have any number of subordinate elements. This is a segment of the Dewey Decimal Classification structure, the entirety of which would not fit on a single page. Some titles have been shortened. Any given subject is expected to fit into one and only one category and then all superordinate categories are implied. Ambiguities, however, exist, as in the difference between mathematics and statistics.

length. Such *ratio measures* can be highly exact in the sense that they can record the finest degree of variation.

Sometimes we can measure only nearness to distinct points, such as the integral number of feet of a person’s height or the letter scale (a, b, etc.) used as a grade in school, where it is understood that the intervals between points on the scale are supposed to be all alike, i.e., that the difference between the grades A and B is the same magnitude as the difference between B and C. This is called *interval scale measurement*.

“Measures” like GOOD and BAD in behavior, LEFT and RIGHT in politics, or GOOD, BETTER, AND BEST as often found in mail order catalogs, imply an ordering of the values but no precise indication of the distance between them; hence, such measures are called *ordinal*. An ordinal scale serves only to place a set of values in order relative to each other, but not to measure their distance apart.

A fourth class of measurement is called *nominal*. It refers to the use of values that have no ordering among them. Typical examples are the terms used to

describe a person’s race, gender, or religion. In a typical data form, there will be a set of possible values for each of these attributes, but there is no established meaning, for example, to relative distance or ordering among Baptists, Anglicans, and Roman Catholics

or between males and females. Usually, we simply resort to alphabetic ordering in such cases.

Measures are used to achieve uniqueness in description, to discriminate, to identify similarity, and to minimize ambiguity. How descriptive they are depends on the scales used. A common procedure in restaurants is to invite patrons to evaluate service using an ordinal scale containing the points FAIR, GOOD, and EXCELLENT. The absence of POOR quite limits descriptiveness.

### 3.4.3. Nominal Descriptors

Nominal descriptors are often used to describe the content of bodies of natural-language text to make it easier for searchers to discriminate between records on their subject and others. These may be individual words selected from the text, called *keywords*, or phrases with some hierarchical relationships among them, called *subject headings*. Their use makes writing a query easier for searchers because there are fewer ways to represent a subject in this “language” than in natural language. Typically, subject-describing terms are limited within any database, i.e., there is a defined vocabulary, and both the creators of records and searchers are expected to use these terms only. That requirement imposes the further requirement that all users learn the language or face the likelihood of unsatisfactory search results.

Nominal values are often limited by a published list of allowable values, as in the classification schedules for book cataloging or the model types in automobile inventories. An attribute such as a person’s name is also a nominal descriptor. It would hardly be possible to devise a complete list of permissible names to be held by employees in a large company or students at a university. Nonetheless, each name is a valid descriptor of the person, and almost any combination of letters might be used in some part of the world as a name. Although we could maintain a list of authorized bibliographic descriptors, and a computer can assist users by rejecting invalid ones, we cannot use any kind of authority list to validate a name, unless we are dealing with a limited entity population, such as the set of authors of books cataloged by the Library of Congress or of persons listed in the Manhattan telephone directory.

One rule that is normally imposed is that only one form of any given term will be used as a nominal descriptor. We might specify the singular form for nouns and the participle form for verbs, e.g., use WORD, but not WORDS, word’s, WORDING, WORDED, etc. More important, in cases like the spelling of authors’ names, it may be necessary to establish an *authority file* showing the form deemed correct within a community. This could overcome variations in transliteration of names from one language or culture to another, the use or non-use of first name or initials, or pseudonyms.

### 3.4.4. Inflected Language

The use of prefixes and suffixes, even when not using full natural language, creates many problems for information systems, at both the record creation stage and the search stage. Inflections show similarities and differences and can add to ambiguity. The usual

method of control over this form of ambiguity is to impose rigid controls, such as:

Alphabetize all names using MC or MAC prefixes as if spelled MAC.

Alphabetize names beginning with VON, VAN, or VAN DER by what follows, i.e., do not include the prefixes in the name for this purpose (or *always* use the prefix for alphabetizing)

Omit punctuation: o'malley becomes OMALLEY or the separate words O and MALLEY.

Use only one form of a word with multiple endings (see the WORD example above).

### 3.4.5. Full Text

As we pointed out earlier, if the content of an attribute is represented in unrestricted natural language, there are many ways to say the same thing, and any given word may have more than one meaning. Relationships between records can be well established if word usage is similar. Dictionaries and thesauri can be used to establish a link between different words or phrases having similar meaning.

Long ago, Luhn (1958) recognized that words occurring frequently in a text, apart from the “common” words like A, THE, OF, etc., characterize the subject matter and can be used to relate similar texts. This basic notion has been expanded on and refined by such researchers as Salton (1989), Croft and Thompson (1987), and Liddy *et al.* (1995), now giving us reliable ways to relate a natural-language question to records stored in a database, or to relate one database record to another. Both relationships are established through similarity in the pattern of frequency of use of words or word families, i.e., a word and its synonyms or other words frequently co-occurring with it. We shall pursue this at greater length in Chapter 9 and show a practical example in Chapter 15.

### 3.4.6. Explicit Pointers and Links

The final means of recording a relationship between records is to establish an attribute of each record that “points” explicitly to related records. This means that the author of the record, or some computer program, has recognized the nature of relationships that could exist and has set up something like a “see also” pointer. In this case, each record indicates the location of the next record in order (whatever

The ordering basis may be): the next closest record in meaning, a subordinate record, a superordinate record, a record cited by the entity of this record, etc.

This is the basis for the design of a database using one of the newest (Nelson, 1981; Parsaye *et al.*, 1989, p. 223-291; Shneiderman and Kearsely, 1989), and yet oldest techniques of database ordering, now called *hypertext*



(Englebart, 1963; Englebart and English, 1968; Nelson, 1965). Essentially this means of ordering was called *linked lists*, *threaded lists*, or simply *list structures* (Lefkovitz, 1969, pp. 127-129). In hypertext a series of records is linked by multiple chains or pointers. The reader or searcher may follow any pointer at any time. If the text of this book were written in hypertext, a reader might choose to follow this paragraph with the next one mentioning the word *Hypertext*, or the first paragraph in the next chapter, or the bibliographic entry for the third reference, or for the next paragraph in the order written by the author. A bibliographic citation is a form of pointer. Most commonly it means, "The source of the information just stated is . . ." It can also indicate a source for more information

than was presented in the citing work. This is, of course, the basis for the World Wide Web in which documents are linked by pointers which go from document to document and can represent any relationship of the one to the other.

### 3.5. Summary

In all of IR, we are continually trying (1) to find records with unique attributes, (2) to separate one set of records from the others, based on attribute values, and (3) to bring together those records with similar attributes, i.e., to create sets of similar records. The success of the mechanics of these activities depends heavily on the manner in which attributes are represented. If one record identifies the color of an object as GREEN and another identifies the same object as RED, because the two humans who created these records disagreed in their perceptions owing to color blindness in one of them, there is almost no way to relate the records on the attribute color. We can, however, relate the records if we can refer to another source that establishes RED and GREEN as synonyms for one particular record author, or if we can find other attributes that may have values in common. It is in this search for alternatives that we sometimes find the greatest creativity exhibited by searchers or by the designers of computer programs to assist users. One of the most challenging problems of IR is that an author of a query may express a concept differently from the way the authors of documents express the same concept.

## Chapter 4

# The Information Retrieval System Interface

### 4.1. General Model of Message Flow

As pointed out in Chapter 1, an information retrieval system is a communication process. The purpose of this chapter is to review this process, not in terms of commands issued or feedback received, but in terms of the exchange of messages, their transformations, and the possibilities for both useful information and errors that these exchanges create. It is important to realize that not everyone will use an IRS in the same way, that we all have different degrees of skill or patience. Hence, while some generalizations about how to communicate are possible and they do aid understanding, they must not be taken as rigid rules.

An interface is something that lies between two entities and may be part of either or both. The interface affects passage of anything that flows between the entities. When two optical lenses or prisms share a common face, that interface marks the beginning of a different angle of refraction of a light ray moving from one lens to the other. An international border is an interface and travelers crossing it may find that they are treated differently on one side than the other.

Every computer system that interacts with people has an interface. This is the combination of hardware and software that enables the user to use the system and it may offer help in doing so. Some interfaces are highly interactive, some make extensive use of graphic images, some simply passively await a valid command from a user. We shall discuss these below, but when considering communications, it is important to recognize that some features are primarily communications-oriented and others are primarily retrieval functions. In this chapter we concentrate on the latter. Extensive coverage of interfaces in general is found in such works as Shneiderman (1986, 1998). Using menus instead of commands is an example of the former—this is a communication technique. Having a front end program alter a query on the basis of the result of an earlier query is an example of retrieval functionality.

We can characterize the flow of information between user and system as cyclic, involving the types of information we have previously identified in Chapter 1. The use of an information system typically begins with recognizing or sensing the existence of an ASK, anomalous state of, or lack of knowledge. It can take many forms, from an explicit recognition that a specific fact is needed to a general desire to find out what is new or interesting. The ASK is then converted, successively, into a statement of information need, a query, a set of retrieval parameters, a set of retrieved records or portions thereof, and feedback from the system about the search. Between each of these message types there is some sort of translation or conversion process, such as the translation of the information

need into a query, or of search parameters into retrieved records. While the retrieval of records is often seen as the very heart of an IRS, the other conversions are no less important.

Feedback from IRS to user, the last of the message types, is used to revise the search and initiate repetition of the sequence, or to decide to stop. Revising the search traditionally means revising the query in accordance with the information need and the results of the previous iteration. But it can also mean recognizing that satisfying the stated information need may not resolve the ASK and that starting all over again is necessary. How a given iteration is handled depends on what the user learns from early tries at the process.

We show this model in Fig. 14.1. Numbers are keyed to numbers in the figure.

1. *ASK to information need translation*—This is not a formal translation and may be the weakest link in the entire structure. Since the ASK often cannot be fully identified, even by the user, its translation into an information need statement cannot be verified or evaluated by an external person or device.

2. *Information need to query translation*—If the end user is also the searcher, then there is probably no intermediary and it is likely there will be no explicit information need statement. In this case, the ASK is directly translated into a query, with the same risks of verifiability as in ASK-to-need translation. If there is an intermediary, then translation to query form is normally done by that person. The intermediary's role is the subject of Section 14.3.

3. *Query to search parameters*—If the query language is a low-level command language, then this translation is largely a matter of parsing well-defined commands. There is little interpretation or decision-making.

If, on the other hand, the query is in natural language or a very high-order command language that does not explicitly identify each attribute-value combination to be used for set definition, then this is a key translation step. It is possible to evaluate the effectiveness of the translation, at least to some extent, because there is a formal input statement (query) and a formal output statement (search results) and it is possible to compare the results of this translation with results of other forms of translation. An example of this sort of translation is the use of any of the word association techniques of Section 9.5.

4. *Search parameters to retrieved information*—This is a relatively deterministic translation, i.e., not subject to much variation as a result of the translation process, unless the IRS is expected to find records *related* to those retrieved by



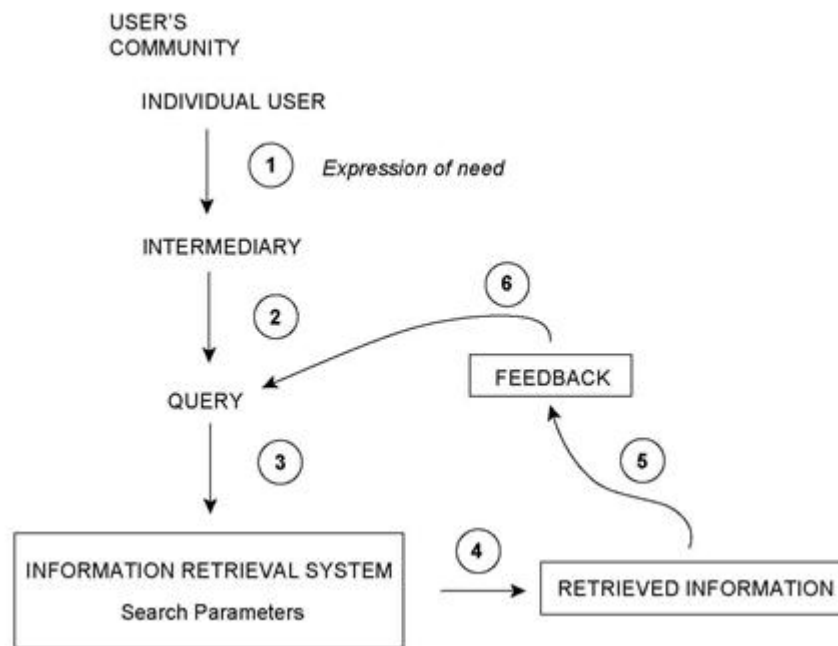


Figure 4.1

Steps in the retrieval process. This simplified sketch of the retrieval process shows points where errors or misunderstandings could occur and where feedback might help. Step 1 is expression of the information need to an intermediary, 2 is conversion of that statement into a formal query, 3 is interpretation of the query by the system. Step 4 involves selecting useful records from among those retrieved, 5 deciding what feedback to provide the user concerning what was found relevant, what not, and 6 shows the process being repeated.

the query, but which did not themselves satisfy the query. This might involve the use of record association techniques such as those described in Section 9.6.

**5. Retrieved information to feedback**—If performed by an intermediary, this translation is *another* of that person's major functions. It involves analyzing the pattern of retrieved output or messages from the system (errors, null sets, etc.) and deciding what to recommend to the user. If no intermediary is present, the user must perform this analysis without assistance, a difficult proposition for the inexperienced searcher who must decide upon the next step.

If performed by a program, the key decisions on what to look for and how to do the analysis are done at the *time of designing the program*. The program may have to decide, at execution time, which of several possible and relevant messages to send. Too much feedback, i.e., telling the user about every conceivable error made, can be as bothersome as not enough.

**6. Feedback to reformulation**—The nature of the feedback to the user was covered in *Chapter 11*.

In the following sections we discuss the main problems of designing or serving as an

interface: ambiguity in statements made to the interface, the role of an intermediary, automated intermediaries, and the general role of intermediaries, particularly noting that all computer programs have some sort of interface, even if not well suited to some particular class of user.

#### 4.2. Sources of Ambiguity

Regretfully, communication between user and intermediary is not always clear. The role of an intermediary or interface is ideally to help—to explain, offer suggestions, or identify the cause and nature of errors. A few decades ago, a computer system might simply have responded “Syntax error” if it got an input it did not recognize. It would then have been left to the user to determine the nature of the error. One step up from such responses would be a primitive attempt to identify the cause of the error, which could be confusing if wrong. For example, in some computer programming languages the statement  $X = Y = Z$  is acceptable. It is treated as two consecutive commands  $Y = Z$  and  $X = Z$ . If such a statement is presented to a programming language compiler that does not allow for the usage, it could reply, “This usage not permitted in this language. Make separate statements” or could say “Syntax error” and place a question mark under the second equal sign. The latter may be helpful to a person who had intended to say  $X = Y - Z$  and who would then see a computer response of

$$X = Y = Z$$

?

The ? points to the character that the query interpreter or compiler did not expect and could not understand. On the other hand, a response about making separate commands could be baffling to a beginning programmer who knows no other languages. What is right? There is no fixed answer.

Similar problems arise with IR systems. Those that allow natural language queries do not always provide meaningful help to users trying to decide how to improve outcome. Such systems *appear* to offer communication in natural language, as would another person, but are unable to deal with answers to such questions as, “What do you mean?” The user who truly believes he or she is communicating in natural language may be quite frustrated that the system seems not to understand but does not explain itself

We seem to have to settle for a general principle. A system interface should be designed to assume certain user characteristics, but should also be able to cope when users who do not have these characteristics are using the system. What is meaningful to an experienced person may be baffling to the novice. And *novice* may refer to state of knowledge or experience in using a system, using a database, or communicating in the argot or jargon of a discipline. It may require as much investment in development time to deal with such a range of users as it takes to provide the basic functionality of the system. Some retailers have learned that the service that goes with a product may be valued more highly by customers than the product itself, especially if it does not differ much from its competition. Chapters 3 and 4 have, we hope, shown the considerable linguistic ambiguity

that exists in all retrieval systems and the attempts to overcome it with the use of stemming, authority files, and controlled vocabularies of various kinds. The better interfaces will incorporate these devices.

#### 4.3. The Role of a Search Intermediary

When most effective, an intermediary works as a consultant to the end user, usually complementing the user's knowledge of subject matter with knowledge of how to search. The user brings to the interaction with the intermediary an understanding of the problem to be solved and the context of the problem. The intermediary should bring a range of knowledge and skills. These include:

1. Help with understanding of the range of databases, Web sites, or search engines available, noting that different SEs may index different sites.
2. Knowledge of database content and structure.
3. Knowledge of search engine variations and communication procedures.
4. Post-processing functions available and how to use them.
5. The human communication skill to be able to elicit the statement of information need from the user and help the user to realistically interpret and evaluate the output.

Usually such a person is a librarian, but the skills may be acquired through other career paths. Sometimes, highly specialized libraries put more emphasis on the intermediary's subject knowledge than on procedural or general library skills. The reason for this is to take advantage of the intermediary's ability to help the user express the information need and interpret the results. An intermediary may also be a colleague who has more experience at searching and likes to help people. There is a large literature on the role of the intermediary, which begins before the user arrives on the scene, when decisions are made about what information to collect or what search services to subscribe to. Several works on intermediaries are: Auster (1990), Cochrane (1981), de Stricker and Dysart (1989), Harter (1986), Meadow (1979), Meadow and Cochrane (1981, 25-38), and Taylor (1968).

There is some evidence (Marchionini *et al.*, 1990) that professional searchers and end users plan and conduct searches differently. The two groups' behavioral patterns reflect their different knowledge and objectives: one knowing the database subject matter and knowing what the objective knowledge is, the other knowing how to plan and conduct an effective search.

The actual tasks performed, which will of course vary from time to time, are described below.

##### 4.3.1. Establishing the Information Need



Our entire educational system deals only to a limited extent with the broad question of how to search for and find information. Most of us are given library tours and lectures but are rarely taught how to define information needs, evaluate results, or make most effective use of the resources available, especially not how to enlist the librarian as a resource. Many are now taught, as early as in elementary school, the mechanical rudiments of online searching. In the workplace, potential users may resist using a library or asking for help from anyone. Reasons for this may include a reluctance to admit not knowing something, or a perception that those doing the actual search are of lesser status than the end user, or to avoid repetition of some unpleasant experience in a library in the past.

Thus, the intermediary must be a skilled interviewer, able to understand the user's information needs and patterns of behavior. The intermediary must proceed from the knowledge that the user often does not know what information is necessary in order to begin a search or to end it. The intermediary knows it is pointless to proceed without a meaningful need statement, but may begin a vaguely defined search to enable the user to browse and thereby to help establish what aspect of a subject is important. For example, a user in a library asking for information about automobile tires may be confused or inarticulate about what aspects of the subject are considered important. One approach is for the intermediary to point out that there is information on how to make tires, their effect on automobile performance, their role in the national economy, or evaluations of tire performance for the retail buyer. This gives the user something tangible to use as the basis for refining the question. In the public library, characterized by a wide range of users and types of information, the primary intermediary skill may be human interaction. In the special library, the intermediary's ability to pose questions in the user's professional terms may be paramount.

#### 4.3.2. Development of a Search Strategy

The next step for an intermediary, having achieved a statement of information need, is to devise a search strategy. Chapter 13 covered this process in detail.

#### 4.3.3. Translation of the Need Statement into a Query

The query should reflect the strategy—such as an attempt to retrieve a large, all-encompassing set or a minimal information seed pearl and the need for feedback information. This translation is particularly difficult for end users because knowledge of command languages tends to atrophy if not frequently used. If the language was not learned well in the first place, it may be difficult to frame a complex subject in unfamiliar terms. An intermediary, on the other hand, may be current in several retrieval system languages.

In the Web based search engine environment the intermediary will have some knowledge of the best search engines for coverage of particular areas, and should be able to utilize the various advanced search capabilities without confusion or delay. The intermediary's knowledge of the search engine's defaults will allow an efficient query formulation.

#### 4.3.4. Interpretation and Evaluation of Output

Evaluation will be discussed at length in Chapter 16. For now, let us recognize that this may be a critical function in the joint undertaking of the user and intermediary. The user can probably easily assess the *value* of retrieved records to himself or herself but may not be skilled at articulating the *characteristics* of the highly valued records that distinguish them from others. The intermediary, on the other hand, once shown two sets of records, one highly valued, one not, probably *can* express the differences in terms meaningful to the IRS, i.e., in terms of record attributes and values. This, of course, is a form of non-mechanized relevance feedback.

It is possible that an end user would not be present during a search, as was often the case when questions for traditional systems were collected and searched by the intermediary is then likely to evaluate output on the basis of subject relatedness rather than value, because he or she has to make the judgment based on the stated information need, and cannot know what else, left unstated, was in the mind of the user.

#### 4.3.5. Search Iteration within the Strategic Plan

Generally, understanding the value of iteration, especially *planned* iteration, comes with experience, hence represents another area in which an intermediary can be of great value. It is as important to know when to stop as when to continue.

#### 4.3.6. Change of Strategy When Necessary

Still another area where an inexperienced user may be stymied is in recognizing that the initial search strategy, and possibly the statement of information need, must be revised. Perhaps there is not enough information in the database, or there is so much that a far more specific need must be established, or simply a recognition that the language of the database is different from the language of the user and there must be a translation. Such is the case when a lay person uses a highly technical database or a searcher encounters many documents in a foreign language not understood. Helping an inexperienced searcher who is knowledgeable in the general subject matter of the search but not the database, and who has had an inconclusive search result, may be among the most difficult tasks for an intermediary. What to do next is rarely clear in the case of an inconclusive result, and a user who does not appreciate the need for a change in direction, or who cannot point to the new direction, adds to the problem.

#### 4.3.7. Help in Using an IRS

Modern software tends to come with built-in help and tutorial programs. Some even have a form of help entitled something like, "How Do I . . . ?" Unfortunately, a great deal of help is based on the assumption that the user will be able to ask a question phrased in the system's vocabulary. If the problem is that the user does not know either how to perform some task or even what the system's designers called the task, the result can be quite frustrating. Here, again, an intermediary can be of great help. If the intermediary is familiar with the software the conversation can take place in a different manner than that

between user and system. The intermediary need not understand the content aspects of the problem to render this form of assistance.

#### 4.4. Automated Search Mediation

It has long been a dream of computer aficionados to have a computer that could converse with a human as another human can, providing information on demand, and asking insightful questions when necessary to understand the original question. This, of course, has never happened, but we have made some progress in that direction. Generally, success has been more likely when the scope is limited. Here, we review first some background and then the functions of a computer intermediary for information retrieval.

##### 4.4.1. Early Development

Around 1980 we saw the first of what became a large number of computer programs that performed some part of the intermediary role in a database search (e.g., Benson and Weinberg, 1988; Marcus, 1987a, 1987b; Meadow *et al.*, 1989; Williams, 1984). A program playing this role has some advantages over a person: it is always at hand when the user wants to do a search; it does some functions fully automatically, not requiring the kinds of detailed input some users find frustrating or embarrassing to be asked for; and it can be less expensive than a professional intermediary. On the other hand, it is hardly credible that a program can better *understand* the user than can a trained intermediary operating in a known subject domain. When the problem is really difficult, that level of understanding may be necessary; if the search problem is relatively trivial, it may not be. A human intermediary should be far better, for example, at recognizing and correcting an unpromising approach to a problem or recognizing uncertainty and lack of clarity on the part of a client. It is most difficult for a program to recognize that the problem it has been asked to solve may be the wrong one. For a program to do this, it would require, at minimum, that the program know the true search objective—the ASK—as well as the user's state of knowledge and comprehension of the material being retrieved. When they first appeared on the scene, computer intermediaries were generally programs that helped a person to conduct a search, but not to formulate an information need statement or evaluate the retrieval set.

Another form of intermediary was generally called a *gateway*. Such a program would connect a user to any of several remote retrieval systems, possibly also offering help in deciding which one to choose in formulating a query. After some initial success in the early 1980s, these faded out of use, but have returned with the Web, first as search programs or home pages that will link a user to another site or program, and more recently as, again, programs that take a user's query to one or more other search systems and combine results. These meta-search engines were briefly discussed in Section 12.2.1.

##### 4.4.2. Fully Automatic Intermediary Functions

Intermediaries typically may perform the following functions fully automatically on



behalf of the user.

1. *Translation of commands*—Most intermediaries simplify target system commands or make use of a single command language for all possible targets of one intermediary.
2. *Displays*—An intermediary program can provide enhanced displays, showing the same information as provided by the target system, but in a more meaningful format or in a manner to make browsing easier.
3. *Cost computation*—Some intermediary programs compute the cost of a search session on behalf of the user, a function particularly useful when the searcher is an intermediary who must charge his or her clients for the cost of the search.
4. *Saving and recalling of searches*—An intermediary program may simplify saving, recalling, and editing of searches, or standard search components, whether in the user's personal computer or in the target system's computer. Saving, recalling, and editing in a local computer can save the time and cost of performing the same functions online, in the target system.
5. *Post-processing*—This term covers a wide range of functions that may be performed on data retrieved by an IRS. They may range from formatting records for easier reading by the ultimate recipients to performing statistical analyses on the content of records. Moving these functions into an IRS, or enabling an IRS to communicate with other software that performs these functions, enables users or intermediaries to specify, all at one time, what data is to be retrieved and what is to be done with it—how it is to be put into a form that solves the original problem. As an example, a tenure committee could use software that provides statistics on the number of publications of a candidate and the average number of citations to those publications, by processing the results of a search on the particular individual's name in author and cited author fields.

#### 4.4.3. Interactive Intermediary Functions

The more advanced automated intermediaries, each in its own way, work with the user to help produce a successful outcome. In this section we describe a number of functions, recognizing that little is standard from one intermediary to another.

- 1- *Selection of a database*—for the professional searcher, selection of a database is not a highly challenging problem. Knowledge of the various databases is part of the basic stock in trade. For the inexperienced searcher, however, it can be a daunting prospect because there are so many and because documentation of the differences among similar ones is skimpy. Hence, selection of the database is a function performed by some intermediary programs. It might be done by comparing a list of words in a natural language statement of the information need with a list from a natural language description of the databases, i.e., computing the degree of record association between the need statement and the database description. If the first selection were not appropriate, the process would have to be repeated. The difficulty, of course, is that determining what is appropriate requires

knowledge of the database or outcome of the search or both.

The problem of database selection admits of no easy solution. There are so many factors that bear on it, some being rather subtle, that it is not practical to expect an inexperienced user to have either the patience or knowledge to make the decision. As a practical matter, it makes more sense to limit the choice. This is not entirely satisfactory because failure to retrieve useful information *can* be the fault of the database choice, but also because the user who has not made an informed decision may never know it. For this reason many intermediary programs offer only a limited number of databases, usually restricted to a single domain, such as medicine. A major problem of Web searching is that it is not easy, for example, to limit a search only to the medical or computer science literature.

2- *Formulation and editing of a query*—This is the heart of most computer intermediaries. A number of methods are possible:

(a)The query is written in natural language and the input is converted by the program into another form of query.

(b)The query is written in a simplified or standardized command language which is then translated into the command language of the target system.

(c)An expert system embedded in the intermediary provides a series of prompts, using its own hierarchy, user-provided key words and phrases, and its knowledge of the subject matter, to play a sort of “twenty questions” game with the searcher, as in Vickery and Brooks (1987). Thus, the user need not explicitly write the query as a single statement or series of statements.

(d)The intermediary offers a simplified structure, such as the concept map, and assists in its completion by offering categories of search facets (subject, author, etc.), automated look-up of terms in online indexes and, while no guarantee against error, a reduction in the probability of the user creating a query that cannot be executed (Meadow *et al.*, 1989).

(e)The intermediary offers direct advice to the user on how to modify a previous iteration of a query, based on information it elicits from the user as to the latter’s goal: more retrieved records, fewer retrieved records, a higher precision ratio, etc. (Marcus, 1987a, 1987b).

(f)Use of retrieved records to compose the next query formulation. This is a form of relevance feedback. The user identifies complete records or, ideally, portions of records, which the system converts into weighted key words for a query.

3- *Interpretation of results*—The closest we have come to having a program evaluate results, or help a user to do so on the basis of content, is the use of the related techniques of relevance feedback (Section 11.5), set analysis (Section 8.3.1),

and the use of retrieved records for reformulating a query (Section 11.3.4). The help is indirect, but it does focus the user's attention on which records are preferred

and in some cases why, i.e., what attribute-value combinations may be of use.

It is important and useful for a user to make some kind of statement about the output of a query: what was valued and what was not, and why. A skilled human intermediary can greatly help the process and can probably outperform a program doing the same task but it is not always possible for the user to find an intermediary skilled in the user's special field of interest. Hence, work continues on the development of ever more intelligent search assistance systems.

- 4- *Revision of a query*—In relevance feedback, the user is asked which records are most relevant or valuable. In most implementations reported, the assumption is made that it is subject that is of primary importance, hence subject-defining words and combinations are weighted and used to revise the query. But it might have been the author or place that was of interest to the user.

Set analysis provides users with the raw data to make their own decisions as to what attributes are most important, and provides information about values of those attributes. The user, seeing information about the composition of a retrieved set, then can decide what changes to make in the query, although more detailed effort is required.

Use of a retrieved record for reformulation is much like relevance feedback, but the user does not rate each individual record. Instead, a single model record is selected and possibly only a portion of it. Thereafter, the system's actions are much like those in the relevance feedback method. Terms or values must be extracted from the indicated portion of a record and used as the basis for the new query. The user does not explicitly identify words or individual values of high or low weight. Some Web search engines incorporate spell-checking software and suggest possible alternatives to unrecognized entries ("George Quashington. Did you mean George Washington?"), as form of assistance not traditionally available.

- 5- *Help in understanding the retrieval system*—Only relatively recently in the history of computers has serious attention been paid to the need to inform a user of what is happening with the hardware and software, i.e., how the system is operating. The modern approach (Forsyth, 1984) is to have an expert system explain its actions as it performs them. Thus, if the intermediary is selecting a database or deciding how to revise a query, it would explain its actions, teaching the user as it did so. The intermediary expert system could also be used to diagnose the manner of use of a system and offer help in explaining to the user why certain actions had the results they did.

#### 4.5. The User Interface as a Component of All Systems

In most systems, especially the older ones, there is only one way by which the user and the IRS communicate with each other. There is one language and one set of feedback



messages. A few older systems offered minor variations, such as allowing for shortened versions of commands or abbreviated system response messages, at the user's option. Some offer a choice of a command language or a menu system and at least one offered users the option to rename the commands of one language to make them look like another. The reason why independent front ends exist is the demand by users, or perceived demand by developers, for more options.

Regardless of the degree of user option, however, every IRS has a user interface in that it has an associated language, a language or query interpreter program, and a set of messages providing system responses and feedback. All that is fundamentally new in front end software is the set of options given to users. Many of the front end or gateway functions described here could easily be provided by the target systems.

Separate front ends have never become large scale commercial successes, and there is no apparent trend in that direction. Most Web search front ends offer two interfaces: a simple term entry box and an advanced form that provides traditional user controlled Boolean operations. Some offer controlled term choices as a prior limitation to the simple search. A possible direction for future development is for all information retrieval software to be so designed that an independently developed user interface can be attached easily. In effect, this means designing software that has a standard interface with other software. In that way, a user could buy (or rent by the minute) any retrieval software and also buy or rent any user interface. While such interchangeability among software products has never been a characteristic of the computer industry, comparable interchangeability is found in other industries, from automobiles (standardized sizes for tires, spark plugs, and the like) to air travel (interline transfers and exchange of tickets).

Many a teacher has tried to explain to technically oriented students who disdain to learn expository composition that it is of little value to have good ideas if you cannot make others understand them. So, we may reach the point where software developers come to the same conclusion: good functionality is not of much benefit to people who cannot understand how to use it. Nonetheless, the trend appears to be toward simplicity from reduced options, rather than clever interfaces. That brings in more business; does it get better results?

#### 4.6. The User Interface in Web Search Engines

Since Web search engines are designed for the untrained end user they tend to be simple in the extreme. Web pages that provide an interface to existing traditional IRS tend to be more complex since they need to provide the capabilities of the original command language. However, the Web search engine standard of simplicity has its effect on these interfaces as well.

For example, while the popular AltaVista (<http://www.altavista.com/>) and Google (<http://www.google.com/>) engines provide a simple uncluttered screen that allows the searcher to enter terms in a box for searching, and provide links to advanced search and some other speciality services, the PubMed page (<http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?DB=pubmed>) of the National Library of Medicine,

which allows searching of the MEDLINE database, among others, is too large to view in a single display without scrolling. However, it does provide a single simple box at the top of the page for term entry. Beneath the box are these instructions:

Enter one or more search terms, or click Preview/Index for advanced searching.

Enter author names as smith. Initials are optional. Database to find journal titles.

Above these instructions are tabs which allow one to set limits by dates, languages, publication types, etc., move into advanced search where searches are assigned searchable set numbers and Boolean operators may be used, get a history of searches run so far, and move desirable items to a clipboard. These menu capabilities essentially allow one to duplicate the power of the original command driven interface.

The Google ([http://www.google.com/advanced\\_search?hl=en](http://www.google.com/advanced_search?hl=en)) and Alta Vista (<http://www.altavista.com/Web/adv>) advanced search pages also require scrolling and supply most of the capabilities of a traditional IRS like MEDLINE or Dialog, that is to say, stemming, phrase search, date limitation and of course, the use of Boolean operators. These capabilities will be described in more detail in the next chapter.



## Chapter 5

# Information Retrieval System Capabilities

- 5.1 Search Capabilities
- 5.2 Browse Capabilities
- 5.3 Miscellaneous Capabilities
- 5.4 Standards
- 5.5 Summary



This chapter discusses the major functions that are available in an Information Retrieval System. Search and browse capabilities are crucial to assist the user in locating relevant items. The search capabilities address both Boolean and Natural Language queries. The algorithms used for searching are called Boolean, natural language processing and probabilistic. Probabilistic algorithms use frequency of occurrence of processing tokens (words) in determining similarities between queries and items and also in predictors on the potential relevance of the found item to the searcher. Chapter 4 discusses in detail the data structures used to support the algorithms, and Chapters 5 and 7 describe the algorithms. The majority of existing commercial systems are based upon Boolean query and search capabilities. The newer systems such as TOPIC, RetrievalWare, and INQUERY all allow for natural language queries. With the introduction of multimedia searches comes new problems in how to represent queries in the different modalities and how to present the results of a search.

Given the imprecise nature of the search algorithms, Browse functions to assist the user in filtering the search results to find relevant information are very important. To allow different systems to inter-operate there are evolving standards in both the language and architecture areas. Standardization of the interfaces between systems will have the same effect on information systems that acceptance of the Structured Query Language (SQL) has had in the Database Management System field. It will allow independent service providers to develop tools and augmentations that will be applicable to any Information Retrieval System accelerating the development of functions needed by the user. Examples of these functions are information visualization tools and query expansion tools.

### 5.1 Search Capabilities

The objective of the search capability is to allow for a mapping between a user's specified need and the items in the information database that will answer that need. The



search query statement is the means that the user employs to communicate a description of the needed information to the system. It can consist of natural language text in composition style and/or query terms (referred to as terms in this book) with Boolean logic indicators between them. How the system translates the search query into processing steps to find the potential relevant items is described in later chapters. One concept that has occasionally been implemented in commercial systems (e.g., RetrievalWare), and holds significant potential for assisting in the location and ranking of relevant items, is the “weighting” of search terms. This would allow a user to indicate the importance of search terms in either a Boolean or natural language interface. Given the following natural language query statement where the importance of a particular search term is indicated by a value in parenthesis between 0.0 and 1.0 with 1.0 being the most important:

Find articles that discuss automobile emissions (.9) or sulfur dioxide(.3) on the farming industry.

the system would recognize in its importance ranking and item selection process that automobile emissions are far more important than items discussing sulfur dioxide problems.

The search statement may apply to the complete item or contain additional parameters limiting it to a logical division of the item (i.e., to a zone). As discussed in Chapter 1, this restriction is useful in reducing retrieval of non-relevant items by limiting the search to those subsets of the item whose use of a particular word is consistent with the user’s search objective. Finding a name in a Bibliography does not necessarily mean the item is about that person. Recent research has shown that for longer items, restricting a query statement to be satisfied within a contiguous subset of the document (passage searching) provides improved precision (Buckley-95, Wilkinson-95). Rather than allowing the search statement to be satisfied anywhere within a document it may be required to be satisfied within a 100 word contiguous subset of the item (Callan-94).

Based upon the algorithms used in a system many different functions are associated with the system’s understanding the search statement. The functions define the relationships between the terms in the search statement (e.g., Boolean, Natural Language, Proximity, Contiguous Word Phrases, and Fuzzy Searches) and the interpretation of a particular word (e.g., Term Masking, Numeric and Date Range, Contiguous Word Phrases, and Concept/Thesaurus expansion).

Rather than continuing the use of the term processing token to represent the searchable units extracted from an item, the terminology “word” or “term” is also used in some contexts as an approximation that is intuitively more meaningful to the reader.

### 5.1.1. Boolean Logic

Boolean logic allows a user to logically relate multiple concepts together to define what information is needed. Typically the Boolean functions apply to processing tokens identified anywhere within an item. The typical Boolean operators are **AND**, **OR**, and **NOT**. These operations are implemented using set intersection, set union and set difference procedures. A few systems introduced the concept of “exclusive or” but it is equivalent to a

slightly more complex query using the other operators and is not generally useful to users since most users do not understand it. Placing portions of the search statement in parentheses are used to overtly specify the order of Boolean operations (i.e., nesting function). If parentheses are not used, the system follows a default precedence ordering of operations (e.g., typically NOT then AND then OR). In the examples of effects of Boolean operators given in Figure 2.1, no precedence order is given to the operators and queries are processed Left to Right unless parentheses are included. Most commercial systems do not allow weighting of Boolean queries. A technique to allow weighting Boolean queries is described in Chapter 7. Some of the deficiencies of use of Boolean operators in information systems are summarized by Belkin and Croft (Belkin-89).

A special type of Boolean search is called “M of N” logic. The user lists a set of possible search terms and identifies, as acceptable, any item that contains a subset of the terms. For example, “Find any item containing any two of the following terms: “AA,” “BB,” “CC.” This can be expanded into a Boolean search that performs an AND between all combinations of two terms and “OR”s the results together ((AA AND BB) or (AA AND CC) or (BB AND CC)). Some search examples and their meanings are given in Figure 2.1. Most Information Retrieval Systems allow Boolean operations as well as allowing for the natural language interfaces discussed in Section 2.1.8. As noted in Chapter 1, very little attention has been focused on integrating the Boolean search functions and weighted information retrieval techniques into a single search result.

### 5.1.2. Proximity

Proximity is used to restrict the distance allowed within an item between two search terms. The semantic concept is that the closer two terms are

| <u>SEARCH STATEMENT</u>                 | <u>SYSTEM OPERATION</u>                                                                                  |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------|
| COMPUTER OR PROCESSOR NOT MAINFRAME     | Select all items discussing Computers and/or Processors that do not discuss Mainframes                   |
| COMPUTER OR (PROCESSOR NOT MAINFRAME)   | Select all items discussing Computers and/or items that discuss Processors and do not discuss Mainframes |
| COMPUTER AND NOT PROCESSOR OR MAINFRAME | Select all items that discuss computers and not processors or mainframes in the item                     |

Figure 2.1 Uses of Boolean Operators

found in a text the more likely they are related in the description of a particular concept. Proximity is used to increase the precision of a search. If the terms COMPUTER and DESIGN are found within a few words of each other than the item is more likely to be discussing the design of computers than if the words are paragraphs apart. The typical

format for proximity is:

TERM1 within “m” “units” of TERM2

The distance operator “m” is an integer number and units are in Characters, Words, Sentences, or Paragraphs. Certain items may have other semantic units that would prove useful in specifying the proximity operation. For very structured items, distances in characters prove useful. For items containing imbedded images (e.g., digital photographs), text between the images could help in precision when the objective is in locating a certain image. Sometimes the proximity relationship contains a direction operator indicating the direction (before or after) that the second term must be found within the number of units specified. The default is either direction. A special case of the Proximity operator is the Adjacent (ADJ) operator that normally has a distance operator of one and a forward only direction (i.e., in WAIS). Another special case is where the distance is set to zero meaning within the same semantic unit. Some proximity search statement examples and their meanings are given in Figure 2.2.

### 5.1.3. Contiguous Word Phrases

A Contiguous Word Phrase (CWP) is both a way of specifying a query term and a special search operator. A Contiguous Word Phrase is two or more words that are treated as a single semantic unit. An example of a CWP is “United States of America.” It is four words that specify a search term representing a

#### SEARCH STATEMENT SYSTEM OPERATION

| <u>SEARCH STATEMENT</u>                        | <u>SYSTEM OPERATION</u>                                                                                                                               |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| “Venetian” ADJ “Blind”                         | would find items that mention a Venetian Blind on a window but not items discussing a Blind Venetian                                                  |
| “United” within five words of “American”       | would hit on “United States and American interests,” “United Airlines and American Airlines” not on “United States of America and the American dream” |
| “Nuclear” within zero paragraphs of “clean-up” | would find items that have “nuclear” and “clean-up” in the same paragraph.                                                                            |

single specific semantic concept (a country) that can be used with any of the operators discussed above. Thus a query could specify “manufacturing” AND “United States of America” which returns any item that contains the word “manufacturing” and the contiguous words “United States of America.”

A contiguous word phrase also acts like a special search operator that is similar to the



proximity (Adjacency) operator but allows for additional specificity. If two terms are specified, the contiguous word phrase and the proximity operator using directional one word parameters or the Adjacent operator are identical. For contiguous word phrases of more than two terms the only way of creating an equivalent search statement using proximity and Boolean operators is via nested Adjacencies which are not found in most commercial systems. This is because Proximity and Boolean operators are binary operators but contiguous word phrases are an “N”ary operator where “N” is the number of words in the CWP.

Contiguous Word Phrases are called Literal Strings in WAIS and Exact Phrases in RetrievalWare. In WAIS multiple Adjacency (ADJ) operators are used to define a Literal String (e.g., “United” ADJ “States” ADJ “of” ADJ “America”).

#### 5.1.4. Fuzzy Searches

Fuzzy Searches provide the capability to locate spellings of words that are similar to the entered search term. This function is primarily used to compensate for errors in spelling of words. Fuzzy searching increases recall at the expense of decreasing precision (i.e., it can erroneously identify terms as the search term). In the process of expanding a query term fuzzy searching includes other terms that have similar spellings, giving more weight (in systems that rank output) to words in the database that have similar word lengths and position of the characters as the entered term. A Fuzzy Search on the term “computer” would automatically include the following words from the information database: “computer,” “computer,” “computer,” “computer,” “compute.” An additional enhancement may look up the proposed alternative spelling and if it is a valid word with a different meaning, include it in the search with a low ranking or not include it at all (e.g., “commuter”). Systems allow the specification of the maximum number of new terms that the expansion includes in the query. In this case the alternate spellings that are “closest” to the query term is included. “Closest” is a heuristic function that is system specific.

Fuzzy searching has its maximum utilization in systems that accept items that have been Optical Character Read (Cored). In the OCR process a hardcopy item is scanned into a binary image (usually at a resolution of 300 dots per inch or more). The OCR process is a pattern recognition process that segments the scanned in image into meaningful sub regions, often considering a segment the area defining a single character. The OCR process will then determine the character and translate it to an internal computer encoding (e.g., ASCII or some other standard for other than Latin based languages). Based upon the original quality of the hardcopy this process introduces errors in recognizing characters. With decent quality input, systems achieves in the 90 - 99 per cent range of accuracy. Since these are character errors throughout the text, fuzzy searching allows location of items of interest compensating for the erroneous characters.

#### 5.1.5. Term Masking

Term masking is the ability to expand a query term by masking a portion of the term and accepting as valid any processing token that maps to the unmasked portion of the term. The value of term masking is much higher in systems that do not perform stemming or only

provide a very simple stemming algorithm. There are two types of search term masking: fixed length and variable length. Sometimes they are called fixed and variable length “don’t care” functions.

Fixed length masking is a single position mask. It masks out any symbol in a particular position or the lack of that position in a word. Figure 2.3 gives an example of fixed term masking. It not only allows any character in the masked position, but also accepts words where the position does not exist. Fixed length term masking is not frequently used and typically not critical to a system.

Variable length “don’t cares” allows masking of any number of characters within a processing token. The masking may be in the front, at the end, at both front and end, or imbedded. The first three of these cases are called suffix search, prefix search and imbedded character string search, respectively. The use of an imbedded variable length don’t care is seldom used. Figure 2.3 provides examples of the use of variable length term masking. If “\*” represents a variable length don’t care then the following are examples of its use:

|              |                        |
|--------------|------------------------|
| “*COMPUTER”  | Suffix Search          |
| “COMPUTER*”  | Prefix Search          |
| “*COMPUTER*” | Imbedded String Search |

Of the options discussed, trailing “don’t care” (prefix searches) are by far the most common. In operational systems they are used in 80-90 per cent of the search terms (Kracsony-81) and in many cases are a default without the user having to specify it.

#### 5.1.6. Numeric and Date Ranges

Term masking is useful when applied to words, but does not work for finding ranges of numbers or numeric dates. To find numbers larger than “125,” using a term “125\*” will not find any number except those that begin with the

#### SEARCH STATEMENT SYSTEM OPERATION

| <u>SEARCH STATEMENT</u> | <u>SYSTEM OPERATION</u>                                                                                                           |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| multi\$national         | Matches “multi-national,” “multinational,” “multinational” but does not match “multi national” since it is two processing tokens. |
| *computer*              | Matches, “minicomputer” “microcomputer” or “computer”                                                                             |
| comput*                 | Matches “computers,” “computing,” “computes”                                                                                      |
| *comput*                | Matches “microcomputers” “minicomputing,” “compute”                                                                               |

Figure 2.3 Term Masking

digits “125.” Systems, as part of their normalization process, characterizes words as numbers or dates. This allows for specialized numeric or date range processing against those words. A user could enter inclusive (e.g., “125-425” or “4/2/93-5/2/95” for numbers and dates) to infinite ranges (“>125,” “<=233,” representing “Greater Than” or “Less Than or Equal”) as part of a query.

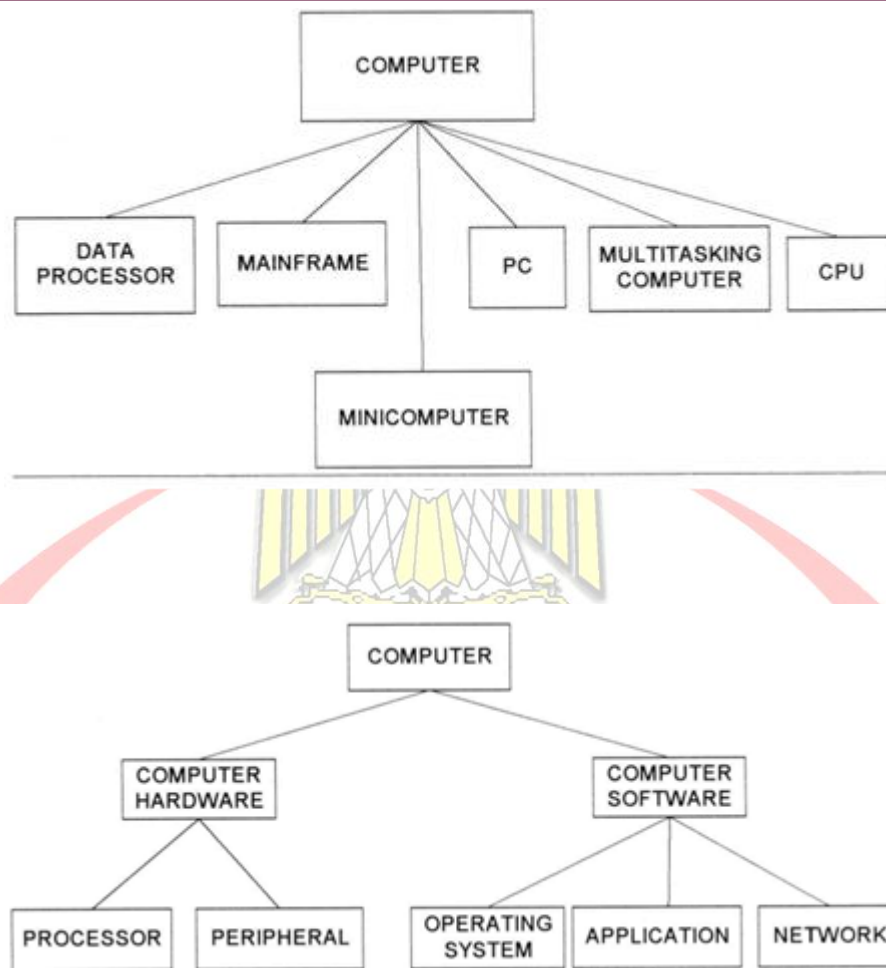
#### 5.1.7. Concept/Thesaurus Expansion

Associated with both Boolean and Natural Language Queries is the ability to expand the search terms via Thesaurus or Concept Class database reference tool. A Thesaurus is typically a one-level or two-level expansion of a term to other terms that are similar in meaning. A Concept Class is a tree structure that expands each meaning of a word into potential concepts that are related to the initial term (e.g., in the TOPIC system). Concept classes are sometimes implemented as a network structure that links word stems (e.g., in the RetrievalWare system). An example of Thesaurus and Concept Class structures are shown in Figure 2.4 (Thesaurus-93) and Figure 2.5. Concept class representations assist a user who has minimal knowledge of a concept domain by allowing the user to expand upon a particular concept showing related concepts. A concept based database shows associations that are not normally found in a language based thesaurus. For example, “negative advertising” may be linked to “elections” in a concept database, but are hopefully not synonyms to be found in a thesaurus. Generalization is associated with the user viewing concepts logically higher in a hierarchy that have more general meaning. Specificity is going lower in the thesaurus looking at concepts that are more specific.

Thesauri are either semantic or based upon statistics. A semantic thesaurus is a listing of words and then other words that are semantically similar. Electronic versions of thesauri are commercially available and are language based (e.g., English, Spanish, etc.). Systems such as RetrievalWare and TOPIC provide them as part of the search system. In executing a query, a term can be expanded to all related terms in the thesaurus or concept tree. Optionally, the user may display the thesaurus or concept tree and indicate which related terms should be used in a query. This function is essential to eliminate synonyms which introduce meanings that are not in the user’s search statement. For example, a user searching on “pasture lands” and “fields” would not want all of the terms associated with “magnetic fields” included in the expanded search statement.

The capability usually exists to browse the thesaurus or concept trees and add additional terms and term relationships in the case of concept trees. This allows users to enhance the thesaurus or concept tree with jargon specific to their area of interest.





what caused its creation (i.e., there is not a semantic basis defining the classes). As such, statistical thesauri are frequently used as automatic expansions of user's searches without the user directly interacting with the thesaurus.

Theoretically thesauri and concept trees could be used to either expand a search statement with additional terms or make it more specific but substituting more specific terms. From this perspective expanding the terms increases the recall of the search with a possible decrease in precision. Going to more specific terms increases precision and possibly reduce recall. In most cases the generalization process is used in expanding a search statement with more terms.

#### 5.1.8. Natural Language Queries

Rather than having the user enter a specific Boolean query by specifying search terms and the logic between them, Natural Language Queries allow a user to enter a prose statement that describes the information that the user wants to find. The longer the prose, the more accurate the results returned. The most difficult logic case associated with Natural Language Queries is the ability to specify negation in the search statement and have the system recognize it as negation. The system searches and finds those items most like the query statement entered. The techniques for locating items similar to the search statement (described in Chapters 5 and 7) are suited for finding items like other items but do not have inherent techniques to exclude items that are like a certain portion of the

search statement. For many users, this type of an interface provides a natural extension to asking someone to perform a search. In this case the discourse is with the computer. An example of a Natural Language Query is:

Find for me all the items that discuss oil reserves and current attempts to find new oil reserves. Include any items that discuss the international financial aspects of the oil production process. Do not include items about the oil industry in the United States.

The way a system uses this input for a search is described in Chapter 7. The problem with many techniques and systems is to understand the negation concept of excluding items about the oil industry in the United States.

When this capability has been made available, users have a tendency to enter sentence fragments that reflect their search need rather than complete sentences. This is predictable because the users want to minimize use of the human resource (their time). The likely input for the above example is:

oil reserves and attempts to find new oil reserves, international financial aspects of oil production, not United States oil industry

This usage pattern is important because sentence fragments make morphological analysis of the natural language query difficult and may limit the system's ability to perform term disambiguation (e.g., understand which meaning of a word is meant).

Using the same search statement, a Boolean query attempting to find the same information might appear:

("locate" AND "new" and "oil reserves") OR ("international" AND "financ\*" AND "oil production") NOT ("oil industry" AND "United States")

Associated with natural language queries is a function called relevance feedback. The natural language does not have to be input by the user but just identified by the user. This introduces the concept of finding items that "are like" other items. Thus a user could identify a particular item(s) in the database or text segments within item(s) and use that as the search statement. This is discussed in detail in Chapter 7.

To accommodate the negation function and provide users with a transition to the natural language systems, most commercial systems have a user interface that provides both a natural language and Boolean logic capability. Negation is handled by the Boolean portion of a search. The integration of these two search statement types has not been achieved in Information Retrieval Systems. Natural language interfaces improve the recall of systems with a decrease in precision when negation is required.

#### 5.1.9. Multimedia Queries

The user interface becomes far more complex with the introduction of the availability of multimedia items. All of the previous discussions still apply for search of the textual portions of a multimedia database. But in addition, the user has to be able to specify search terms for the other modalities. The current systems only focus on specification of still images as another search criteria. The still image could be used to search images that are part of an item. They also could be used to locate a specific scene in a video product. As described later, in the video modality, scene changes are extracted to represent changes in the information presentation. The scene changes are represented as a series of images. Additionally, where there is static text in the video, the current technology allows for OCRing the text (e.g., in the latest release of the VIRAGE system). The ability to search for audio as a match makes less sense as a user specification. To adequately perform the search you would have to simulate the audio segment and then look for a match. Instead audio sources are converted to searchable text via audio transcription. This allows queries to be applied to the text. But, like Optical Character Reading (OCR) output, the transcribed audio will contain many errors (accuracies of 85-90% are the best that can be achieved from news broadcasts, conversational speech is in the range of 60%). Thus the search algorithms must allow for errors in the data. The errors are very different compared to OCR. OCR errors will usually create a text string that is not a valid word. In automatic speech recognition (ASR), all errors are other valid words since ASR selects entries ONLY from a dictionary of words. Audio also allows the user to search on specific speakers, since speaker identification is relatively accurate against audio sources.

The correlation between different parts of a query against different modalities is usually based upon time or location. The most common example would be on time. For example if a video news program has been indexed, the user could have access to the scene changes, the transcribed audio, the closed captioning and the index terms that a user has assigned while displaying the video. The query could be "Find where Bill Clinton is discussing Cuban refugees and there is a picture of a boat". All of the separate tracks of information are correlated on a time basis. The system would return those locations where Bill Clinton is identified as the speaker (user the audio track and speaker identification), where in any of the text streams (OCR'd text from the video, transcribed audio, closed captioning, or index terms) there is discussion of refugees and Cuba, and finally during that time segment there is at least one scene change that includes a boat.

## 5.2 Browse Capabilities

Once the search is complete, Browse capabilities provide the user with the capability to determine which items are of interest and select those to be displayed. There are two ways of displaying a summary of the items that are associated with a query: line item status and data visualization. From these summary displays, the user can select the specific items and zones within the items for display. The system also allows for easy transitioning between the summary displays and review of specific items. If searches resulted in high precision, then the importance of the browse capabilities would be lessened. Since searches return many items that are not relevant to the user's information need, browse capabilities can assist the user in focusing on items that have the highest likelihood in meeting his need.



### 5.2.1. Ranking

Under Boolean systems, the status display is a count of the number of items found by the query. Every one of the items meet all aspects of the Boolean query. The reasons why an item was selected can easily be traced to and displayed (e.g., via highlighting) in the retrieved items. Hits are retrieved in either a sorted order (e.g., sort by Title) or in time order from the newest to the oldest item. With the introduction of ranking based upon predicted relevance values, the status summary displays the relevance score associated with the item along with a brief descriptor of the item (usually both fit on one display screen line). The relevance

score is an estimate of the search system on how closely the item satisfies the search statement. Typically relevance scores are normalized to a value between 0.0 and 1.0. The highest value of 1.0 is interpreted that the system is sure that the item is relevant to the search statement. This allows the user to determine at what point to stop reviewing items because of reduced likelihood of relevance. Theoretically every item in the system could be returned but many of the items will have a relevance value of 0.0 (not relevant). Practically, systems have a default minimum value which the user can modify that stops returning items that have a relevance value below the specified value. In addition to ranking based upon the characteristics of the item and the database, in many circumstances collaborative filtering is providing an option for selecting and ordering output. In this case, users when reviewing items provide feedback to the system on the relative value of the item being accessed. The system accumulates the various user rankings and uses this information to order the output for other user queries that are similar. Collaborative filtering has been very successful in sites such as AMAZON.COM MovieFinder.com, and CDNow.com in deciding what products to display to users based upon their queries (Herlocker-99)

Since one line is usually dedicated per item in a summary display, part of a zone truncated by allocated space on the display is typically displayed with the relevance weight of the item. This zone is frequently the Title and provides the user with additional information with the relevance weight to avoid selecting non-relevant items for review. Presenting the actual relevance number seems to be more confusing to the user than presenting a category that the number falls in. For example, some systems create relevance categories and indicate, by displaying items in different colors, which category an item belongs to. Other systems uses a nomenclature such as High, Medium High, Medium, Low, and Non-relevant. The color technique removes the need for written indication of an item's relevance, thereby providing additional positions in a line to display more of the title but causes problems with users that suffer from partial or total color blindness.

Rather than limiting the number of items that can be assessed by the number of lines on a screen, other graphical visualization techniques showing the relevance relationships of the hit items can be used. For example, a two or three dimensional graph can be displayed where points on the graph represent items and the location of the points represent their relative relationship between each other and the user's query. In some cases color is also used in this representation. This technique allows a user to see the clustering of items by

topics and browse through a cluster or move to another topical cluster. This has an analogy of moving through the stacks at a library. In a single image the user can see the effects of his search statement rather than displaying a few items at a time.

Information visualization is also being used in displaying individual items and the terms that contributed to the item's selection. This graphical display assists the user in determining how to reformulate his query to improve finding the information the user requires. Chapter 8 describes information visualization.

### 5.2.2. Zoning

When the user displays a particular item, the objective of minimization of overhead still applies. The user wants to see the minimum information needed to determine if the item is relevant. Once the determination is made an item is possibly relevant, the user wants to display the complete item for detailed review. Limited display screen sizes require selectability of what portions of an item a user needs to see to make the relevance determination. For example, display of the Title and Abstract may be sufficient information for a user to predict the potential relevance of an item. Limiting the display of each item to these two zones allows multiple items to be displayed on a single display screen. This makes maximum use of the speed of the user's cognitive process in scanning the single image and understanding the potential relevance of the multiple items on the screen.

Related to zoning for use in minimizing what an end user needs to review from a hit item is the idea of locality and passage based search and retrieval. In this case the basic search unit is not the complete item, but an algorithmic defined subdivision of the item. This has been known as passage retrieval where the item is divided into uniform-sized passages that are indexed (Kaskiel-97, Knaus-95, Zobel-95) and locality based retrieval where the passage boundaries can be dynamic (Kretser-99.) In these cases the system can display the particular passage or locality that caused the item to be found rather than the complete item. The system would also provide an expand capability to retrieve the complete item as an option.

### 5.2.3. Highlighting

Another display aid is an indication of why an item was selected. This indication, frequently highlighting, lets the user quickly focus on the potentially relevant parts of the text to scan for item relevance. Different strengths of highlighting indicates how strongly the highlighted word participated in the selection of the item. Most systems allow the display of an item to begin with the first highlight within the item and allow subsequent jumping to the next highlight. Another capability, which is gaining strong acceptance, is for the system to determine the passage in the document most relevant to the query and position the browse to start at that passage. The DCARS system that acts as a user frontend to the RetrievalWare search system allows the user to browse an item in the order of the paragraphs or individual words that contributed most to the rank value associated with the item.

Highlighting has always been useful in Boolean systems to indicate the cause of the

retrieval. This is because of the direct mapping between the terms in the search and the terms in the item. Using Natural Language Processing, automatic expansion of terms via thesauri, and the similarity ranking algorithms discussed in detail later in this book, highlighting loses some of its value. The terms being highlighted that caused a particular item to be returned may not have direct or obvious mapping to any of the search terms entered. This causes frustration by the user trying to guess why a particular item was retrieved and how to use that information in reformulating the search statement to make it more exact. In a ranking system different terms can contribute to different degrees to the decision to retrieve an item. The highlighting may vary by introducing colors and intensities to indicate the relative importance of a particular word in the item in the decision to retrieve the item. Information visualization appears to be a better display process to assist in helping the user formulate his query than highlights in items.

### 5.3 Miscellaneous Capabilities

There are many additional functions that facilitate the user's ability to input queries, reducing the time it takes to generate the queries, and reducing *a priori* the probability of entering a poor query. Vocabulary browse provides knowledge on the processing tokens available in the searchable database and their distribution in terms of items within the database. Iterative searching and search history logs summarize previous search activities by the user allowing access to previous results from the current user session. Canned queries allow access to queries generated and saved in previous user sessions.

#### 5.3.1. Vocabulary Browse

Vocabulary Browse provides the capability to display in alphabetical sorted order words from the document database. Logically, all unique words (processing tokens) in the database are kept in sorted order along with a count of the number of unique items in which the word is found. The user can enter a word or word fragment and the system will begin to display the dictionary around the entered text. Figure 2.6 shows what is seen in vocabulary browse if the user enters "comput." The system indicates what word fragmentation the user entered and then alphabetically displays other words found in the database in collating sequence on either side of the entered term. The user can continue scrolling in either direction reviewing additional terms in the database. Vocabulary browse provides information on the exact words in the database. It helps the user determine the impact of using a fixed or variable length mask on a search term and potential mis-spellings. The user can determine that entering the search term "compul\*" in effect is searching for "compulsion" or "compulsive" or "compulsory." It also shows that someone probably entered the word "computen" when they really meant "computer." It provides insight on the impact of using terms in a search. By vocabulary browsing, a term may be seen to exist in a large number of documents which could make it a poor candidate as an ORed term requiring additional ANDed terms to focus on items of interest. The search term "computer" would return an excessive number of hits if used as an "OR" term.



| TERM          | OCCURRENCES |
|---------------|-------------|
| compromise    | 53          |
| comptroller   | 18          |
| compulsion    | 5           |
| compulsive    | 22          |
| compulsory    | 4           |
| <b>comput</b> |             |
| computation   | 265         |
| compute       | 1245        |
| computen      | 1           |
| computer      | 10,800      |
| computerize   | 18          |
| computes      | 29          |



Figure 2.6 Vocabulary Browse List with entered term “comput”

### 5.3.2. Iterative Search and Search History Log

Frequently a search returns a Hit file containing many more items than the user wants to review. Rather than typing in a complete new query, the results of the previous search can be used as a constraining list to create a new query that is applied against it. This has the same effect as taking the original query and adding additional search statement against it in an AND condition. This process of refining the results of a previous search to focus on relevant items is called iterative search. This also applies when a user uses relevance feedback to enhance a previous search.

During a login session, a user could execute many queries to locate the needed information. To facilitate locating previous searches as starting points for new searches, search history logs are available. The search history log is the capability to display all the previous searches that were executed during the current session. The query along with the search completion status showing number of hits is displayed.

### 5.3.3. Canned Query

The capability to name a query and store it to be retrieved and executed during a later user session is called canned or stored queries. Users tend to have areas of interest within which they execute their searches on a regular basis. A canned query allows a user to create and refine a search that focuses on the user's general area of interest one time and then retrieve it to add additional search criteria to retrieve data that is currently needed. For example, a user may be responsible for European investments. Rather than always having to create a query that limits the search to European geographic search terms and then the specific question requiring resolution, a canned query can be created with all the needed geographic terms and used as a starting point for additional query specification. Significant effort can go into making the canned query into a comprehensive and focused

search since it is created once but used many times with additional search terms for specific information needs. Queries that start with a canned query are significantly larger than ad hoc queries. Canned query features also allow for variables to be inserted into the query and bound to specific values at execution time.

#### 5.3.4. Multimedia

Once a list of potential items that satisfy the query are discovered, the techniques for displaying them when they are multimedia introduces new challenges. The discussions under Ranking, above, suggest that typically hits are listed in the likely order of importance with one hit per line. The rationale is to present the most information possible to the user on the largest number of item to allow the user to select the items to be retrieved. To display more aggregate data, textual interfaces sometimes allow for clustering of the hits and then use of graphical display to show a higher level view of the information. Neither of these techniques lend themselves well when the information is multimodal. The textual aspect of the multimedia can be used to apply all of the techniques described above. But using the example in 2.1.9 above, how does the system also include the image of the boat that was part of the query. Typically a "thumbnail" of the image is displayed with the hit item. But this has the disadvantage of using more than one line per hit and reducing the number of hits that a user can select from on a single screen. If the source is audio, then other problems associated with the human linear processing of audio becomes major issues. In the textual domain, users can visually scan text very fast and via peripheral processing can maintain the context of what they are reading. If the output against audio searches was audio hit files, the user processing rate associated with listening drops dramatically. This is another reason why the transcribed audio (even if it is errorful) becomes a critical augmentation in users reviewing audio files. Thus in addition to listening to the audio, the user can visually be following the transcribed text. This provides a mechanism for the user to perceive the context and additionally provides a quick scanning option to look ahead at upcoming information to be used in conjunction with the audio processing of the original source. In tests recently performed (May 2000 in paper to be published) it was shown that the combination of errorful transcribed audio used in conjunction with the having the original audio also available reduced the processing time of items by 50 per cent from audio only (this was the initial testing, experience would drop it even further). The transcribed text could also be used as a navigation technique through the audio (Whittaker-99.) They appropriately label this new paradigm What You See Is (Almost) What You Hear (WYSIAWYH). They also noted that even though the transcribed text could be used as an index into future retrieval of the audio source, most users also needed the ability to annotate the transcriptions (note taking) to allow them to include other audio information such as in tonal information provided by the original speech.

The second area of complexity comes from any attempt at ranking such an output. In the textual only domain, there has been significant research into algorithms and heuristics associated with weighting algorithms for textual items. But when the hit is composed of different modalities, (e.g., text, image, audio), how did you create an aggregate weight of each "hit region". Little research has gone into the derivation of the weight assigned to an item that combines the weights assigned to how well each modality satisfied its portion of

the query.

#### 5.4 Z39.50 and WAIS Standards

There are two potential sources for any standard, those agreed to by an official standards organization and a de facto standard based upon high usage across a large user population. The standards organization most involved in Information Systems standards in the United States is the American National Standards Institute/ National Information Standards Organization (ANSI/NISO) in generating its Z39.50, Information Retrieval Application Service Definition and Protocol Specification. NISO is the only organization accredited by ANSI to approve and maintain standards for information services, libraries and publishers. This standard is one of many standards they generated to support interconnection of computer systems. Its relationship to other standards can be seen in the Open Systems Interconnection (OSI) basic reference model (ISO 7498). In addition to the formal standard, a second de facto standard is the WAIS standard based upon its usage in the INTERNET.

In addition to standards associated with specific language interfaces of Information Retrieval Systems, there are attempts being made to standardize the architecture of information systems. The largest de facto information system is the Internet and the Internet Engineering Task Force is focusing on how the architecture of the Internet should be modified to allow future scalability and addressability of items on the Internet. This architecture directly affects commercial information systems usage on the Internet and the integratability of future research into this environment.

The Z39.50 standard does not specify an implementation, but the capabilities within an application (Application Service) and the protocol used to communicate between applications (Information Retrieval Application Protocol). It is a computer to computer communications standard for database searching and record retrieval. Its objective is to overcome different system incompatibilities associated with multiple database searching (e.g., unique user interfaces, command language, and basic search functions). The first version of Z39.50 was approved in 1992. An international version of Z39.50, called the Search and Retrieve Standard (SR), was approved by the International Organization for Standardization (ISO) in 1991. Z39.50-1995, the latest version of Z39.50, replaces SR as the international information retrieval standard.

The standard describes eight operation types: Init (initialization), Search, Present, Delete, Scan, Sort, Resource-report, and Extended Services. There are five types of queries (Types 0, 1, 2, 100, 101, and 102). Type 101 extends previous types allowing for proximity and Type-102 is a Ranked List Query (yet to be defined). The Extended Services include saving query results, queries, and updating the database.

The client is identified as the "Origin" and performs the communications functions relating to initiating a search, translation of the query into a standardized format, sending a query, and requesting return records. The server is identified as the "Target" and interfaces to the database at the remote responding to requests from the Origin (e.g., pass query to database, return records in a standardized format and status). The end user does



not have to be aware of the details of the standard since the Origin function performs the mapping from the user's query interface into Z39.50 format. This makes the dissimilarities of different database systems transparent to the user and facilitates issuing one query against multiple databases at different sites returning to the user a single integrated Hit file.

The communications between the Origin and Target utilize a dialogue known as Z39.50 association. Z39.50 not only standardizes the messages to be exchanged between the Origin and Target systems, but also the structure and the semantics of the search query, the sequence of message exchange, and the mechanism for returning records (Turner-95, Kunze-95). The 1992 version of Z39.50 was focused on library functions such as database searching, cataloguing and interlibrary loan (primarily MARC bibliographic record structure). Z39.50 version 3 (in Z39.50-1995) addresses additional functions to support non-bibliographic data such as full text documents and images. It also begins to address some of the functions being defined as necessary to information systems such as ranking values. Z39.50-1995 has just been approved by NISO and is published as the new standard (the latest information on Z39.50 is available on the Worldwide Web at <http://ds.internic.net/z3950/z3950.html>).

Wide Area Information Service (WAIS) is the de facto standard for many search environments on the INTERNET. WAIS was developed by a project started in 1989 by three commercial companies (Apple, Thinking Machines, and Dow Jones). The original idea was to create a program that would act as a personal librarian. It would act like a personal agent keeping track of significant amounts of data and filtering it for the information most relevant to the user. The interface concept was user entered natural language statements of topics the user had interest. In addition it provided the capability to communicate to the computer that a particular item was of interest and have the computer automatically find similar items (i.e., relevance feedback). The original corporate interest was in Apple providing the user interface, Thinking Machines providing the processing power, and Dow Jones providing the data.

The developers of WAIS pursued a generalized system of information retrieval that could access data collections all around the world (Hahn-94, Levine-94) on the Internet. Like other Internet services, free versions of WAIS were originally provided. Some of the initial products migrated to a commercial company that sells and supports a WAIS system. A free version of WAIS is still available via the Clearinghouse for Networked Information Discovery and Retrieval (CINDIR) called "FreeWAIS."

The original development of WAIS started with the 1988 Z39.50 protocol as a base following the client/server architecture concept. At that time Z39.50 was focused on a bibliographic MARC record structure against structured files. Numerous deficiencies were identified in the Z39.50 protocol forcing the developers of WAIS to vary from the Z39.50 standard (ORION-93). The developers incorporated the information retrieval concepts that allow for ranking, relevance feedback and natural language processing functions that apply to full text searchable databases. Since they were diverging from the standard, they decided to simplify their client server interface. In particular they decided to have a non-

Markovian process. Thus state information is not kept between requests between clients and servers. This is one of the major differences between WAIS and Z39.50 compliant systems.

The availability of FreeWAIS in the public domain made it the method of choice for implementing databases on the INTERNET. The architecture gained significant momentum from the mandate that all U.S. Government Agencies publish their material electronically and make it accessible to the general public. WAIS and the INTERNET became the standard approach for answering the mandate. Additionally, many organizations are using WAIS as the engine to archive and access large amounts of textual information. The appeal for WAIS is that, for public domain software, it represents a well tested and documented product that can be trusted with data. A substantial community continues to test, fix and enhance the basic system. The current trend is away from WAIS and to a standard Internet interface (WEB interface) and using one of the more powerful search systems.

Another activity being driven by the publishing and Digital Libraries efforts is the creation of a unique way of identifying, naming and controlling documents in an information retrieval environment. It has the same paradigm as

the TIPSTER architecture of separating the indexes and surrogate search structures from storing the actual item. The leader in this effort is the Center for National Research Initiatives (CNRI) that is working with the Department of Defense and also the American Association of Publishers (AAP), focusing on an Internet implementation that allows for control of electronic published and copyright material. The CNRI concept is based upon using a “handle” system where the handle is the unique network identifier of a digital object. The AAP prefers the term Digital Object Identifier over the term handle (CNRI-97). Most computer systems identify an item by the location it is stored (e.g., Uniform Resource Locators (URLs) on the Internet - see Section 4.7 for more detail). From a library and large information system perspective it is far more efficient to refer to items by name rather than location. There is still significant debate over whether the name should be just a unique identifier (e.g., a number) or also have semantic content to the user. The term “handle” refers to this unique name. A Handle server (similar to document manager in TIPSTER) ensures persistence, location independence and multiple instance management. Persistence ensures that the handle is available to locate information potentially past the life of the organization that created the item to identify any locations of the item. Location independence allows for the movement of items with a mechanism for knowing their current location. Multiple instance allows keeping track of the location of duplicates of an item. Inherent in this architecture is a unique “Handle” naming authority(s) and servers to assign names and keep track of locations. This is similar to the Domain Name Servers used in networks, but is designed to handle orders of magnitude more objects in efficient fashion.

In addition to the Handle Server architecture, CNRI is also advocating a communications protocol to retrieve items from existing systems. This protocol call Repository Archive Protocol (RAP) defines the mechanisms for Clients to use the handles to retrieve items. It also includes other administrative functions such as privilege validation.

The Handle system is designed to meet the Internet Engineering Task Force (IETF) requirements for naming Internet objects via Uniform Resource Names to replace URLs as defined in the Internet's RFC-1737 (IETF-96).

## 5.5 Summary

Chapter 2 provides an overview of the functions commonly associated with Information Retrieval Systems. These functions define the user's view of the system versus the internal implementation that is described throughout the rest of this book. Until the early 1990s, the pressure on development of new user functions that assist the user in locating relevant items was driven by the academic community. The commercialization of information retrieval functions being driven by the growth of the Internet has changed the basis of development time from "academic years" (i.e., one academic year equals 18 months - the time to define the research, perform it and publish the results) to "Web years" (i.e., one Web year equals three months - demand to get new products up very quickly to be first). The test environment and test databases are changing from small scale academic environments to millions of records with millions of potential users testing new ideas. Even IBM, one of the most traditional, conservative companies, has an "alpha" site available on the Internet which contains the latest visualization and search software that is still in the process of being developing.

The areas to expect Web year changes in capabilities is in functions to assist the user in expanding his query, application of the above functions into a multilingual environment (i.e., the Internet provides information in many languages), and most importantly tools to support information visualization capabilities. The short queries that the typical user enters return too much data. The research community continues to development algorithms that is used to improve the precision and recall of the user's search. Automatic learning from user's queries coupled with large thesauri and concept dictionaries performs the query expansion process. New visualization tools have the most significant impact by allowing use of human cognitive processing to interpret the results of a user's search statement and focus on the items that most likely are relevant. Visualization tools also assist the users in enhancing their queries to find needed information. The basic search capabilities described in this chapter will not change much, but significant improvements can be expected in the browse capabilities. The underlying reason for these advancements is the need to optimize the human resource in finding needed information.



## EXERCISES

1. Describe the rationale why use of proximity will improve precision versus use of just the Boolean functions. Discuss its effect on improvement of recall.
2. Show that the proximity function can not be used to provide an equivalent to a Contiguous Word Phrase.
3. What are the similarities and differences between use of fuzzy searches and term masking? What are the potentials for each to introduce errors?
4. Are thesauri a subclass of concept classes? Justify your answer.
5. Which would users prefer, Boolean queries or Natural Language queries? Why?
6. Ranking is one of the most important concepts in Information Retrieval Systems. What are the difficulties in applying ranking when Boolean queries are used?
7. What is the relationship between vocabulary browse and thesauri/concept classes?
8. Why should researchers in information retrieval care about standards?
9. What problems does multimedia information retrieval introduce? What solutions would you recommend to resolve the problems?

## Chapter 6

# Information Visualization

- 6.1 Introduction to Information Visualization
- 6.2 Cognitive and Perception
- 6.3 Information Visualization Technologies
- 6.4 Summary



The primary focus on Information Retrieval Systems has been in the areas of indexing, searching and clustering versus information display. This has been due to the inability of technology to provide the technical platforms needed for sophisticated display, academic's focusing on the more interesting algorithmic based search aspects of information retrieval, and the multi-disciplinary nature of the human-computer interface (HCI). The core technologies needed to address sophisticated information visualization have matured, supporting productive research and implementation into commercial products. The commercial demand for these technologies is growing with availability of the "information highway." System designers need to treat the display of data as visual computing instead of treating the monitor as a replica of paper. Functions that are available with electronic display and visualization of data that were not previously provided are (Brown-96): modify representations of data and information or the display condition (e.g., changing color scales) use the same representation while showing changes in data (e.g., moving between clusters of items showing new linkages) animate the display to show changes in space and time enable interactive input from the user to allow dynamic movement between information spaces and allow the user to modify data presentation to optimize personal preferences for understanding the data.

Create hyperlinks under user control to establish relationships between data if information retrieval had achieved development of the perfect search algorithm providing close to one hundred per cent precision and recall, the need for advances in information visualization would not be so great. But reality has demonstrated in TREC and other information fora that advancements are not even close to achieving this goal. Thus, any technique that can reduce the user overhead of finding the needed information will supplement algorithmic achievements in finding potential relevant items. Information Visualization addresses how the results of a search may be optimally displayed to the users to facilitate their understanding of what the search has provided and their selection of most likely items of interest to read. Visual displays can consolidate the search results into a form easily processed by the user's cognitive abilities, but in general they do not answer the specific retrieval needs of the user other than suggesting database coverage of the

concept and related concepts.

The theoretical disciplines of cognitive engineering and perception provide a theoretical base for information visualization. Cognitive engineering derives design principles for visualization techniques from what we know about the neural processes involved with attention, memory, imagery and information processing of the human visual system. By 1989 research had determined that mental depiction plays a role in cognition that is different from mental description. Thus, the visual representation of an item plays as important a role as its symbolic definition in cognition.

Cognitive engineering results can be applied to methods of reviewing the concepts contained in items selected by search of an information system. Visualization can be divided into two broad classes: link visualization and attribute (concept) visualization. Link visualization displays relationships among items. Attribute visualization reveals content relationships across large numbers of items. Related to attribute visualization is the capability to provide visual cues on how search terms affected the search results. This assists a user in determining changes required to search statements that will return more relevant items.

## 6.1 Introduction to Information Visualization

The beginnings of the theory of visualization began over 2400 years ago. The philosopher Plato discerned that we perceive objects through the senses, using the mind. Our perception of the real world is a translation from physical energy from our environment into encoded neural signals. The mind is continually interpreting and categorizing our *perception* of our surroundings. Use of a computer is another source of input to the mind's processing functions. Text-only interfaces reduce the complexity of the interface but also restrict use of the more powerful information processing functions the mind has developed since birth.

Information visualization is a relatively new discipline growing out of the debates in the 1970s on the way the brain processes and uses mental images. It required significant advancements in technology and information retrieval techniques to become a possibility. One of the earliest researchers in information visualization was Doyle, who in 1962 discussed the concept of "semantic road maps" that could provide a user a view of the whole database (Doyle-62). The road maps show the items that are related to a specific semantic theme. The user could use this view to focus his query on a specific semantic portion of the database. The concept was extended in the late 1960s, emphasizing a spatial organization that maps to the information in the database (Miller-68). Sammon implemented a non-linear mapping algorithm that could reveal document associations providing the information required to create a road map or spatial organization (Sammons-69).

In the 1990s technical advancements along with exponential growth of available information moved the discipline into practical research and commercialization. Information visualization techniques have the potential to significantly enhance the user's ability to minimize resources expended to locate needed information. The way users



interact with computers changed with the introduction of user interfaces based upon Windows, Icons, Menus, and Pointing devices (WIMPs).

Although movement in the right direction to provide a more natural human interface, the technologies still required humans to perform activities optimized for the computer to understand. A better approach was stated by Donald A. Norman (Rose-96): *people are required to conform to technology. It is time to reverse this trend, time to make technology conform to people*

Norman stresses that to optimize the user's ability to find information, the focus should be on understanding the aspects of the user's interface and processing of information which then can be migrated to a computer interface (Norman-90).

Although using text to present an overview of a significant amount of information makes it difficult for the user to understand the information, it is essential in presenting the details. In information retrieval, the process of getting to the relevant details starts with filtering many items via a search process. The results of this process is still a large number of potentially relevant items. In most systems the results of the search are presented as a textual list of each item perhaps ordered by rank. The user has to read all of the pages of lists of the items to see what is in the Hit list. Understanding the human cognitive process associated with visual data suggests alternative ways of presenting and manipulating information to focus on the likely relevant items. There are many areas that information visualization and presentation can help the user:

- a) reduce the amount of time to understand the results of a search and likely clusters of relevant information
- b) yield information that comes from the relationships between items versus treating each item as independent
- c) perform simple actions that produce sophisticated information search functions

A study was performed by Fox et al. using interviews and user task analysis on professionals in human factors engineering, library science, and computer science to determine the requirements to optimize their work with documents (Fox-93a). Once past the initial requirement for easy access from their office, the researchers' primary objective was the capability to locate and explore patterns in document databases. They wanted visual representations of the patterns and items of interest. There was a consistent theme that the tools should allow the users to view and search documents with the system sensitive to their view of the information space. The users wanted to be able to focus on particular areas of their interest (not generic system interest definitions) and then easily see new topical areas of potential interest to investigate. They sought an interface that permits easy identification of trends, interest in various topics and newly emerging topics. Representing information in a visual mode allows for cognitive parallel processing of multiple facts and data relationships satisfying many of these requirements.

The exponential growth in available information produces large Hit files from most searches. To understand issues with the search statement and retrieved items, the user has to review a significant number of status screens. Even with the review, it is hard to generalize if the search can be improved. Information visualization provides an intuitive interface to the user to aggregate the results of the search into a display that provides a high-level summary and facilitates focusing on likely centers of relevant items. The query logically extracts a virtual workspace (information space) of potential relevant items which can be viewed and manipulated by the user. By representing the aggregate semantics of the workspace, relationships between items become visible. It is impossible for the user to perceive these relationships by viewing the items individually. The aggregate presentation allows the user to manipulate the aggregates to refine the items in the workspace. For example, if the workspace is represented by a set of named clusters (name based upon major semantic content), the user may select a set of clusters that defines the next iteration of the search.

An alternative use of aggregates is to correlate the search terms with items retrieved. Inspecting relevant and non-relevant items in a form that highlights the effect of the expanded search terms provides insights on what terms were the major causes for the results. A user may have thought a particular term was very important. A visual display could show that the term in fact had a minimal effect on the item selection process, suggesting a need to substitute other search terms.

Using a textual display on the results of a search provides no mechanism to display inter-relationships between items. For example, if the user is interested in the development of a polio vaccine, there is no way for a textual listing of found items to show “date” and “researcher” relationships based upon published items.

The textual summary list of the Hit file can only be sorted via one attribute, typically relevance rank.

Aspects of human cognition are the technical basis for understanding the details of information visualization systems. Many techniques are being developed heuristically with the correlation to human cognition and perception analyzed after the techniques are in test. The commercial pressures to provide visualization in delivered systems places the creativity under the intuitive concepts of the developer.

## 6.2 Cognition and Perception

The user-machine interface has primarily focused on a paradigm of a typewriter. As computers displays became ubiquitous, man-machine interfaces focused on treating the display as an extension of paper with the focus on consistency of operations. The advent of WIMP interfaces and simultaneous parallel tasks in the user work environment expanded the complexity of the interface to manipulate the multiple tasks. The evolution of the interface focused on how to represent to the user what is taking place in the computer environment. The advancements in computer technology, information sciences

and understanding human information processing are providing the basis for extending the human computer interface to improve the information flow, thus reducing wasted user overhead in locating needed information. Although the major focus is on enhanced visualization of information, other senses are also being looked at for future interfaces. The audio sense has always been part of simple alerts in computers. Illegal inputs are usually associated with a beep, and more recently users have a spectrum of audio sounds to associate with everything from start-up to shut down. The sounds are now being replaced by speech in both input and output interfaces. Still in the research arena is the value of using audio to encapsulate information (e.g., higher pitch as you move through an information space plus increased relevance). The tactile (touch) sense is being addressed in the experiments using Virtual Reality (VR). For example, VR is used as a training environment for areas such as medical procedures where tactile feedback plays an increasing role. Olfactory and taste are two areas where practical use for information processing or computer interfaces in general has yet to be identified. For Information Retrieval Systems, the primary area of interest is in information visualization.

### 6.2.1 Background

A significant portion of the brain is devoted to vision and supports the maximum information transfer function from the environment to a human being. The center of debates in the 1970s was whether vision should be considered data collection or also has aspects of information processing. In 1969 Arnheim questioned the then current psychological division of cognitive operations of perception and thinking as separate processes (Arnheim-69). Until then perception was considered a data collection task and thinking as a higher level function using the data. He contended that visual perception includes the process of understanding the information, providing an ongoing feedback mechanism between the perception and thinking. He further expanded his views arguing that treating perception and thinking as separate functions treats the mind as a serial automata (Arnheim-86). Under this paradigm, the two mental functions exclude each other, with perception dealing with individual instances versus generalizations. Visualization is the transformation of information into a visual form which enables the user to observe and understand the information. This concept can be extended where the visual images provide a fundamentally different way to understand information that treats the visual input not as discrete facts but as an understanding process. The Gestalt psychologists postulate that the mind follows a set of rules to combine the input stimuli to a mental representation that differs from the sum of the individual inputs (Rock-90):

Proximity - nearby figures are grouped together

Similarity - similar figures are grouped together

Continuity - figures are interpreted as smooth continuous patterns rather than discontinuous concatenations of shapes (e.g., a circle with its diameter drawn is perceived as two continuous shapes, a circle and a line, versus two half circles concatenated together)



Closure - gaps within a figure are filled in to create a whole (e.g., using dashed lines to represent a square does not prevent understanding it as a square)

Connectedness - uniform and linked spots, lines or areas are perceived as a single unit

Shifting the information processing load from slower cognitive processes to faster perceptual systems significantly improves the information-carrying interfaces between humans and computers (Card-96). There are many ways to present information in the visual space. An understanding of the way the cognitive processes work provides insights for the decisions on which of the presentations will maximize the information passing and understanding. There is not a single correct answer on the best way to present information.

## 6.2.2 Aspects of the Visualization Process

One of the first-level cognitive processes is preattention, that is, taking the significant visual information from the photoreceptors and forming primitives.

Primitives are part of the preconscious processes that consist of involuntary lower order information processing (Friedhoff-89). An example of this is the ease with which our visual systems detect borders between changes in orientation of the same object. In Figure 8.1 the visual system detects the difference in orientations between the left and middle portion of the figure and determines the logical border between them. An example of using the conscious processing capabilities of the brain is the detection of the different shaped objects and the border between them shown between the left side and middle of the Figure 8.1. The reader can likely detect the differences in the time it takes to visualize the two different boundaries.

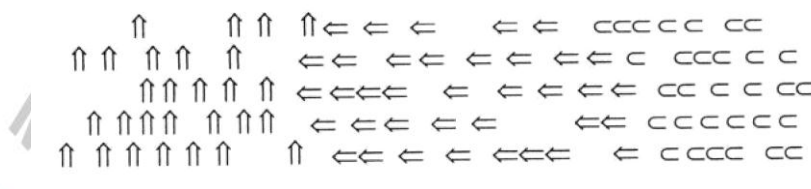


Figure 6.1 Preattentive Detection Mechanism

This suggests that if information semantics are placed in orientations, the mind's clustering aggregate function enables detection of groupings easier than using different objects (assuming the orientations are significant). This approach makes maximum use of the feature detectors in the retina.

The preattentive process can detect the boundaries between orientation groups of the same object. A harder process is to identify the equivalence of rotated objects. For example, a rotated square requires more effort to recognize it as a square. As we migrate into characters, the problem of identification of the character is affected by rotating the character in a direction not normally encountered. It is easier to detect the symmetry when the axis is vertical. Figure

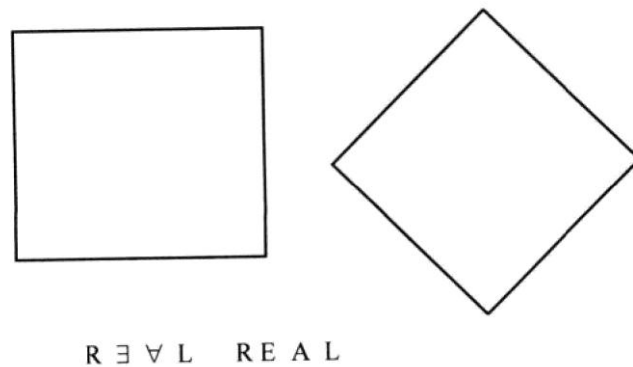


Figure 6.2 Rotating a Square and Reversing Letters in “REAL”

Another visual factor is the optical illusion that makes a light object on a dark background to appear larger than if the item is dark and the background is light. Making use of this factor suggests that a visual display of small objects should use bright colors. An even more complex area is the use of colors. Colors have many attributes that can be modified such as hue, saturation and lightness. Hue is the physiological attribute of color sensation. Saturation is the degree to which a hue is different from a gray line with the same lightness, while lightness is the sensation of the amount of white or black. Complementary colors are two colors that form white or gray when combined (red/green, yellow/blue). Color is one of the most frequently used visualization techniques to organize, classify, and enhance features (Thorell-90). Humans have an innate attraction to the primary colors (red, blue, green and yellow), and their retention of images associated with these colors is longer. But colors also affect emotion, and some people have strong aversion to certain colors. The negative side of use of colors is that some people are color blind to some or many colors. Thus any display that uses colors should have other options available.

Depth, like color, is frequently used for representing visual information. Classified as monocular cues, changes in shading, blurring (proportional to distance), perspective, motion, stereoscopic vision, occlusion and texture depict depth. Most of the cues are affected more by lightness than contrast. Thus, choice of colors that maximizes brightness in contrast to the background can assist in presenting depth as a mechanism for representing information. Depth has the advantage that depth/size recognition are learned early in life and used all of the time. Gibson and Walk showed that six-month-old children already understand depth suggesting that depth may be an innate concept (Gibson-60). The cognitive processes are well developed, and the use of this information in classifying objects is ubiquitous to daily life. The visual information processing system is attuned to processing information using depth and correlating it to real world paradigms.

Another higher level processing technique is the use of configural aspects of a display (Rose-95). A configural effect occurs when arrangements of objects are presented to the user allowing for easy recognition of a high-level abstract condition. Configural clues substitute a lower level visual process for a higher level one that requires more concentration (see preattentive above). These clues are frequently used to detect changes from a normal operating environment such as in monitoring an operational system. An

example is shown in Figure 8.3 where the sides of a regular polygon (e.g., a square in this example) are modified. The visual processing system quickly detects deviations from normally equally sized objects.

Another visual cue that can be used is spatial frequency. The human visual and cognitive system tends towards order and builds an coherent visual image whenever possible. The multiple spatial channel theory proposes that a complex image is constructed from the external inputs, not received as a single image. The final image is constructed from multiple receptors that detect changes

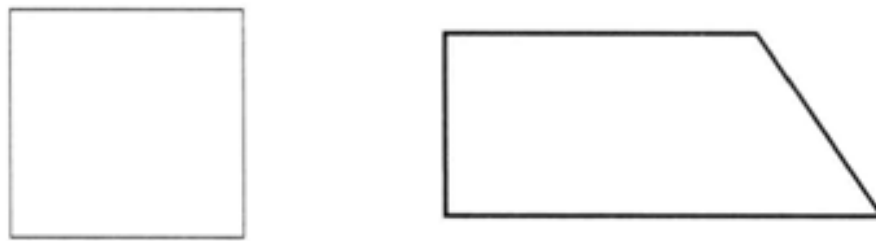


Figure 6.3 Distortions of a Regular Polygon

in spatial frequency, orientation, contrast, and spatial phase. Spatial frequency is an acuity measure relative to regular light-dark changes that are in the visual field or similar channels. A cycle is one complete light-dark change. The spatial frequency is the number of cycles per one degree of visual field. Our visual systems are less sensitive to spatial frequencies of about 5-6 cycles per degree of visual field (NOTE: one degree of visual field is approximately the viewing angle subtended by the width of a finger at arms length). Other animals have significantly more sensitive systems that allow them to detect outlines of camouflaged prey not detected by humans until we focus on the area. Associated with not processing the higher spatial frequencies is a reduction in the cognitive processing time, allowing animals (e.g. cats) to react faster to motion. When looking at a distinct, well defined image versus a blurred image, our visual system will detect motion/changes in the distinct image easier than the blurred image. If motion is being used as a way of aggregating and displaying information, certain spatial frequencies facilitate extraction of patterns of interest. Dr. Mary Kaiser of NASA-AMES is experimenting with perceptually derived displays for aircraft. She is interested in applying the human vision filters such as limits of spatial and temporal resolution, mechanisms of stereopsis, and attentional focus to aircraft (Kaiser-96).

The human sensory systems learn from usage. In deciding upon visual information techniques, parallels need to be made between what is being used to represent information and encountering those techniques in the real world environment. The human system is adept at working with horizontal and vertical references. They are easily detected and processed. Using other orientations requires additional cognitive processes to understand the changes from the expected inputs. The typical color environment is subdued without large areas of bright colors. Thus using an analogous situation, bright colors represent items



to be focused on correlating to normal processing (i.e., noticing brightly colored flowers in a garden). Another example of taking advantage of sensory information that the brain is used to processing is terrain and depth information. Using a graphical representation that uses depth of rectangular objects to represent information is an image that the visual system is used to processing. Movement in that space is more easily interpreted and understood by the cognitive processes than if, for example, a three-dimensional image of a sphere represented a visual information space.

In using cognitive engineering in designing information visualization techniques, a hidden risk is that “understanding is in the eye of the beholder.” The integration of the visual cues into an interpretation of what is being seen is also based upon the user’s background and context of the information. The human mind uses the latest information to assist in interpreting new information. If a particular shape has been representing important information, the mind has a predisposition to interpret new inputs as the same shape. For example, if users have been focusing on clusters of items, they may see clusters in a new presentation that do not exist. This leads to the question of changing visualization presentations to minimize legacy dispositions. Another issue is that our past experiences can affect our interpretation of a graphic. Users may interpret figures according to what is most common in their life experiences rather than what the designer intended.

### 6.3 Information Visualization Technologies

The theories associated with information visualization are being applied in commercial and experimental systems to determine the best way to improve the user interface, facilitating the localization of information. They have been applied to many different situations and environments (e.g., weather forecasting to architectural design). The ones focused on Information Retrieval Systems are investigating how best to display the results of searches, structured data from DBMSs and the results of link analysis correlating data. The goals for displaying the result from searches fall into two major classes: document clustering and search statement analysis. The goal of document clustering is to present the user with a visual representation of the document space constrained by the search criteria. Within this constrained space there exist clusters of documents defined by the document content. Visualization tools in this area attempt to display the clusters, with an indication of their size and topic, as a basis for users to navigate to items of interest. This is equivalent to searching the index at a library and then pursuing all the books on the different shelf locations that are retrieved by the search. The second goal is to assist the user in understanding why items were retrieved, thereby providing information needed to refine the query. Unlike the traditional Boolean systems where the user can easily correlate the query to the retrieved set of items, modern search algorithms and their associated ranking techniques make it difficult to understand the impacts of the expanded words in the search statement. Visualization techniques approach this problem by displaying the total set of terms, including additional terms from relevance feedback or thesaurus expansion, along with documents retrieved and indicate the importance of the term to the retrieval and ranking process.

Structured databases are important to information retrieval because structured

tiles are the best implementation to hold certain citation and semantic data that describe documents. Link analysis is also important because it provides aggregate-level information within an information system. Rather than treating each item as independent, link analysis considers information flowing between documents with value in the correlation between multiple documents. For example, a time/event link analysis correlates multiple documents discussing a oil spill caused by a tanker. Even if all of the items retrieved on the topic are relevant, displaying the documents correlated by time may show dependencies of events that are of information importance and are not described in any specific document. This section summarizes some of the major techniques being applied. This can assist in correlating the theory of visual perception to the practice of implementing systems.

One way of organizing information is hierarchical. A tree structure is useful in representing information that ranges over time (e.g., genealogical lineage), constituents of a larger unit (e.g., organization structures, mechanical device definitions) and aggregates from the higher to lower level (e.g., hierarchical clustering of documents). A two-dimensional representation becomes difficult for a user to understand as the hierarchy becomes large. One of the earliest experiments in information visualization was the Information Visualizer developed by XEROX PARC. It incorporates various visualization formats such as DataMap, InfoGrid, ConeTree, and the Perspective wall. The Cone-Tree is a 3-Dimensional representation of data, where one node of the tree is represented at the apex and all the information subordinate to it is arranged in a circular structure at its base. Any child node may also be the parent of another cone. Selecting a particular node, rotates it to the front of the display. Compared to other hierarchical representations (e.g., node and link trees) the cone makes the maximum information available to the user providing a perspective on size of each of the subtrees (Gershon-95a, Robertson-93). An example of a Cone-Tree is shown in Figure 8.4. The squares at the leaf nodes in tree are the actual documents. Higher level nodes can be considered centroids representing the semantic of the child nodes. Where the database is large, the boxes may represent a cluster of related items versus a single item. These clusters could be expanded to lower levels of the tree. The perspective wall divides the information into three visual areas with the area being focused on in the front and other areas out of focus to each side (see Figure 8.5). This allows the user to keep all of the information in perspective while focusing on a particular area.

Another technique used in display of hierarchical information is tree maps (Johnson-91). This technique makes maximum use of the display screen space by using rectangular boxes that are recursively subdivided based upon parent-child relationships between the data. A particular information work space focused on articles on computers may appear as shown in Figure 8.6. The size of the boxes can represent the number of items on a particular topic. The location of the boxes can indicate a relationship between the topics. In Figure 8.6, the CPU, OS, Memory, and Network management articles are all related to a general category of computer operating systems versus computer applications which are shown in the rest of the figure.

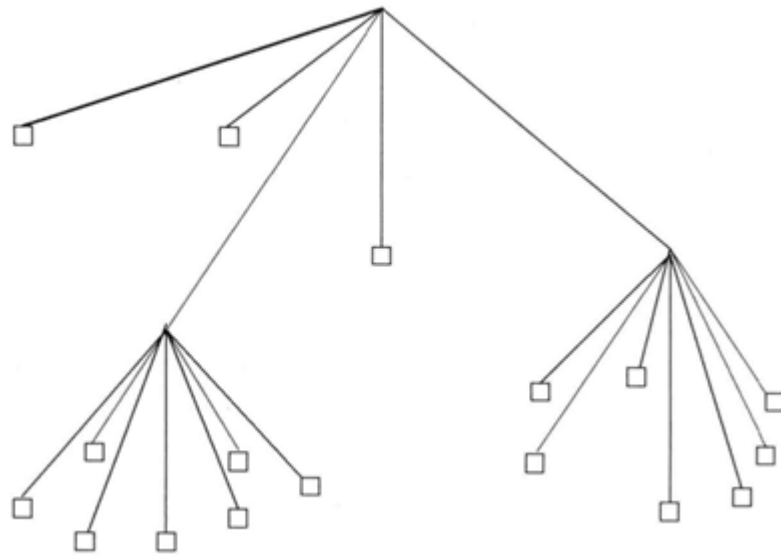


Figure 6.4 Cone Tree

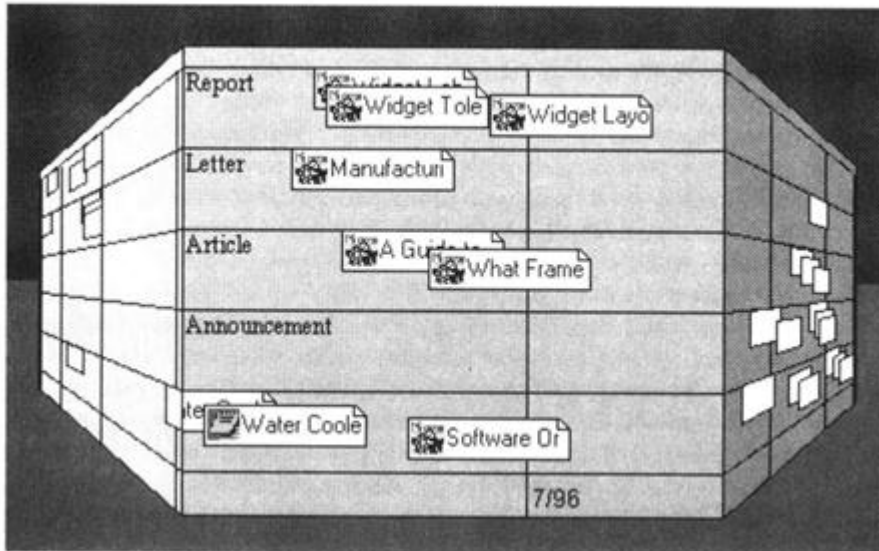


Figure 6.5 Perspective Wall From inXight web site - [www.inxight.com](http://www.inxight.com)

|                                   |    |                             |
|-----------------------------------|----|-----------------------------|
| CPU articles                      | OS | Network management articles |
| Memory articles                   |    |                             |
| Word Processing software articles |    |                             |
| Spreadsheet software articles     |    |                             |
| Powerpoint software articles      |    |                             |
| Drawing software articles         |    |                             |

Figure 6.6 Tree Map

When the information has network-type relationships, an approach using clusters can



be shown via a semantic scatterplot. Both the Vineta and Bead systems display clustering patterns using a three-dimensional scatterplot (Krohn-95, Chalmers-92 respectively). Battelle Pacific Northwest National Laboratory correlates documents and performs multidimensional scaling to plot each as a point in Euclidean vector space. The difficulty of representing all of the axis is overcome by projecting the space onto a plane and using elevation to indicate the frequency of occurrence and importance of a theme (concept) creating a semantic landscape (Wise-95 Card-96). The detailed relationships between items and their composite themes can be seen by the valleys, cliffs and ranges shown on the terrain map. One way of overcoming the multidimensional space representation in a hierarchical environment is to create embedded coordinate spaces. In this technique the larger coordinate space is redefined with coordinates inside of other coordinates. Thus a six-dimensional coordinate space may have three of the coordinates defined as a subspace within the other three coordinate spaces. This has been called Feiner's "worlds within worlds" approach (Feiner-90). Other techniques suggested to solve this representation problem can be found in semantic regions suggested by Kohonen, linked trees or graphs in the Narcissus system, or a non-Euclidean landscape suggested by Lamping and Rao (Munzner-95, Lin-91 and Lin-92, Hendley-95, Lamping-95 respectively). When searches are used to define the user's info space of interest and provide additional focusing of semantic interest, the "information crystal" (similar to a VENN diagram) assists the user in detecting patterns of term relationships in the constrained Hit file (Spoerri-93). The Cyber World system constrains its clustering visualization to a three-dimensional sphere (Hemmje-94). Another clustering system that uses statistical information for a small number of items (50 - 120) to show term relationships via spatial positioning is the VIBE system (Olsen-93). The VIBE system allows users to associate query terms with different locations in the visual display. Documents are distributed to show their relevance to the different terms. Lin has taken the self-organization concept further by using Kohonen's algorithm to automatically determine a table of contents (TOC) and display the results in a map display (Lin-96).

The goal of many visualization techniques is to show the semantic relationships between individual items to assist the user in locating those groups of items of interest. Another objective of visualization is in assisting the users in refining their search statements. It is difficult for users in systems using similarity measures to determine what are the primary causes for the selection and ranking of items in a Hit file. The automatic expansion of terms and intricacies of the similarity algorithms can make it difficult to determine the effects that the various words in the search statement are having on creating the Hit file. Visualization tools need to assist the user in understanding the effects of his search statement even to the level of identifying important terms that are not contributing to the

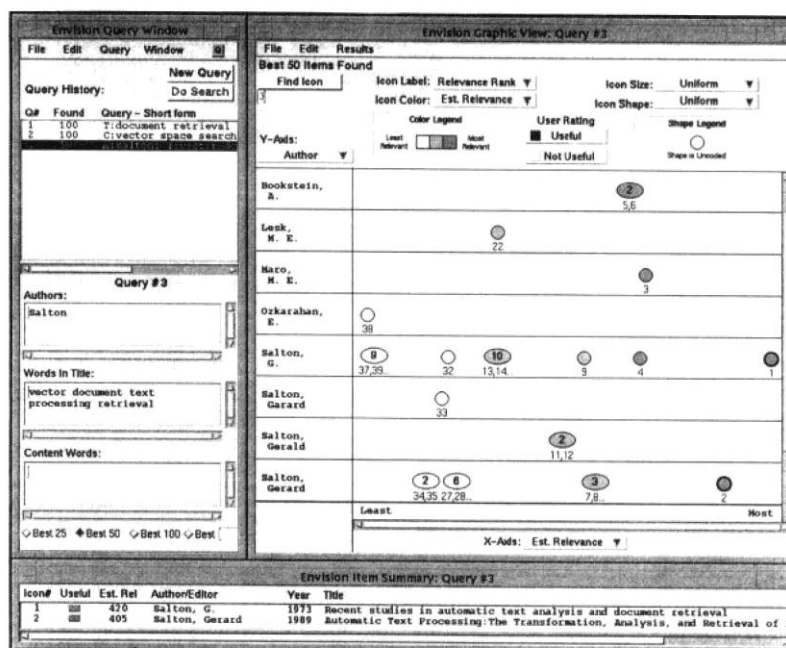


Figure 6.7 Envision Interface (from SIGIR 96, page 68)

search process. One solution is a graphical display of the characteristics of the retrieved items which contributed to their selection. This is effected in the Envision system when index terms are selected as an axis.

The Envision system not only displays the relevance rank and estimated relevance of each item found by a query, but also simultaneously presents other query information. The design is intentionally graphical and simple using two-

dimensional visualization. This allows a larger variety of user computer platforms to have access to their system (Nowell-96). Figure 8.7 shows Envisions three interactive windows to display search results: Query window, Graphic View window, and Item Summary window. The Query window provides an editable version of the query. The Item Summary window provides bibliographic citation information on items selected in the Graphic View window. The Graphic View window is similar to scatterplot graphs. Each item in the Hit file is represented by an icon in the window. Selecting an item in the window provides bibliographic information on the same display. Circles represent single items with the relevance weights displayed below them. Ellipses represent clusters of multiple items that are located at the same point in the scatterplot with the number of items in the

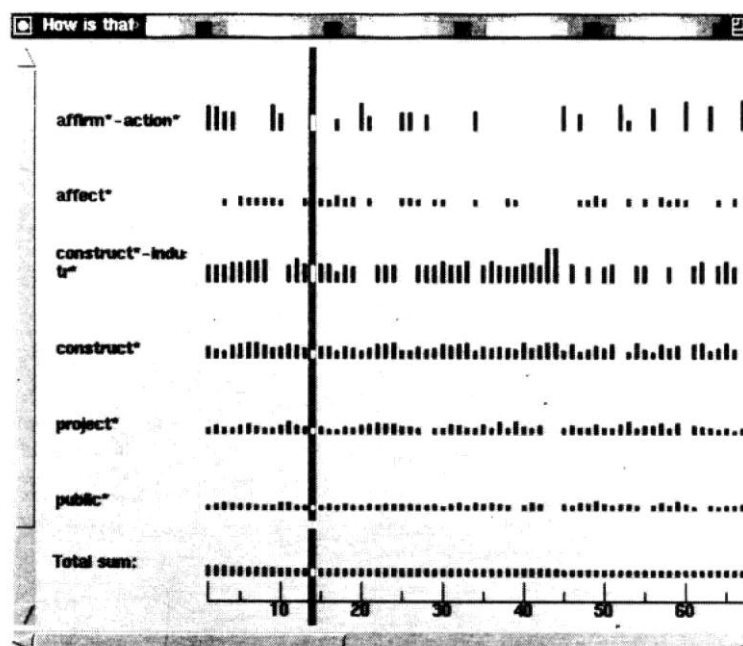


Figure 6.8 Visualization of Results (from SIGIR 96, page 88)

ellipse and their weights below the ellipse. In this example, estimated relevance is on the X-axis and author's name is on the Y-axis. This type of interface provides a very user friendly environment but encounters problems when the number of relevant items and entries for an axis becomes very large. Envision plans to address this issue by a “zoom” feature that will allow seeing larger areas of the scatterplot at lesser detail.

A similar technique is used by Veerasamy and Belkin (Veerasamy-96). They use a series of vertical columns of bars. The columns of bars represent documents, and the rows represent index terms. The height of the bar corresponds to the weight of the corresponding term (row) in the corresponding item (column). In addition to the query terms, the system shows the additional words added to the system by relevance feedback. Figure 8.8 provides an example for a search statement of “How affirmative action affected the construction industry.” This approach quickly allows a user to determine which terms had the most effect on retrieving a specific item (i.e. by scanning down the column). It also allows the user to determine how the various terms contributed to the retrieval process (i.e. by scanning a row). This latter process is very important because it allows a user to determine if what he considers to be an important search term is not contributing strongly or not found at all in the items being retrieved. It also shows search terms that are causing items to be retrieved allowing their removal or reduction in query weight if they are causing false hits. In the Boolean environment this function was accomplished by vocabulary browsing (see Chapter 2) that allows for a user to see the number of items a particular term is in prior to including it in a search.



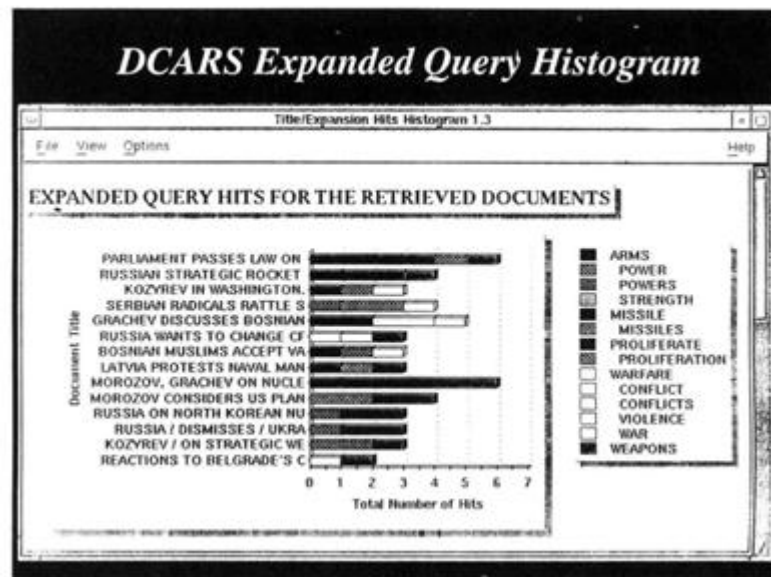


Figure 6.9 Example of DCARS Query Histogram (from briefing by CALSPAN)

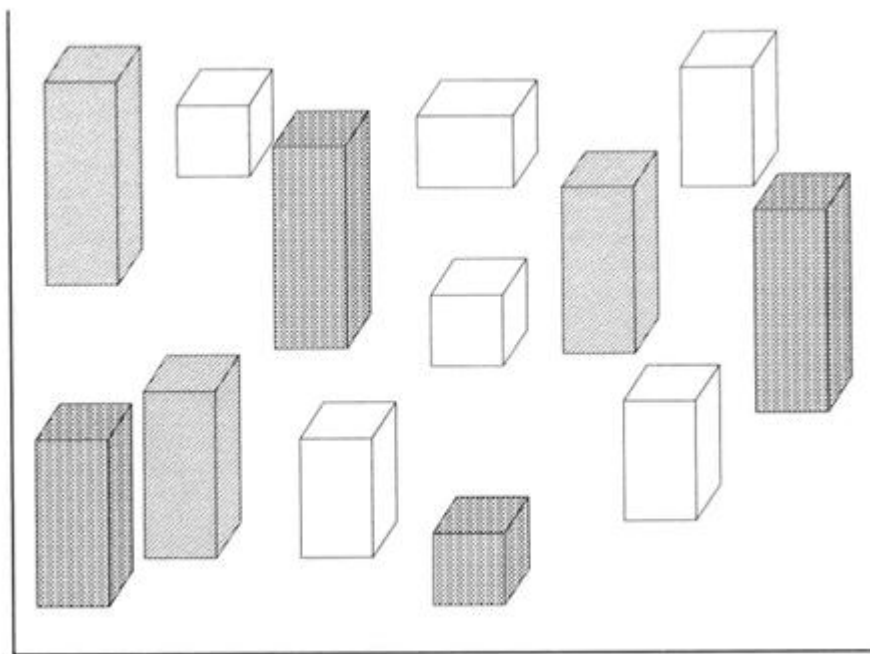


Figure 6.10 Cityscape Example

A slightly different commercial version having properties similar to the systems above is the Document Content Analysis and Retrieval System (DCARS) being developed by Calspan Advanced Technology Center. Their system is designed to augment the RetrievalWare search product. They display the query results as a histogram with the items as rows and each term's contribution to the selection indicated by the width of a tile bar on the row (see Figure 8.9). DCARS provides a friendly user interface that indicates why a particular item was found, but it is much harder to use the information in determining how to modify search statements to improve them.

Another representation that is widely used for both hierarchical and network related

information is the “cityscape” which uses the metaphor of movement within a city. In lieu of using hills, as in the terrain approach, skyscrapers represent the theme (concept) area as shown in Figure 8.10. This is similar to extending bar charts to three dimensions. Buildings can be connected by lines which can vary in representation to describe interrelationships between themes. Colors or fill designs can be used for the visualization presenting another layer of information (e.g., the building having the same color may be members of a higher concept). Movement within the cityscape (or terrain) of the viewer perspective allows zooming in on specific information areas that will bring into view additional structures that might have been hidden by the previous viewpoint.

An easily understood metaphor for users is that of a library. Information content can be represented as areas within a library that the user can navigate through. Once in a particular “information room” the user can view the virtual “books” available within that space as if they are sitting on a bookshelf. Once the book is accessed, the user can scan a group of related items with each item represented as a page within the book. The user can fan the pages out. This is exemplified by the WebBook (Card-96a).

Correlating items or words within items was described in detail in Chapter 6 to cluster items or create statistical thesauri. When the complete term relationship method is used, a very large matrix is created. Each cell in the matrix defines the similarity between two terms (or items). Meaningful display of the table is not possible in table form. Mitre Corporation has developed an interface that displays the complete matrix using clusters of dots to represent correlation's. Once the user zooms in on a particular area of correlation, the specific words become visible along with clusters showing their correlation to other words (Gershon-96). Another approach to representing thesaurus and contents is being tested by Zizi and Pediotakis (Zizi-96). They build a thesaurus automatically from the abstracts of the items extracting both single and two-word expressions. They create a presentation view and a document view. They divide the display space, based upon thesaurus classes, into regions. Each area is sized proportionally to the importance of the class for the collection. Once the presentation view is defined, the document view is created. The documents are placed on ellipses corresponding to the presentation view, and the weight of the document is reflected by the radius of the ellipse.

Another task in information systems is the visualization of specific text within an item versus between items. In some situations, items are allowed to change over time via editing. Thus, there is both the static representation and a time varying representation. Text changing representations are very important when the text being represented is a software program of millions of lines of code. AT&T Bell laboratories created the SeeSoft system which uses columns and color codes to show when different lines of code have been changed. This technique was used as a basis for a similar code visualization tool, DEC FUSE/SoftVis (Zaremba-95). They created small pictures of files that represent the code in the file with the size of the picture scaled to the number of lines of code in the file. Color coding indicates different characteristics of the code (e.g., green is comments). A user can quickly see the relative structure of all of the code files composing a system along with the complexity of each of the modules. The TileBars tool from Xerox PARC provides the user with a visualization of the distribution of query terms within each item in a Hit file. Using

this tool, the user can quickly locate the section of the item that is most likely to be of interest.

Although information retrieval focuses on the unstructured text, another aspect of informational items is the citation data and structured aspects of indexing items. This data structure can be manipulated via structured databases as well as traditional information systems. Visualization tools have also been constructed for databases. The first visualization tool was the Query By Example user interface developed by IBM (Zloof-75). The interface presented the user with two-dimensional tables on the display screen and based upon the user is defining values of interest on the tables, the system would complete the search. Current visual query languages use visual representations of the database structure and contents (Catarci-96). The Information Visualization and Exploration Environment (IVEE) makes use of the three dimensional representation of the structured database as constrained by the user's search statement (Ahlberg-95). In one representation a three-dimensional box represents a larger space and smaller boxes within the space represent realizations of specific values (e.g., the box represents a department and the smaller boxes represent employees in the department). It has additional visualizations of data as maps and star fields. The user is provided with sliders and toggles to manipulate the search. Another specialized tool for displaying homes for sale in a database is the Home Finder system. It presents a star field display of homes for sale and overlays it with a city map showing the geographic location for each icon that represents a home (Ahlberg-94).

When hyperlinks are used as the information retrieval basis for locating relevant items, the user encounters orientation problems associated with the path the user followed to get to the current location. This is effect getting "lost in cyberspace." One solution is providing the user with a view of the information space. The user can user a pointing device to indicate the item the user would like to navigate to. MITRE Corporation has developed a tool used with web browsers that enables a user to see a tree structure visual representation of the information space they have navigated through (Gershon-96).

Another area in information visualization is the representation of pattern and linkage analysis. A system that incorporates many information visualization techniques including those used to represent linkage analysis is the Pathfinder Project sponsored by the Army (Rose-96). It contains the Document Browser, CAMEO, Counts, Cross Field Matrix, OILSTOCK and SPIRE tools. The Document Browser uses different colors and their density for words in the text of items to indicate the relative importance of the item to their profile of interest. CAMEO models an analytic process by creating nodes and links to represent a problem. Queries are associated with the nodes. The color of the nodes change based on how well the found items satisfy the query. Counts uses statistical information on words and phrases and plots them over time. Time is used as a parameter to show trends in development of events. The display uses a three-dimensional cityscape representation of the data. The Cross Field Matrix creates a two-dimensional matrix of two fields in a dataset. Selected values in each of the datasets will be in each row and column for the two fields. Colors are used to represent time span. For example, countries could be on one axis and products on the other axis. The colors would indicate how long the country has been producing a particular product. Intersection can be used to access all the items that supported the



particular product in a particular country. OILSTOCK allows the placement of data on a geographic mapping tool. The relationship of data to maps is a different use of information visualization. Discussion of this area is left to the many sources on Geographic Information Systems (GIS). The SPIRE tool is a type of scattergraph of information. Items are clustered and displayed in a star chart configuration. Distance between two points is indicative of their similarity based upon concurrence of terms.

#### 6.4 Summary

Information visualization is not a new concept. The well known saying that “a picture is worth a thousand words” is part of our daily life. Everything from advertisements to briefings make use of visual aides to significantly increase the amount of information presented and provide maximum impact on the audience. The significant amount of “noise” (non-relevant items) in interactions with information systems requires use of user interface aides to maximize the information being presented to the user. Pure textual interfaces provide no capabilities for aggregation of data, allowing a user to see an overview of the results of a search. Viewing the results of a search using a hierarchical paradigm allows higher levels of abstraction showing overall results of searches before the details consume the display.

Visualization techniques attempt to represent aggregate information using a metaphor (e.g., peaks, valleys, cityscapes) to highlight the major concepts of the aggregation and relationships between them. This allows the user to put into perspective the total information before pursuing the details. It also allows major pruning of areas of non-relevant information. A close analogy is when searching on “fields” in the index at a library, the book shelves on horticulture would be ignored if magnetic fields was the information need. By having the data visualized constrained by the users search, the display is focused on the user’s areas of interest. Relationships between data and effectiveness of the search become obvious to the user before the details of individual items hide the higher level relationships.

Cognitive engineering suggests that alternative representations are needed to take maximum advantage of different physiological and cultural experiences of the user. Colors are useless to a color blind user. A person who grew up on a farm living in the country may have more trouble understanding a “city-scape” than a New York city resident. Using visual cues that a person has developed over his life-experience can facilitate the mapping of the visual metaphor to the information it is representing.

As the algorithms and automatic search expansion techniques become more complex, use of visualization will take on additional responsibilities in clarifying to the user, not only what information is being retrieved, but the relationship between the search statement and

the items. Showing relationships between items has had limited use in systems and been focused on data mining type efforts. The growth of hypertext linkages will require new visualization tools to present the network relationships between linked items and assist the user in navigating this new structure.



## Chapter 7

# Electronic Health Record Systems

- 
- 7.1 What is the definition of an electronic health record (EHR)
  - 7.2 How does an EHR differ from the paper record?
  - 7.3 What are the functional components of an EHR?
  - 7.4 What are the benefits of an EHR?
  - 7.5 What are the impediments to development and use of an EHR?
  - 7.1. What Is An Electronic Health Record?

The preceding chapters introduced the conceptual basis for the field of biomedical informatics, including the use of patient data in clinical practice and research. We now focus attention on the **patient record**, commonly referred to as the patient's **chart** or **medical record**. The patient record is an amalgam of all the data acquired and created during a patient's course through the health care system. The use of medical data was covered extensively in Chapter 2. We also discussed the limitations of the paper record in serving the many users of patient information. In this chapter, we examine the definition and use of computer-based patient record systems, discuss their potential benefits and costs, and describe the remaining challenges to address in their dissemination.

### 7.1.1. Purpose of a Patient Record

Stanley Reiser (1991) wrote that the purpose of a patient record is “to recall observations, to inform others, to instruct students, to gain knowledge, to monitor performance, and to justify interventions.” The many uses described in this statement, although diverse, have a single goal—to further the application of health sciences in ways that improve the well-being of patients, including the conduct of research and public health activities that address population health. Yet, observational studies of physicians' use of the paper-based record find that logistical, organizational, and other practical limitations reduce the effectiveness of traditional records for storing and organizing an ever-increasing number of diverse data. An electronic health record (EHR) is designed to overcome many of these limitations, as well as to provide additional benefits that cannot be attained by a static view of events.

**An electronic health record (EHR)** is a repository of electronically maintained information about an individual's lifetime health status and health care, stored such that it can



serve the multiple legitimate users of the record. Traditionally, the patient record was a record of care provided when a patient is ill. Managed care (discussed in Chapter 23) encourages health care providers to focus on the continuum of health and health care from wellness to illness and recovery. Consequently, the record must integrate elements regarding a patient's health and illness acquired by multiple providers across diverse settings. In addition, the data should be stored such that different views of those data can be presented to serve the many uses described in Chapter 2.

A **computer-based patient-record (EHR) system** adds information management tools to provide clinical reminders and alerts, linkages with knowledge sources for health care decision support, and analysis of aggregate data both for care management and for research. To use a paper-based patient record, the reader must manipulate data either mentally or on paper to glean important clinical information. In contrast, an EHR system provides computer-based tools to help the reader organize, interpret, and react to data. Examples of tools provided in current EHR systems are discussed in Section 12.3.

A number of large institutions have installed EHRs and described their many different approaches and the lessons learned (Pryor et al., 1983; Giuse and Mickish, 1996; Halamka et al., 1998; Hripcsak et al., 1999; McDonald et al., 1999; Slack and Bleich, 1999; Teich et al., 1999; Yamazaki and Satomura, 2000; Cheung et al., 2001; Duncan et al., 2001; Brown et al., 2003).

#### 7.1.2. Ways in Which an Electronic Health Record Differs from a Paper-Based Record

In contrast to a traditional patient record, whose functionality is tethered by the static nature of paper—a single copy of the data stored in a single format for data entry and retrieval—an EHR is flexible and adaptable. Data may be entered in a format that simplifies the input process (which includes electronic interfaces to other computers where patient data are stored) and displayed in different formats suitable for their interpretation. Further, the EHR can integrate multimedia information such as radiology images and echocardiographic video loops that were never part of the traditional medical record. Data can be used to guide care for a single patient or in aggregate form to help administrators develop policies for a population. Hence, when considering the functions of an EHR, we do not confine discussion to the uses of a single, serial recording of provider-patient encounters. An EHR system extends the usefulness of patient data by applying information-management tools to the data.

**Inaccessibility** is a common drawback of paper records. In large organizations, the traditional record may be unavailable to others for days while the clinician finishes documentation of an encounter. For example, paper records are often sequestered in a medical records department until the discharge summary is completed and every document is signed. During this time, special permission and extra effort are required to locate and retrieve the record. Individual physicians often borrow records for their convenience, with the same effect. With computer-stored records, all authorized personnel can access patient data immediately as the need arises. Remote access to EHRs also is possible.

When the data are stored on a secure network, authorized clinicians with a need to

know can access them from the office, home, or emergency room, to make timely informed decisions. At the same time that EHR systems make data more available to authorized users for legitimate uses, they also provide the tools needed to control and track access to patient records to enforce the privacy policies required by the Health Insurance Portability and Accountability Act (HIPAA; see Chapter 10).

Documentation in an EHR can be more legible because it is recorded as printed text rather than as handwriting, and it is better organized because structure is imposed on input. The computer can even improve completeness and quality by automatically applying validity and required field checks on data as they are entered. For example, numerical results can be checked against reference ranges. Typographical errors can be detected via spell checkers and restricted input menus. Moreover, an interactive system can prompt the user for additional information. In this case, the data repository not only stores data but also enhances their completeness.

Data entered into a computer can be reused. For example, a physician could cut and paste parts of their visit note into a letter to a referring physician and into an admission note. Reusability of data is one way that an EHR increases the provider's efficiency. Data entered as part of the patient care process can also be reused in reports that support patient safety, quality improvement, and regulatory or accreditation requirements.

The degree to which a particular EHR demonstrates these benefits depends on several factors:

- 1- *Comprehensiveness of information.* Does the EHR contain information about health as well as illness? Does it include information from all clinicians who participated in a patient's care? Does it cover all settings in which care was delivered (e.g., office practice, hospital)? Does it include the full spectrum of clinical data, including clinicians' notes, laboratory test results, medication details, and so on?
- 2- *Duration of use and retention of data.* A record that has accumulated patient data over 5 years will be more valuable than one that contains only the last month's worth of records.
- 3- *Degree of structure of data.* Medical data that are stored simply as narrative text entries will be more legible and accessible than similar entries in a paper medical record. Uncoded information, however, is not standardized (see Chapter 7), and inconsistent use of medical terminology limits the ability to search for data, but as anyone who has used an Internet search tool knows, clever use of synonyms and statistics improves the hit rates. Use of a controlled, predefined vocabulary (see Chapter 7) facilitates computer-supported decision making and clinical research, but at the cost of "coding" time to the provider who is entering such data.
- 4- *Ubiquity of access.* A system that is accessible from a few sites will be less valuable than one accessible from any computer by an authorized user (see Chapter 5).

A computer-stored medical record system has disadvantages. It requires a larger initial



investment than its paper counterpart due to hardware, software, training, and support costs. The human and organizational factors often dominate the technical challenges. Physicians and other key personnel will have to take time from their work to learn how to use the system and to redesign their work flow to use the system efficiently.

Another risk associated with computer-based systems is the potential for subtle as well as catastrophic failures. If the computer system fails, stored information may be unavailable for an indeterminate time. Paper records fail one chart at a time. On the other hand, if the average paper chart is unavailable up to 10 percent of the time, that would be equivalent to a 10 percent downtime for the average patient. Furthermore, modern computers with fully redundant components, *swappable memory* and computers and disk units and *mirrored servers*, have very high reliability. Yet, nothing provides complete protection; so contingency plans must be developed for handling brief or longer computer outages. Physicians record large amounts of clinical information in their history, physical examination, and progress notes. Capture of this information directly from the physician is a major goal of medical informatics because it provides the most timely, accurate, and useful content. However, the goal is elusive. The time cost of physician input can be high, so the physician input may be infeasible in some settings. Although new input devices are introduced or improved each year (e.g., pen-based entry, speech input), none of these have become time-competitive with dictation or handwriting. The wireless personal digital assistants (PDAs) and slate computers improve accessibility compared with fixed workstations, but battery life and physical attributes still limit their widespread applicability.

Some institutions are scanning selected parts of the patient chart (including the physicians' notes) into the computer (Teich, 1997). Scanning and storing chart notes into an EHR does solve the availability problems of the paper chart, and the solution can be applied to any kind of document. Indeed, many hospitals now scan in the entire paper chart at discharge for easy retrieval. However, a typical scanned page occupies 50 to 80 kilobytes, so high-speed communication links are required for quick display. Further no option exists for searching or analyzing the content of a scanned document without an abstraction step.

Although it takes time to learn how to use the system and to change workflows, there is mounting recognition that an EHR system is required to support the care process, as well as the regulatory and business side of health care. One of the largest health maintenance organizations (HMOs) recently committed \$1.8 billion to implementing an EHR throughout its health system.

## 7.2. Historical Perspective

The historical development of the medical record parallels the development of science in clinical care. The development of automated systems for dealing with health care data parallels the need for data to comply with reimbursement requirements. Early health care systems focused on inpatient-charge capture to meet billing requirements in a fee-for-service environment. Contemporary systems need to capture clinical information in a managed care environment focusing on clinical outcomes in ambulatory care.



### 7.2.1. Early Hospital Focus

The Flexner report on medical education was the first formal statement made about the function and contents of the medical record (Flexner, 1910). Mayo clinic had begun to record the diagnoses for every admitted patient 3 years earlier (Melton, 1996). In advo-

cating a scientific approach to medical education, the Flexner report also encouraged physicians to keep a patient-oriented medical record. The contents of medical records in hospitals became the object of scrutiny in the 1940s, when hospital-accrediting bodies began to insist on the availability of accurate, well-organized medical records as a condition for accreditation. Since then, these organizations also have required that hospitals abstract certain information from the medical record and submit that information to national data centers. Such discharge abstracts contain (1) demographic information, (2) admission and discharge diagnoses, (3) length of stay, and (4) major procedures performed. The national centers produce statistical summaries of these case abstracts; an individual hospital can then compare its own statistical profile with that of similar institutions.

In the late 1960s, computer-based hospital information systems (HISs) began to emerge (see also Chapter 13). These systems were intended primarily for communication. They collected orders from nursing stations, routed the orders to various parts of the hospital, and identified all chargeable services. They also gave clinicians electronic access to results of laboratory tests and other diagnostic procedures. Although they contained some clinical information (e.g., test results, drug orders), their major purpose was to capture charges rather than to assist with clinical care. Many of the early HISs stored and presented much of their information as text, which is difficult to analyze. Moreover, these early systems rarely retained the content for long after a patient's discharge.

The introduction of the **problem-oriented medical record (POMR)** by Lawrence Weed (1969) influenced medical thinking about both manual and automated medical records. Weed was among the first to recognize the importance of an internal structure of a medical record, whether stored on paper or in a computer. He suggested that the primary organization of the medical record should be by the medical problem; all diagnostic and therapeutic plans should be linked to a specific problem.

Morris Collen (1972) was an early pioneer in the use of hospital-based systems to store and present laboratory test results as part of preventive care. He also wrote an extensive history of the field (Collen, 1995). Use of computers to screen for early warning signs of illness was a basic tenet of HMOs. Other early university hospital-based systems provided feedback to physicians that affected clinical decisions and ultimately patient outcomes. The HELP system (Pryor, 1988) at LDS Hospital, the CCC system at Beth Israel Deaconess Medical Center (Slack and Bleich, 1999), and the Regenstrief System (Tierney et al., 1993; McDonald et al., 1999) at Wishard Memorial Hospital continue to add more clinical data and decision-support functionality.

### 7.2.2. Influence of Managed Care and the Integrated Delivery System

Until recently, the ambulatory care record has received less attention from the commercial vendors than the hospital record because of differences in financing and regulatory requirements. The status of ambulatory care records was reviewed in a 1982 report (Kuhn et al., 1984). Under the influence of managed care (described in detail in Chapter 13), the reimbursement model has shifted from a **fee-for-service model** (payers pay providers for all services the provider deemed necessary) toward a payment scheme

where providers are paid a **fixed fee** for a specific service (payers pay a fixed amount for services approved by the payer). Information management tools that facilitate effective management of patients outside of the hospital setting help providers manage patients' disease more cost-effectively. The emphasis on ambulatory care brought new attention to the ambulatory care record.

Thirty years ago, a single family physician provided almost all of an individual's medical care. Today, however, responsibility for ambulatory care is shifting to teams of health care professionals in outpatient clinics and HMOs (see Chapter 13). Ambulatory care records may contain lengthy notes written by many different health care providers, large numbers of laboratory test results, and a diverse set of other data elements, such as X-ray examination and pathology reports and hospital discharge summaries. Accordingly, the need for information tools in ambulatory practice has increased. COSTAR (Barnett, 1984), the Regenstrief Medical Record System (RMRS) (McDonald et al., 1975), STOR (Whiting-O'Keefe et al., 1985), and TMR (Stead and Hammond, 1988) are among the early systems that focused on ambulatory care. Costar and RMRS are still in use today.

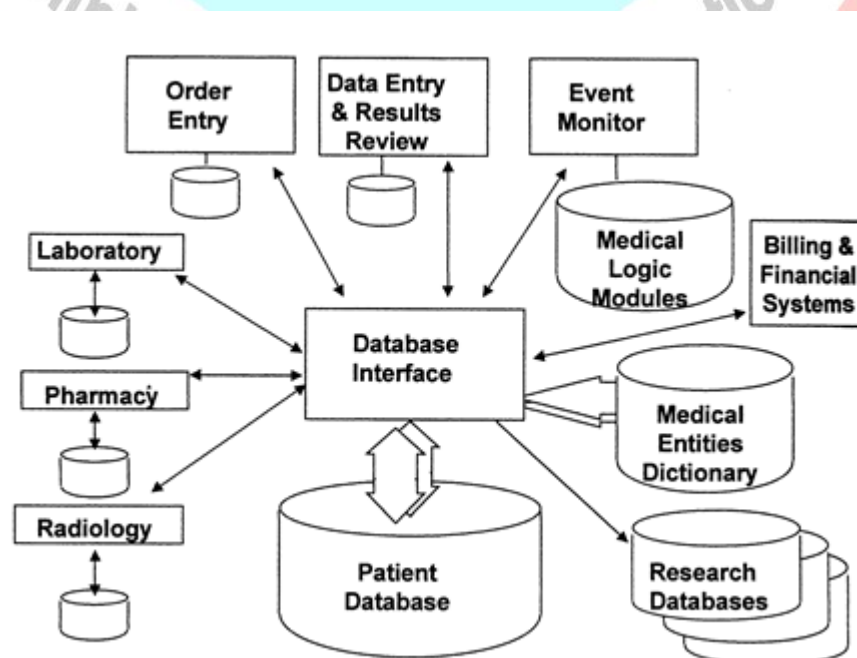
### 7.3. Functional Components of an Electronic Health Record System

As we explained in Section 12.1.2, an EHR is not simply an electronic version of the paper record. When the record is part of a comprehensive EHR system, there are linkages and tools available to facilitate communication and decision making. In Sections 12.3.1 to 12.3.5, we summarize components of a comprehensive EHR system and illustrate functionality with examples from systems currently in use. The five functional components are:]

- Integrated view of patient data
- Clinical decision support
- Clinician order entry
- Access to knowledge resources
- Integrated communication and reporting support

#### 7.3.1. Integrated View of Patient Data

Clearly, providing integrated access to all patient data is the primary purpose of an EHR. Although this task may seem relatively simple, acquisition and organization of these data are major challenges because of the complexity and diversity of the data—ranging from simple numbers to graphs to images to motion images—and the large number and organizationally distributed sources of patient data such as clinical laboratories, radiology departments, free-standing magnetic resonance imaging (MRI) centers, community pharmacies, home health agencies. Furthermore, no unique national patient identifier exists in the United States for linking patient data obtained from many sites (patient indexes to link disparate patient identifiers are discussed in Chapter 10). The fact that different patient data source systems use different identifiers, data content terminologies, and data formats creates substantial work. Administrators of each HER system must revise the message formats and map coding systems from the source system to the format and codes that are acceptable to their EHR system. Today most clinical data sources can deliver the clinical content as **health level 7 (HL7)** messages, but senders deviate from the standard and use local codes as identifiers for clinical observations and orders in these messages. So, some small amount of message tweaking and a large amount of code mapping is usually required. Interface engines facilitate the management and tweaking of the messages (see Chapter 7); Figure 12.1 shows an example of architecture to integrate data from multiple source systems. The database interface depicted not only provides message-handling capability but can also automatically translate codes from the source system to the preferred codes of the receiving EHR. However, human labor is needed to define the mappings that drive this automatic translation. The **interface engine** provides a technical and translation buffer between systems manufactured by different vendors. In this way, organizations can mix different vendors' products and still achieve the goal of integrated access to patient data for the clinician. The idiosyncratic, local terminologies used to identify clinical variables and their values in many source systems represent major barriers to integration of medical record data by EHRs. Code systems such as LOINC (McDonald et al., 2003) and SNOMED (Wang et al., 2002), which have been adopted by the U.S. federal government.





**Figure 7.1.** A block diagram of multiple-source-data systems that contribute patient data ultimately reside in a CPR. The database interface, commonly called an interface engine, may perform a number of functions. It may simply be a router of information to the central database. Alternatively, it may provide more intelligent filtering, translating, and alerting functions. as it dose at Columbia presbyterian Medical Center. (Source: Courtesy of Columbia Presbyterian Medical Center, New York.) agencies for their health-related data and are discussed in Chapter 10, help overcome these barriers.

Clinicians need more than just integrated access to patient data; they also need various **views** of these data (e.g., in chronologic order by report date) so providers can easily find the newest individual results, in a flowsheet format to highlight changes over time across multiple variables, and in focused views tailored to specialties and settings. An example of such a snapshot for general medicine in an outpatient setting visit is shown in Figure 12.2. This summary view of patient data shows the active patient problems, active medications, medication allergies, health maintenance reminders, and other relevant summary information. Such a view presents a current summary of patient context that is updated automatically at every encounter; such updating is not possible in a paper record.

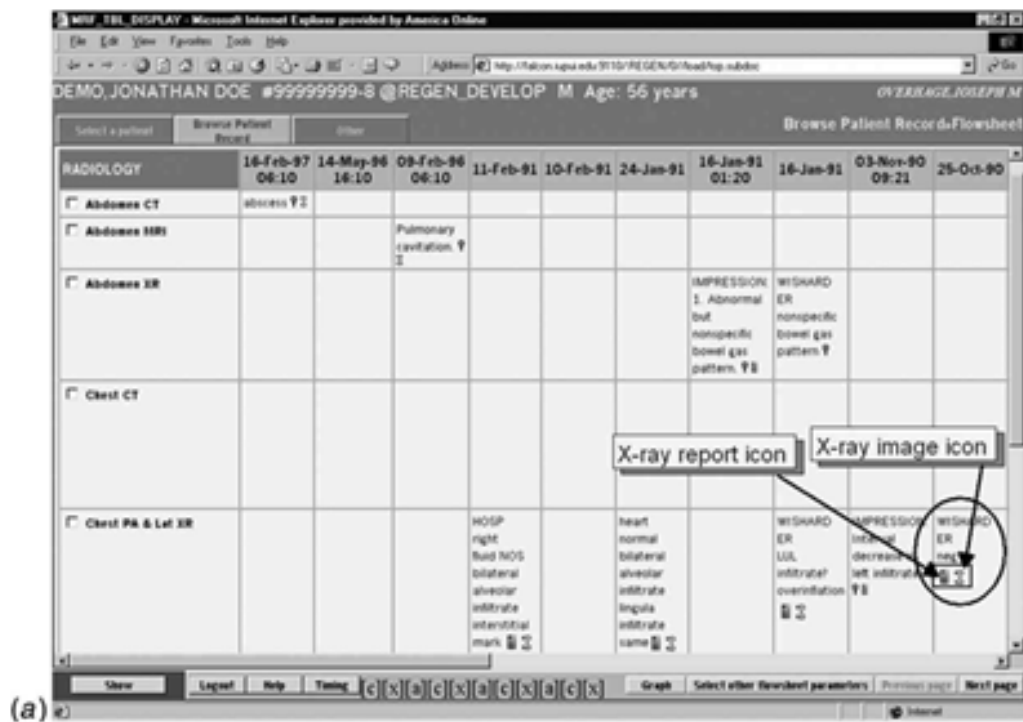
Web browsers for finding and viewing information on the Internet (see Chapters 13 and 24) also provide health care workers with tools to view patient data from remote systems. Advanced security features (e.g., Secure Socket Layer (SSL)) are required to ensure the confidentiality of patient data transmitted over the public Internet. Figure 12.3a shows an integrated view of a flowsheet of the radiology impressions with the rows representing all kinds of radiology examinations and the columns representing Web dates. Clicking on the radiology image icon **8** brings up the radiology images, e.g., the quarter resolution PA and lateral chest X-ray views in Figure 12.3b. An analogous

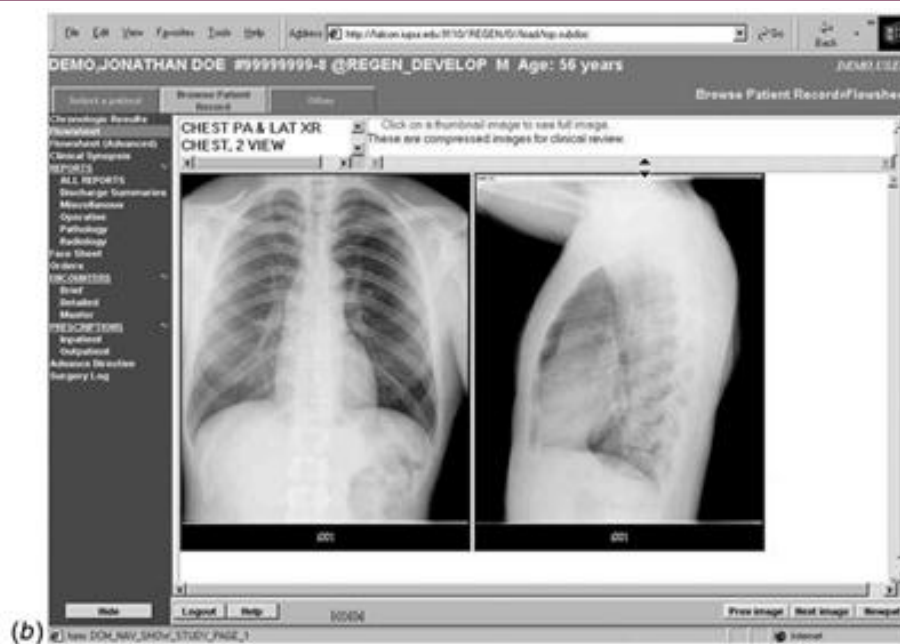
The screenshot shows a web-based patient summary interface for 'Wilson, Stacy'. The patient's age is 65 years, sex is female, and the date of birth is 3/1/1938. The interface is divided into several sections:

- Problem List:** Hypertension, Asthma, Colon polyps, ADVANCED DIRECTIVE, Smoking.
- Medications:** ALBUTEROL 30 MG INALERS, LISINAPRIL 10 MG OR TABS, AZADOCORT 100 MG INALERS.
- Allergies:** PENICILLINS (Allergy: 06/04/1998, Td (Adult)).
- Specialty Report:** Problem Summary.
- Medications (continued):** MAMMOGRAM, PAP-YEARLY, INFLUENZA VACCINE, PNEUMOCOCCAL VACCINE, SIGMOIDOSCOPY, TETANUS IMMUNIZATION.
- Specialty Comments:** (Editable).
- Patient Messages:** Data Message, Family Comments (Editable).
- Family Comments:** Husband recently diagnosed with Alzheimers Disease.

The interface includes navigation links like 'Home', 'Schedule', 'Encounter', 'Patient Info', 'Secure', 'Log Out', and 'Refresh'.

**Figure 7.2.** Quick access to summary information about a patient. The patient's active medical problems, current medications, and drug allergies are among the core data that physicians must keep in mind when making any decision on patient care. This one-page screen provides an instant display of these core clinical data elements as well as reminders about required preventive care. (Source: Courtesy of Epic Systems, Madison, WI.)





**Figure 7.3.** Web resources. (a) Web-browser flowsheet of radiology reports. The rows all report one kind of study, the columns one date. Each cell shows the impression part of the radiology report as a quick summary of the content of that report. The cells include two icons. Clicking on the report icon provides the full radiology report. Clicking on the radiology image icon S provides the images. (b) Shows the chest X-ray images on radiology images obtained by clicking on the “bone” icon. What shows by default is a quarter-sized view of both the PA and lateral chest view X-ray. By clicking on various options, users can obtain up to the full (2,000 x 2,300) resolution, and window and level the images over the 12 bits of a radiographic image, using a control provided by Medical Informatics Engineering (MIE), Fort Wayne Indiana. (Source: Courtesy of Regenstrief Institute, Indianapolis, IN.)

process applies to EKG measurements where clicking on the EKG icon for a particular result brings up the full EKG tracing in Portable Document Format (PDF) form.

### 7.3.2. Clinical Decision Support

Decision support is thought to be most effective when provided at the point of care, while the physician is formulating his or her assessment of the patient’s condition and is making ordering decisions. The most successful decision-support intervention makes complying with the suggested action easy (e.g., simply hitting the “Enter key” or clicking “Accept” with the mouse), while still allowing the physician to control the final decision. Providing access to a brief rationale with the recommendation may increase acceptance of reminders and at the same time educate the care provider.

Figure 12.4 shows the suggestions of a software module in a large HIS. The patient diagnosis uses sophisticated treatment protocols that consider a wide spectrum of clinical information to recommend antibiotic choice, dose, and duration of treatment. Clinicians can view the basis for the recommendations and the logic used. A notable part of this program is its solicitation of feedback when the clinician decides not to follow the recommendations. This feedback is used to improve the clinical protocol and the software



program. Providing online advice on antimicrobial selection has resulted in

| IHC ADULT ANTIBIOTIC ASSISTANT & ORDER PROGRAM                                 |                       |                       |                             |                           |                    |
|--------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------------|---------------------------|--------------------|
| 000000000 Doe, Jane Q                                                          |                       | E606                  | 67yr                        | F                         | Dx: ABD SEPSIS     |
| » Max 24hr WBC=21.0↓ (21.3)                                                    |                       | Admit: 07/27/98 14:55 |                             | Max 24hr                  | Temp=38.7↑ (38.2)  |
| Patient's Diff shows a left shift, max 24hr bands = 22 ↑ (11)                  |                       |                       |                             |                           |                    |
| » RENAL FUNCTION: Decreased, CrCl = 50, Max 24hr Cr= 1.0↓ (1.1)                |                       |                       |                             | IBWeight: 58kg            |                    |
| » ANTI-BIOTIC ALLERGIES: Ampicillin,                                           |                       |                       |                             |                           |                    |
| » CURRENT ANTIBIOTICS:                                                         |                       |                       |                             |                           |                    |
| 1.                                                                             | 07/29/98              | 5DAYS                 | TROVAFLOXACIN (TROVAN),     |                           | VIAL 300. Q 24 hrs |
| 2.                                                                             | 08/01/98              | 2DAYS                 | AMPHOTERICIN B (FUNGIZONE), |                           | VIAL 35. Q 24 hrs  |
| Total amphotericin given = 71mg                                                |                       | K= 3.6mg/dl           | 08/03/98                    | MAG= 2.5mg/dl             | 08/03/98 »»»       |
| IDENTIFIED PATHOGENS                                                           |                       | SITE                  |                             | COLLECTED                 |                    |
| p                                                                              | Gram negative Bacilli | Peritoneal Fluid      |                             | 07/27/98 17:12            |                    |
|                                                                                | Yeast                 | Peritoneal Fluid      |                             | 07/27/98 17:12            |                    |
|                                                                                | Torulopsis glabrata   | Peritoneal Fluid      |                             | 07/27/98 17:12            |                    |
| » THERAPEUTIC SUGGESTION                                                       |                       | DOSAGE                | ROUTE                       | INTERVAL                  |                    |
|                                                                                | Imipenem              | 500mg                 | IV                          | *q12h (infuse over 1hr)   |                    |
|                                                                                | Amphotericin B        | 35mg                  | IV                          | q24h (infuse over 2-4hrs) |                    |
| Suggested Antibiotic Duration: 10 days                                         |                       |                       |                             |                           |                    |
| *Adjusted based on patient's renal function.                                   |                       |                       |                             |                           |                    |
| P-Prelim, Susceptibilities based on antibiogram or same pathogen w/ suscept    |                       |                       |                             |                           |                    |
| <1>Micro <2>OrganismSuscept, <3>Drug Info, <4>ExplainLogic, <5>Empiric Abx,    |                       |                       |                             |                           |                    |
| <6>Abx Hx <7>ID Rnds, <8>Lab/Abx Levels, <9>Xray, <10>Data Input Screen,       |                       |                       |                             |                           |                    |
| <Esc>EXIT, <F1>Help, <O>UserInput, <O>OutpatientModels, <+orF12>Change Patient |                       |                       |                             |                           |                    |
| ↑↓, ORDER: <S>Suggested Abx, <Enter>Other Abx, <D/C>D/C Abx, <->Modify Abx,    |                       |                       |                             |                           |                    |

**Figure 12.4.** Example of the main screen from the Intermountain Health Care Antibiotic Assistant program. The program displays evidence of an infection-relevant patient data (e.g., kidney function, temperature), and recommendations for antibiotics based on the culture results. (Source: Courtesy of R. Scott Evans, Stanley L. Pestotnik, David C. Classen, and John P. Burke, LDS Hospital, Salt Lake City, UT.)

significantly improved clinical and financial outcomes for patients whose infectious diseases were managed through the use of the program.

Reminders and alerts can be raised during outpatient encounters as well. Indeed the outpatient setting is where the most formal reminder studies have been performed (Garg et al., 2005). Figure 12.5 shows how alerts and reminders are included on a preprinted encounter form for use during an outpatient visit. The system searches for applicable decision-support rules and prints relevant reminders on the encounter form during batch printing the night before the scheduled visit. Figure 12.6 shows computer-based suggestions regarding health maintenance topics from the Veterans Administration EHRS. These suggestions were derived from rules that examine the patient's problems

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>SAMPLE, PATIENT</b><br>06-Mar-03 317-630-7400<br>1002 W Tenth St INDIANAPOLIS 46202<br>HCP: none (1) Dr<br>1 N BRIEF, 2 N LIMITED, 3 N INTEND, 4 N EXT, 5 N COMP, 6 INT /C PROC, 7 E RN, 8 E BRIEF, 9 LIMITED OV-EP, 10 E INTEND                                                                                                                                                                                                                                 |  | <b>AM 2 Dr: BIONDICH, P</b><br>Wishard Memorial Hospital<br>1002 W Tenth Street<br>Indianapolis IN 46202                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
| Observations List ---<br>HEIGHT PDS 19.25 IN<br>WEIGHT PDS 11.75 lb<br>TEMP RECTAL<br>TEMP AXILLARY<br>RR<br>PULSE<br>HEAD CIRCUMF 37.00 CM<br>HEAD CIRC NILE<br>WT NILE<br>HEIGHT NILE<br>SYS BP SITTING<br>DIAS BP SITTING<br>TIME COUNSELING<br>TOTAL TIME C FAT 30 ml                                                                                                                                                                                           |  | Age: 2mo<br>Informant: mom/dad<br>Feedings: nl, EW 302 q3-4L<br>Elimination: nl<br>Sleep: nl<br>Concerns: "bumps on his face", "recent congestion"<br>Interim illness: 0<br>Reaction to Previous Vaccines: n/a<br>Reviewed PMHx / SR / PMHx from 2 wk WCC note                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
| d/c --- Diagnoses List ---<br><input type="checkbox"/> 1 466.19 ACU BRNCHITS D/T OTH ORG<br><input type="checkbox"/> 2 URI<br><input checked="" type="checkbox"/> 3 diarrhea<br><input checked="" type="checkbox"/> 4 seborrhea nose<br><input checked="" type="checkbox"/> 5 esophageal reflux<br><input type="checkbox"/> 6<br><input type="checkbox"/> 7                                                                                                         |  | PHYSICAL EXAM:<br>Normal:<br><input checked="" type="checkbox"/> General Condition happy, playful, smiles<br><input type="checkbox"/> Head mild rhinorrhea<br><input type="checkbox"/> Skin CTA (B) 5 W<br><input type="checkbox"/> Eyes/Vision RRR J M<br><input type="checkbox"/> Ears/Hearing<br><input type="checkbox"/> Nose/Throat<br><input type="checkbox"/> Teeth/Gums<br><input type="checkbox"/> Nodes<br><input type="checkbox"/> Chest/Lungs<br><input type="checkbox"/> Heart<br><input type="checkbox"/> Pulses<br><input type="checkbox"/> Abdomen<br><input type="checkbox"/> Ext Genitalia<br><input type="checkbox"/> Hip Stability<br><input type="checkbox"/> Back<br><input type="checkbox"/> Extremities<br><input type="checkbox"/> Neuro/Muscle Tone<br>Significant findings:<br>very mild papular rash  |  |
| DEVELOPMENT:<br><input checked="" type="checkbox"/> Vocalizes/Babbles<br><input checked="" type="checkbox"/> Transfers Objects<br><input checked="" type="checkbox"/> Rolls over<br><input checked="" type="checkbox"/> Pulls to sit - no head lag<br><input checked="" type="checkbox"/> Sits with support<br><input checked="" type="checkbox"/> Supports weight on legs<br><input type="checkbox"/> Vision Observed<br><input type="checkbox"/> Hearing Observed |  | ANTICIPATORY GUIDANCE:<br><input checked="" type="checkbox"/> Nutrition, 2-3 solid meals<br><input type="checkbox"/> No bottle in bed<br><input type="checkbox"/> Flotride if indicated<br><input checked="" type="checkbox"/> Oral health / teething/cleaning<br><input checked="" type="checkbox"/> Sleep practices / Night crying<br><input checked="" type="checkbox"/> Poison control # 962-2323<br><input checked="" type="checkbox"/> Baby proof home/stair gates<br><input checked="" type="checkbox"/> Smoke alarm<br><input checked="" type="checkbox"/> Car seat<br><input checked="" type="checkbox"/> Sun exposure<br><input checked="" type="checkbox"/> Infant stimulation/Stranger Anxiety<br><input checked="" type="checkbox"/> Vaccine risk/Benefit<br><input checked="" type="checkbox"/> Environmental smoke |  |
| ASSESSMENT:<br>2mo WCC<br>mild URI                                                                                                                                                                                                                                                                                                                                                                                                                                  |  | PLANS:<br>immunizations as below<br>✓ RTC in 2mo for WCC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
| 26-Aug-03<br>Encounter Date<br>20030826092009999996                                                                                                                                                                                                                                                                                                                                                                                                                 |  | 26-Aug-03 09:20 AM<br>Printed: 23-Aug-03 Page: 3<br>OPB-8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |

Figure 7.5. Pediatric encounter form. The questions on these forms vary by age. Reminders for routine immunizations appear at the bottom. (Source: Courtesy of Regenstrief Institute, Indianapolis, IN.)

| Available Reminders           | Due Date   | Last Occurrence | Priority |
|-------------------------------|------------|-----------------|----------|
| Advanced Directives Education | DUE NOW    |                 |          |
| Blood Pressure Check          | 08/13/2000 | 08/18/2000      |          |
| Alcohol Abuse Education       | 04/05/2001 | 04/05/2000      |          |
| Ankylos Arteritis             | DUE NOW    |                 |          |
| Applicable                    |            |                 |          |
| Other Categories              |            |                 |          |

| Icon Legend | Templates                                                | Reminders | Notes | Consults |
|-------------|----------------------------------------------------------|-----------|-------|----------|
|             | Reminder Category                                        |           |       |          |
|             | Reminder is Due                                          |           |       |          |
|             | Reminder is not due, but is Applicable                   |           |       |          |
|             | Reminder is Not Applicable                               |           |       |          |
|             | Reminder status has not yet been evaluated               |           |       |          |
|             | Reminder has an associated Reminder Dialog               |           |       |          |
|             | Reminder's associated Reminder Dialog has been processed |           |       |          |

Figure 7.6. A powerful feature of electronic medical records is their use of protocols or algorithms to review the patient's electronic data and provide alerts and reminders about any unmet goals of those protocols to clinical decision makers. The first figure shows adult advisories for preventive services produced by the Veterans Administration computer-based patient record system (CPRS)—released as VISTA open source system (web reference accessed sep 17, 2005 <http://www.vistasoftware.org/>). Clinical users may optionally complete any “due” reminders, browse applicable reminders, or review completed

reminders. The second screen defines the meaning of the icons from that same system functionality. (Source: Courtesy of Veterans Administration, 2003, Salt Lake City, UT.)

and medications and the timing of laboratory test orders. Even for hospitalized patients, reminders about preventive health procedures can improve compliance. In one study, there was a 50-fold increase in influenza immunization orders among eligible patients (Dexter et al., 2001).

### 7.3.3. Clinician Order Entry

If the ultimate goal of an EHR system is to help clinicians make informed decisions, then the system should present relevant information at the time of order entry. Several systems have the capability of providing decision support during the order-entry process (Dexter et al., 2001; Steen, 1996; Evans et al., 1999; Sanders and Miller, 2001; Kuperman et al., 2003). For example, a clinical team in the medical intensive care unit (ICU) at Vanderbilt University Hospital can use an electronic chart rack to view active orders and enter new orders. The WIZ Order screen integrates information about a patient's active orders, clinical alerts based on current data from the electronic patient record, and abstracts of relevant articles from the literature. Clinical alerts attached to a laboratory test result can also include suggestions for appropriate actions (Geissbuhler and Miller, 1996). Figure 12.7 shows computer suggestions for dosing and follow-up of intravenous heparin orders at Vanderbilt. Physician order-entry systems can warn the physician about allergies and drug interactions before they complete a medication order as exemplified by the screen from Partner's outpatient medical record shown in Figure 12.8.

**1) Upon MD stating patient is eligible for protocol, WizOrder calculates heparin dose and makes it easy to order tests associated with guidelines**

**IV heparin for Confirmed PE in Adults**

Guidelines for the treatment of Confirmed PE are listed below with calculated values in RED based on the patient's weight (77 kg)

- Bolus with heparin 80 U/kg I.V. [CONTRAINDICATIONS] [LMW HEPARIN]
- Begin maintenance infusion of heparin at 18 U/kg/hr [CONTRAINDICATIONS] [LMW HEPARIN]
- Check PTT at 6 hour intervals to keep PTT in range of 65 to 110 seconds
- Check platelet count daily [INFO ON HEPARIN-INDUCED THROMBOCYTOPENIA]
- Start warfarin therapy on day 1 at 5 mg and adjust to give INR of 2-3 [CONTRAINDICATIONS]
- Stop heparin therapy after at least 4-6 days of combined therapy when INR is > 2.0 for 2 consecutive days
- Continue warfarin treatment for at least 3 months at INR of

**2) Links to educational materials available in protocol**

**3) MD reviews relevant medications & labs**

Orders you may wish to consider (check to order) - Order only necessary items (duplicate order checking not done on this page).

Current Date and Time: 04/20/2000 09:10 AM

☐ Bolus/rebolus with I.V. heparin (U) 6200 (80 x 77 = 6200 IU)

☐ Begin continuous infusion of I.V. heparin (U/hr) 1390 (18 x 77 = 1390 IU/hr)

☐ Check PTT q6 (starting 6 hours after bolus)

☐ Check platelet count qAM

☐ Begin warfarin p.o. at (mg/day) 5 on (mm/dd/yyyy) 04/12/00

☐ Check PT/INR qAM

I am not doing some/all suggestions above because:

**4) MD selects actions and clicks button to activate guideline-related orders**

| Anticoag Meds         | Dose | Date |
|-----------------------|------|------|
| No Anticoagulant Meds |      |      |

| Labs           | Value          | Date |
|----------------|----------------|------|
| PTT            | None available |      |
| INR            | None available |      |
| Platelet Count | None available |      |
| PCV            | None available |      |

**Figure 12.7.** An example of computer-assisted decision support. At the top left, this screen specifies the actions required for intravenous (IV) heparin drip and provides access to text knowledge about them, and alternatives. To the right, the screen displays computer-relevant laboratory and medication information, and at the bottom, patient-specific protocol-based values for the bolus and infusion rate. The provider can complete the order (with a single click) or can modify the order as needed. (Source: Courtesy of



Vanderbilt University, Nashville, TN.)

Testi, Test

16733347 (BWH) 09/02/1970 (32 yrs.) F

Select Desktop Pt Chart: Medications Oncology Custom Reports Admin Sign Results Resour

**Warning**

**You are ordering: FLUVASTATIN**

**Drug - Allergy Intervention**

**Alert Message**

The patient has a documented allergy: LIPITOR (ATORVASTATIN). Reaction: GI upset.

**Keep New Order - select reason(s)**

☒ Patient does not have this allergy, will D/C pre-existing allergy

**Reasons for override:**

☐ Patient has taken previously without allergic reaction

☐ Low risk cross sensitivity, will monitor

☐ No reasonable alternatives

☐ Other

**Drug - Drug Interaction**

**Alert Message**

Patient is currently on: MVI (MULTIVITAMINS) 1 TAB PO QD

Pt. on HMG-CoA Reductase Inhibitor and Niacin - Severe myopathy or rhabdomyolysis may occur. - Recommend that these drugs not be used together or that LFT's and symptoms be regularly monitored.

**Keep New Order - select reason(s)**

☒ Will D/C pre-existing drug

**Reasons for override:**

☐ Will adjust dose as recommended

☐ Will monitor as recommended

☐ Patient has already tolerated combination

☐ No reasonable alternatives

☐ Other

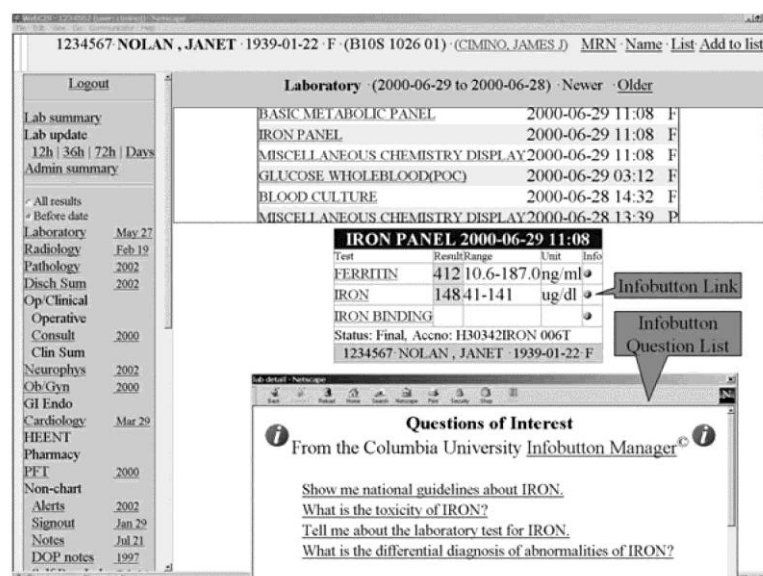
Continue New Order Cancel

**Figure 7.8.** Drug-alert display screen from Partners outpatient medical record application (Longitudinal Medical Record, LMR). The screen shows a drug-allergy alert and a drug-drug interaction alert in response to a proposed prescription for fluvastatin. (Source: Courtesy of Partners Health Care System, Chestnut Hill, MA.)

Once a physician order-entry system is adopted into the practice culture, simply changing the default drug or dosing based on the latest scientific evidence can significantly change the physician's ordering behavior. Clinical quality and financial costs can be changed virtually overnight.

#### 7.3.4. Access to Knowledge Resources

Most queries of knowledge resources, whether they are satisfied by consulting another human colleague or by searching through reference materials or the literature, are conducted in the context of a specific patient (Covell et al., 1985). Consequently, the most effective time to provide access to knowledge resources is at the time decisions or orders are being contemplated by the clinician. Today a rich selection of knowledge sources ranging from the National Library of Medicine's free literature search site, PubMed to full-text resources such as OVID and online references such as Up-To-Date are available for perusal. Consequently, it is relatively easy for physicians to get medical knowledge while reviewing results or writing notes or orders online. However, active presentation of literature relevant to a particular clinical situation, such as an "Infobutton" would increase the chance that the knowledge will influence clinicians' decisions (Cimino et al., 2003) (Figure 12.9).



**Figure 7.9.** A Computer-based patient record (CPR) linked to knowledge resources so that context-specific information can be displayed at the time of clinical decision making. This figure shows the use of Columbia University's info buttons - during results review. Clicking on the info button adjacent to the Iron result generates a window (below) with a menu of questions. When the user clicks on one of the questions—he/she gets the answers. (Source: Courtesy of Columbia Presbyterian Medical Center, New York.)

### 7.3.5. Integrated Communication and Reporting Support

As the care function becomes increasingly distributed among multidisciplinary health care professionals, the effectiveness and efficiency of communication among the team members affect the overall coordination and timeliness of care provided. Most messages are associated with a specific patient. Thus, communication tools should be integrated with the EHR system such that messages (including system messages or laboratory test results) are electronically attached to a patient's record, i.e., the patient's record should be available at the touch of a button. Geographic separation of team members creates the demand for networked communication that reaches all sites where providers make decisions on patient care. These sites include the providers' offices, the hospital, the emergency room, and the home. Connectivity to the patient's home will provide an important vehicle for monitoring health (e.g., home blood-glucose monitoring, health status indicators) and for enabling routine communication. Communication also can be "pushed" to the user via e-mail and or pager services (Major et al., 2002; Poon et al., 2002) or "pulled" by providers at their routine interactions with the computer. Figure 12.10 shows a screen that can be viewed after the immediate notification via pager about a critically low serum potassium level in a patient taking digitalis.

An EHR system can also help with routine patient handoffs, where the responsibility for care is transferred from one clinician to another. Typically, a brief verbal or written exchange helps the covering clinician understand the patient's problems, as is important for making decisions when the primary clinician is unavailable. An example of a screen

```

View PtLookup
Patient: H...E 65F 06215446 Adm: 08/17/94 Room: 12B-351
Time: 10/19/94 05:39 AM Alert#38269 12B phone: x7865
Alert: DANGEROUSLY LOW SERUM POTASSIUM
Reason: (BLOOD) K = 3.2 at 04:22 AM, 10/19/94
Patient is currently on DIGOXIN.

Relevant medications: LASIX 20 MG IV BID Starting on 10/18 (10/17)
DIGOXIN EVEN days:.125; ODD days:.25 PO (09/28)

Alert Details

Actions:
[ JA D/C or EDIT relevant medications
[ JB Order POTASSIUM CHLORIDE IV
[ JC Order POTASSIUM CHLORIDE PO
[ JD Order set: STAT EKG
[ JE Order set: STAT K

Ganesan, Shridar, M.D., Ph.D. Bp#2710 was paged on 05:40 AM Oct 19, 1994
Covering M.D.: Morrow, David Andrew, M.D. Bp#5336 pAge M.D.
done <done, Go to OE> comments Logic
Press Alt-O or Alt-G to exit and acknowledge alert.

```

**Figure 7.10.** When such a laboratory test is completed, the computer identifies critically abnormal results of special importance then sends a message to the ordering physician's pager announcing the result. The physician can find details on a nearby computer screen (shown in figure) that displays the relevant facts, and suggests remedial actions, such as an order for potassium replacement. The example in the figure is based on a rule that checks for serum potassium concentrations of less than 3.3 mEq/L when a patient is also taking dioxin. (Source: Courtesy of Brigham and Women's Hospital, Boston, MA.)

The screenshot shows a medical software interface with a menu bar at the top (Patient, Schedule, Exam, Review, Charting, Other Enc, Options, In Basket, Desk, Message, Pathways, Secure, Help, Exit). Below the menu bar is a patient information section for "CORMICK, MADELINE K" (DOB: 10/23/56, Age: 38, Sex: F, Patient Type: FFS, Primary Provider: CURMAN, MICHAEL, Primary Center: MAIN EPIC). The "Results" section displays a table of test results:

| Status  | Visit | Patient             | Test                | Provider        | Results            |
|---------|-------|---------------------|---------------------|-----------------|--------------------|
| Pending | 06/28 | Panis, Roberto L    | STREP. RAPID SCREEN | Seeger, Marty S | Final result       |
| Pending | 06/01 | Danvers, James      | HEMATOCRIT          | Seeger, Marty S | Final result       |
| Pending | 05/26 | Able, Sarah         | URINALYSIS          | Seeger, Marty S | Final result       |
| Pending | 10/10 | Cormick, Madeline K | PAP SMEAR           | Seeger, Marty S | Final result       |
| Pending | 07/13 | Pfeiler, Simon      | UPPER GI SERIES     | Seeger, Marty S | Preliminary result |

Below the results table, there is a section for "Status of other order(s) for this encounter:" listing several orders (URINALYSIS, MAMMOGRAM, CONSULT, SURGERY, CBC) with their respective order numbers and status (e.g., "No Result, Ordered"). On the right side of the interface, there is a "Quick Browser" section with buttons for "List", "View", "Print", "Forward", "Sort", "Dial", "Quick Note", "Flowcharts", "Letter", "Telephone Enc", "Review", and "Close".

**Figure 7.11.** Prompt notification of laboratory test results. When a messaging system is integrated with the CPR system, test results can be directed to the provider's in-basket as soon as they are available. By clicking on the Review button at the lower right corner, the clinician can retrieve the patient's CPR instantly and with it any relevant information that he/she reviewed before acting on the most recent result or message. Telephone messages and other patient-related information can be handled in the same manner. (Source: Courtesy of Epic Systems, Madison, WI.)



that contains instructions from the primary physician, as well as system-provided information (e.g., recent laboratory test results), is shown in Figure 12.11.

Although a patient encounter is usually defined by a face-to-face visit (e.g., outpatient visit, inpatient bedside visit, home health visit), provider decision making occurs in response to other events such as patient telephone calls about new symptoms or prescription renewal requests and the arrival of test results. Ideally, the clinician or a key office staff should be notified of these events and have the EHR tools needed to respond, including the patient's electronic chart content, mechanisms for electronic renewal authorizations, templates for producing reports to patients about normal test results, and back-to-work forms as shown in Figure 12.11. In addition, when the physician asks the patient to schedule a diagnostic test such as a mammogram, an EHR system can keep track of the time since the order was written and can notify the physician that a test result has not appeared in a specified time. This tracking function prevents diagnostic plans from falling through the cracks.

EHRs are usually bounded by the institution in which they reside. The National Health Information Infrastructure (NHII) (NCVHS, 2001) reaches beyond the single institution with an infrastructure that would permit an authorized provider caring for Mr. Jones to automatically obtain relevant medical data about that patient from other organizations within their community, e.g., from the hospital across town. Examples of such community-based "EHRs" are operating in Indianapolis (McDonald et al., 2005) and Santa Barbara to serve emergency, and other urgent, care needs. The Boston NEHEN Collaborative has created an analogous community-wide system for managing eligibility, preauthorization, and claim status information ([www.nehen.net/technol-ogy.htm](http://www.nehen.net/technol-ogy.htm)) (NEHEN, 2003).

Communication tools that support timely and efficient communication between patients and the health care team can enhance coordination of care and more effective disease management. eHealth applications can provide patients with secure access to their EHR and provide integrated communication tools to ask medical questions or perform other clinical (e.g., renew a prescription) or administrative tasks (e.g., schedule an appointment) conveniently online (Tang, 2003).

In addition to supporting communication among health care professionals and patients, community-based EHRs can facilitate the efficient creation and transmission of reports that support patient safety, quality improvement, public health, research, and other health care operations. (NCVHS, 2001) The creation of aggregate reports from such data, however, requires standardization as described in Chapter 10.

#### 7.4. Fundamental Issues for Computer-Based Patient Record Systems

All medical record systems must serve the same functions, whether they be automated or manual. From a user's perspective, the two approaches differ fundamentally in the way data are entered into and information is extracted from the record. In this section, we explore the issues and alternatives related to data entry and then describe the options for displaying and retrieving information from an EHR.

### 7.4.1. Data Capture

Data entry of information into the EHR remains a formidable challenge, because it requires significant time and effort on the part of its users. There are two general methods of **data capture** in EHRs: electronic interfaces from systems, such as laboratory systems that are already fully automated and manual data entry, when no such electronic source exists. A third method, scanning paper reports, is available, but unless OCR methods are used the resulting format is not searchable or understandable to the computer.

#### - Electronic Interfaces

The preferred method of capturing EHR data is to implement an electronic interface between the EHR and existing electronic data sources (e.g., laboratory systems, electronic instruments, registration systems, scheduling systems).

Though interfaces can require real effort as described under Section 12.3.1, they provide near-instant availability of the clinical data and avoid the labor costs and potential errors of manual transcription. Generally, the information technology department will invest resources needed to develop these electronic interfaces when the volume of data makes it worth the initial investment. This typically occurs when the organization that owns the EHR system also owns or is tightly affiliated with the owner of the source system. Efforts to capture data from systems outside the organizational boundary can be more difficult. Solutions require personal negotiations with sites that frequently provide care to a practice's patients and place extra work on the part of the practice.

The above discussion focuses on data produced by the home organization and/or produced by an outside organization in response to an order from the home organization. But even when all of the patient data produced, or ordered, by the home organization are captured, important clinical information produced by other organizations will not be included. For example, if the organization is a hospital system, it would not automatically capture pediatric immunizations. That information would be scattered across the pediatric and public health offices around town. So, the hospital would have to develop special procedures to collect immunization records if they wanted to help to improve immunization rates for children cared for in their clinics. The trend toward larger, more integrated, and more self-contained health care systems and community-based health information infrastructures described by the NHII report will tend to reduce such problems of data capture. Nevertheless, the standards required to move data faithfully and automatically from source systems to EHRs within a health care delivery system remain a significant challenge (see Chapters 7 and 13).

#### - Data Entry

The data-entry step is burdensome because of the personnel time required. People must interpret or translate the data, as well as enter them into the computer. Data may be entered in free-text form, in coded form, or in a form that combines both **free text** and **codes**. Trade-offs between the use of codes and narrative text exist.

The major advantages of coding are that data are classified and standardized, thus facilitating selective retrieval of patient data for clinical research, quality improvement, and administration. Coding lets the computer “understand” the data and thereby process them more intelligently. Immediate coding by physicians yields codes that the EHR can use to guide physicians’ decisions. If selection methods are carefully designed, physician coding can be more accurate than coding by other personnel. The major disadvantage of coding is the human time it takes to translate the source text into valid codes. There also is the potential for coding errors, which, in contrast to errors in free-text entry, are difficult to detect, because coded information lacks the internal redundancy of text. For example, a transposition error causing a substitution of code 392 for 329 may not be detected unless the computer displays the associated text and the data-entry operator notices the error. Natural-language processing offers hope for automatic encoding of narrative text (Hripcsak et al., 2004).

In Chapter 7, we described alternative schemes for classifying diagnoses and medical procedures. Various computer sources of coded data, including laboratory systems, pharmacy systems, and electrocardiogram (ECG) carts, exist in health care settings.

Data from these systems can flow automatically to the EHR through message standards such as HL7 (described in Chapter 7). Here, the challenge is the variety of local coding systems. The solution is to use standard coding systems—such as LOINC for identification of laboratory tests and clinical measurements (McDonald et al., 2003), and RxNorm NDFRT (Nelson et al., 2002; RxNorm Overview, A guide to RxNorm. Accessed Sep 17, 2005 [http://www.nlm.nih.gov/research/nmls/rxnorm\\_overview.htm/](http://www.nlm.nih.gov/research/nmls/rxnorm_overview.htm/)) or one of the commercial drug database codes for the identification of drug products, and SNOMED for diagnosis and findings (see Chapter 7).

#### - Physician-Entered Data

Physician-gathered patient information requires special comment because it presents the most difficult challenge to developers and operators of EHR systems. Physicians’ notes can be entered via one of three general mechanisms: transcription of dictated or written notes, entry of data recorded on structured encounter forms, or direct data entry by physicians (direct entry may include use of electronic templates or macros). Dictation and **transcription** is a common option for data entry of textual information into EHRs because it is widely and comfortably used by physicians. This method is especially attractive when the practice has already invested in dictation services, because then the cost of keying already has been absorbed. If physicians dictate their reports using standard formats (e.g., present illness, past history, physical examinations, and treatment plan), then the transcriptionist maintains this structure in the transcribed document. Voice recognition software offers a very attractive option for “dictating” without the cost or delay of transcription. The technology is improving as hardware processing speeds increase. However, even with accuracy rates of 95 to 98 percent, finding and correcting the few errors takes time and inhibits the use of this technology.

Although dictation is a preferred means of note writing, in many settings, physicians



are recognizing the benefits of direct data entry because it eliminates the turnaround time required for the dictation to be transcribed and for someone to review, correct, sign, and file the transcribed document. All these steps leave room for errors and delays in completing the record. The lack of encoding further reduces the benefits of using an EHR. The problem with direct physician entry of their notes via templates and typing is cost of physician time.

The second data-entry method is to have physicians use a **structured encounter form** from which their notes are transcribed and possibly scanned in which case some of the content can be automatically encoded via an optical character reader (OCR) and/or mark-sense reading while some is stored as scanned image as is done at Regenstrief (Downs et al., 2005) and Mayo (Hagen et al., 1998). The handwritten numbers on the left side and the computer-printed numbers at the bottom left of Figure 12.5 are read by OCR techniques.

The third alternative is the **direct entry** of data, by a care provider, via a computer. This alternative has the advantage that the computer can immediately check the entry for consistency with previously stored information and can generate reminders about further questions that should be asked based on the information just entered.

Physician orders are specially important for the medical record because they specify what is to be done for the patient. Because of the many potential advantages for care

quality and efficiency, organizations are being encouraged to move to direct computerized physician order entry (CPOE). CPOE can be facilitated by custom menus that contain standing orders for specific problems (e.g., postoperative orders for patients who undergo coronary artery bypass operations). Menus must be carefully structured. They must not contain lists that are too long, require scrolling, or impose a rigid hierarchy (Kuhn et al., 1984). Direct entry of the patient's history, physical findings, and progress notes has been challenging because of the extra time it takes the physician to enter such information into the computer compared with scribbling a note. Yet some physicians do enter all of their notes into the computer in some EHRs, while transcribed dictation remains an important source of physicians' notes in many EHR implementations.

#### - Data Validation

Because of the chance of transcription errors occurring when clinical information is entered into the computer, EHR systems must apply **validity checks** scrupulously. A number of different kinds of checks apply to clinical data (Schwartz et al., 1985). **Range checks** can detect or prevent entry of values that are out of range (e.g., a serum potassium level of 50.0 mEq/L—the normal range for healthy individuals is 3.5 to 5.0 mEq/L). **Pattern checks** can verify that the entered data have a required pattern (e.g., the three digits, hyphen, and four digits of a local telephone number). **Computed checks** can verify that values have the correct mathematical relationship (e.g., white blood cell differential counts, reported as percentages, must sum to 100). **Consistency checks** can detect errors by comparing entered data (e.g., the recording of cancer of the prostate as the diagnosis for a female patient). **Delta checks** warn of large and unlikely differences between the values of a new result and of the previous observations (e.g., a recorded weight that changes by 100 pounds

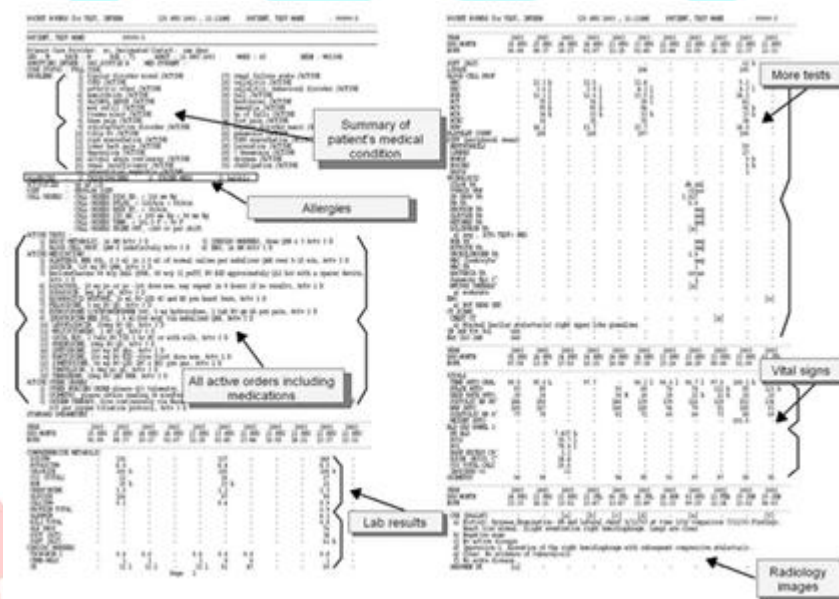
in 2 weeks). **Spelling checks** verify the spelling of individual words. No such syntactic checks can catch all errors.

#### 7.4.2. Data Display

Once stored in the computer, data can be presented in numerous formats for different purposes without further entry work. In addition, computer-stored records can be produced in novel formats that are unavailable in manual systems. We discuss a few helpful formats.

##### - Flowsheets of Patient Data

A **flowsheet** is similar to a spreadsheet; it organizes patient data according to the time that they were generated, thus emphasizing changes over time. Figure 12.12 shows the popular pocket rounds report that provides laboratory and nursing measurements as a very compact flowsheet that fits in a white coat pocket (McDonald et al., 1992). A flowsheet used to monitor patients who have hypertension (high blood pressure) might contain values for weight, blood pressure, heart rate, and doses of medications that control hypertension. Other pertinent information could be added, such as results of laboratory tests that monitor complications associated with hypertension, or complications of



**Figure 7.12.** The very popular Pocket rounds report - so called because it fits in the clinician's white coat pocket when folded in half. The report is a very dense (16 lines per inch 2 characters per inch) printed in landscape mode on one 8 1/2x11 inch page) summary of the patient's medical state, and includes the all active orders (including medications), recent laboratory results, vital signs and the summary impressions of radiology, endoscopy, and cardiology reports (*Source: Courtesy of Regenstrief Institute, Indianapolis, IN.*)

medications used to control hypertension. Flowsheets are potentially problem-, patient-, or specialty-specific. The time granularity may change from one setting to another. For example, when a patient is in the ICU, minute-to-minute changes in the patient's clinical state may be of interest. On the other hand, an outpatient physician is

more likely to want to know how that patient's data have changed over weeks or months. For convenience of human review, the temporal granularity should be appropriate to the intensity of care. Thus, of the hundreds of blood pressure measurements taken every 20 minutes in an ICU not all are of interest to the physician taking care of the same patient later in the clinic after the patient's condition has stabilized.

#### - Summaries and Abstracts

EHRs can highlight important components (e.g., active allergies, active problems, active treatments, and recent observations) in a clinical summary (Tang et al., 1999). In the future, we can expect more sophisticated summarizing strategies, such as automated detection of adverse events (Bates, 2003) or automated time-series events (e.g., cancer chemotherapy cycles). We may also see reports that distinguish abnormal changes that have been treated from those that have not and displays that dynamically organize the supporting evidence for existing problems (Tang, 1994a). Ultimately, computers should be able to produce concise and flowing summary reports that are like an experienced physician's hospital discharge summary.

#### - Dynamic Displays

Anyone who has reviewed a patient's chart knows how hard it can be to find a particular piece of information. From 10 percent (Fries, 1974) to 81 percent (Tang et al., 1994b) of the time, physicians do not find patient information that has been previously recorded and belongs in the medical record. Furthermore, the questions clinicians routinely ask are often the ones that are difficult to answer from perusal of a paper-based record. Common questions include whether a specific test has ever been performed, what kinds of medications have been tried in the past, and how the patient has responded to particular treatments (e.g., a class of medications) in the past. Physicians constantly ask these questions as they flip back and forth in the chart searching for the facts to support or refute one in a series of evolving hypotheses. Search tools help the physician to locate relevant data, and specialized presentation formats (e.g., flowsheets or graphics) make it easier for them to glean information from the data. Special displays can identify problem-specific parameters to help the physician retrieve relevant information, and a graphical presentation can help the physician to assimilate the information quickly and to draw conclusions (Fafchamps et al., 1991; Tang et al., 1994a; Starren and Johnson, 2000).

#### 7.4.3. Query and Surveillance Systems

The **query** and **surveillance** capabilities of computer-stored records have no counterpart in manual systems. Medical personnel, quality and patient safety professionals, and administrators can use these capabilities to analyze patient outcomes and practice patterns. Public health professionals can use the reporting functions of computer-stored records for surveillance, looking for emergence of new diseases or other health threats that warrant medical attention.

Although these functions are different, their internal logic is similar. In both, the central procedure is to examine a patient's medical record and, if the record meets pre-



specified criteria, to generate an appropriate output. Queries generally address a large subset, or all, of a patient population; the output is a tabular report of selected raw data on all the patient records retrieved or a statistical summary of the values contained in the records. Surveillance generally addresses only those patients under active care; its output is an **alert or reminder message** (McDonald, 1976). Query and surveillance systems can be used for clinical care, clinical research, retrospective studies, and administration.

- **Clinical Care**

Computer reminders have increased substantially physicians' use of preventive care for eligible patients. Surveillance systems can identify patients who are due for periodic screening examinations such as immunizations, mammograms, and cervical Pap tests and can remind physicians to perform these procedures during the next visit. For example, physicians given computer reminders quadrupled the use of certain vaccines in eligible patients compared with those who did not receive reminders (McDonald et al., 1984a; McPhee et al., 1991; Hunt, 1998; Teich, 2000). Query systems are particularly useful for conducting ad hoc searches such as those required to identify and notify patients who have been receiving a recalled drug. Such systems can also facilitate quality management and patient safety activities. They can identify candidate patients for concurrent review and can gather many of the data required to complete such audits.

- **Clinical Research**

Query systems can be used to identify patients who meet eligibility requirements for prospective clinical trials. For example, an investigator could identify all patients seen in a medical clinic who have a specific diagnosis and meet eligibility requirements while not having any exclusionary conditions. Surveillance facilities can support the execution of a study by tracking patients through their visits and by following the steps of a clinical trial as described in the study protocol to ensure that treatments are given and measurements obtained when required (Tierney and McDonald, 1991). Some existing systems use reminder logic to request the provider's permission to invite a patient into an ongoing study. The entry of back pain as the visit diagnosis, for example, could trigger a request about an ongoing trial of physical therapy for back pain. If the physician agrees, the computer sends an electronic page to the nurse recruiter for that study so that he/she can get to the clinic to invite the patient into the study.

- **Retrospective Studies**

Randomized **prospective studies** are the gold standard for clinical investigations, but **retrospective studies** of existing data have contributed much to medical progress. Retrospective studies can obtain answers at a small fraction of the time and cost of comparable prospective studies.

EHR systems can provide many of the data required for a retrospective study. They can, for example, identify study cases and comparable control cases, and they can perform the statistical analyses needed to compare the two groups (McDonald et al., 2005; Bleich et al., 1989; Safran, 1991; Mahon et al., 2000).

Computer-stored records do not eliminate all the work required to complete an epidemiologic study; chart reviews and patient interviews may still be necessary if some patient information is recorded as narrative text. The more information that can be retrieved from the computer record, however, the less frequently and less intensively such time-consuming tasks must be conducted. Computer-stored records are likely to be most complete and accurate with respect to drugs administered, laboratory test results, and visit diagnoses, especially if the first two types of data are entered directly from automated laboratory and pharmacy systems. Consequently, computer-stored records are most likely to contribute to research on a physician's practice patterns, on the efficacy of tests and treatments, and on the toxicity of drugs.

#### - Administration

In the past, administrators had to rely on data from billing systems to understand practice patterns and resource utilization. However, claims data are notoriously unreliable for understanding clinical practice because the source data are often entered by non-clinical personnel not directly involved with the care decisions. Medical query systems can provide information about the relationships among diagnoses, indices of severity of illness, and resource consumption. Thus, query systems are important tools for administrators who wish to make informed decisions in the increasingly cost-sensitive world of health care.

### 7.5. Challenges Ahead

Although many commercial products are labeled as EHR systems, they do not all satisfy the criteria that we defined at the beginning of this chapter. Even beyond matters of definition, however, it is important to recognize that the concept of an EHR is neither unified nor static. As the capability of technology evolves, the function of the EHR will expand. A review of current products would be obsolete by the time that it was published. We have included examples from various systems in this chapter, both developed by their users and commercially available, to illustrate a portion of the functionality of EHR systems currently in use.

The future of EHR systems depends on both technical and nontechnical considerations. Hardware technology will continue to advance, with processing power doubling every 2 years according to Moore's law. Software will improve with more powerful applications, better user interfaces, and more integrated decision support. Perhaps the greater need for leadership and action will be in the social and organizational foundations that must be laid if EHRs are to serve as the information infrastructure for health care. We touch briefly on these challenges in this final section.

#### 7.5.1. Users' Information Needs

We discussed the importance of clinicians directly using the EHR system to achieve

maximum benefit from computer-supported decision making. On the one hand, organizations that require providers to enter all of their order, notes, and data directly into the EHR will gain the most efficiency and care quality. But it will be the physicians who will bear the time costs of entering this information and face disruptions to their routines. So, some balance between the organization's and providers' interests must be struck, and this balance is easiest to reach when the physicians are part of the organization's leadership.

Developers of EHR systems must thoroughly understand clinicians' information needs and workflows in the various settings where health care is delivered. The most successful systems have been developed either by clinicians or through close collaborations with practicing clinicians.

Studies of clinicians' information needs reveal that common questions that physicians ask concerning patient information (e.g., Is there evidence to support a specific patient diagnosis? Has a patient ever had a specific test? Has there been any follow up because of a particular laboratory test result?) are difficult to answer from the perusal of the paper-based chart (Tang et al., 1994b). Regrettably, most clinical systems in use now cannot answer many of the common questions that clinicians ask. Developers of EHR systems must have a thorough grasp of users' needs and workflows if they are to produce systems that help health care providers use these tools efficiently to deliver care effectively.

### 7.5.2. User Interfaces

**An intuitive and efficient user interface is an important part of the system.** Designers must understand the cognitive aspects of the human and computer interaction if they are to build interfaces that are easy-to-learn and easy-to-use. Improving human-computer interfaces will require changes not only in how the system behaves but also in how humans interact with the system. What information the provider needs and what tasks the provider performs should influence what and how information is presented. Development of human-interface technology that matches the data-processing power of computers with the cognitive capability of humans to formulate insightful questions and to interpret data is still a rate-limiting step (Tang and Patel, 1993). User interface requirements of clinicians entering patient data are different from the user interfaces developed for clerks entering patient charges. Among the most potent functions of EHRs is CPOE. To facilitate use by busy health care professionals, health care applications developers must focus on clinicians' unique information needs.

### 7.5.3. Standards

We alluded to the importance of standards earlier in this chapter, when we discussed the architectural requirements of integrating data from multiple sources. Standards were discussed in Chapter 7. Here, we stress the critical importance of national standards in the development, implementation, and use of EHR systems (McDonald et al., 1997b). Health information should follow patients as they interact with different providers in different care settings. Uniform standards are essential for systems to interoperate and exchange data in meaningful ways. Having standards reduces development costs, increases integration, and facilitates the collection of meaningful aggregate data for quality



improvement and health policy development. The HIPAA legislation has mandated standards for administrative messages, privacy, security, and clinical data. Regulations based on this legislation have been promulgated for the first three of these categories. These can be obtained at <http://www.cms.hhs.gov/hipaa/hipaa2/regulations/default.asp>. A Consolidated Health Informatics (CHI) group within the U.S. federal government is selecting standards that will be required for transmission of clinical data within federal agencies. They have selected HL7 Version. 2.4 as the primary message standard, DICOM for radiology images, LOINC for laboratory test observations. They are working on a host of additional selections to cover units of measures, test answers, medication identifier codes, general clinical terminology (likely SNOMED), and more (see Chapter 7). The interested reader can follow their progress at the Web site <http://www.hhs.gov/healthit/chi.html>. A public-private consortium, connecting for Health, is also making recommendations that have influenced CHI. Standards for clinical data transmission will not be mandated in the United States but they will be encouraged and incented. Those who wish to implement EHRs should promote and adopt clinical information standards.

#### 7.5.4. Legal and Social Issues

In addition to legislation on standards, federal laws and policies regarding other aspects of the use of EHR systems must be established before widespread adoption will occur. HIPAA has also established key regulations to protect the confidentiality of individually identifiable health information and the security of computer systems that store and transmit patient information (see <http://cms.hhs.gov/hipaa>). With appropriate laws and policies computer-stored data can be more secure and confidential than those data maintained in paper-based records (Barrows and Clayton, 1996).

#### 7.5.5. Costs and Benefits

The Institute of Medicine declared the EHR an essential infrastructure for the delivery of health care (Institute of Medicine Committee on Improving the Patient Record, 1997) and the protection of patient safety (IOM, 2001). Like any infrastructure project, the benefits specifically attributable to infrastructure are difficult to establish; an infrastructure plays an enabling role in all projects that take advantage of it. Part of the difficulty in comparing costs and benefits of an EHR is our inability to measure accurately the actual costs and opportunity costs of using paper-based records. Many randomized controlled clinical studies have shown that computer-based decision-support systems that are integrated in an EHR reduce costs and improve quality compared with usual care supported with a paper medical record (Tierney et al., 1993; Bates et al., 1997, 2003; Classen et al., 1997). It is difficult, however, to determine the scalability and longevity of such benefits.

Because of the significant resources needed and the significant broad-based potential benefits, the decision to implement an EHR system is a strategic one. Hence, the evaluation of the costs and benefits must consider the effects on the organization's strategic goals, as well as the objectives for individual health care. Recently, the federal government and professional organizations have both expressed interest in the **Open Source** options for EHR software (McDonald et al., 2003).

### 7.5.6. Leadership

Leaders from all segments of the health care industry must work together to articulate the needs, to define the standards, to fund the development, to implement the social change, and to write the laws to accelerate the development and routine use of EHR systems in health care. Because of the prominent role of the federal government in health care—as a payer, provider, policymaker, and regulator—federal leadership to create incentives for developing and adopting standards and for promoting the implementation and use of EHRs is crucial. Technological change will continue to occur at a rapid pace, driven by consumer demand for entertainment, games, and business tools. Nurturing the use of information technology in health care requires leaders who promote the use of EHR systems and work to overcome the obstacles that impede widespread use of computers for the benefit of health care.

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